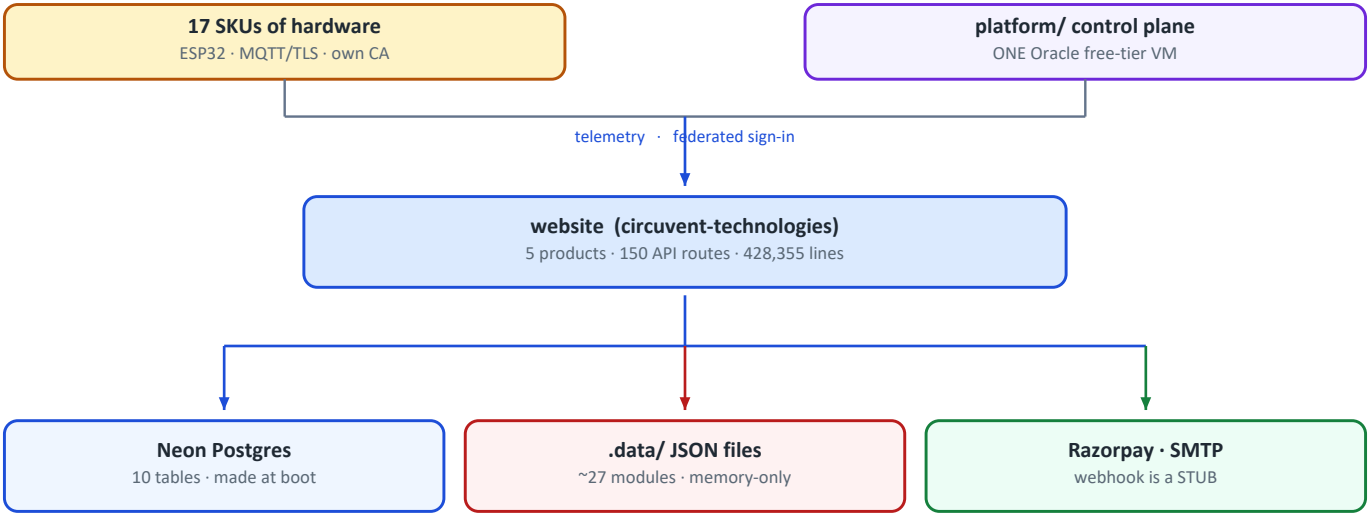


Circuvent Technologies

Architecture & Technical Audit

Circuvent Technologies
Next.js 16 · Neon · ESP32 · MQTT/TLS · Expo · KiCad
Generated 2026-08-20



It is called "website". It is a company: a storefront, an IoT platform, firmware for a 17-product line, PCB designs and a shipped mobile app.

Architecture	Five web products, firmware, hardware and two native apps in one repository
Scale	23,184 files · 1,810 src TypeScript · 428,355 lines · 150 API routes · 108 pages
Database	Neon Postgres, HTTP driver · 10 tables · CREATED AT RUNTIME · no migrations
Sessions	Home-grown HMAC-SHA256 — no JWT library is used anywhere in src/
Devices	17 retail SKUs · ESP32 · MQTT/TLS on an own broker with an own CA
Tests	236 suites · 4,328 tests · 1 failing · 49 seconds
CI/CD	14 steps, 13 hard gates — and 27 of 27 runs failed to start
Primary risk	~27 storage modules are memory-only in production, passkeys among them

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PART 1

01 · System Overview

Audience: everyone. An intern reads §1–§5; a CTO reads §1 and §11. **System:** the repository called **website** — which is a marketing site, an e-commerce store, an admin console, an IoT SaaS platform, a developer portal, firmware, hardware designs and two native apps.

1. Executive summary

THE FIRST THING TO KNOW

README.md says:
"Premium portfolio and services website... showcasing 53+ projects across 6 technology domains"
"src/app – Next.js App Router pages (18 routes)"
"components/ – 50+ React components"

THE REPOSITORY ACTUALLY CONTAINS:
23,184 files · 428,355 lines · 150 API routes · 108 pages
a full e-commerce storefront with payments
an 83-route admin back office
a smart-home / IoT SaaS console with 60+ sections
a public developer API portal
13,003 files of firmware · hardware designs
two separate native mobile codebases

Someone onboarding from the README would misunderstand this system by roughly an order of magnitude.

At a glance

Type	A company monorepo — five web products plus firmware, hardware and native apps
Framework	Next.js 16.2.11 App Router · React 19.2.3 · TypeScript 5 strict · Tailwind 4
Scale	23,184 files · 1,810 src TypeScript files · 428,355 lines · 150 API routes · 108 pages
src/lib	288 modules
Database	Neon Postgres, HTTP driver · 10 tables, created at runtime · no migrations · no ORM
Sessions	Home-grown HMAC-SHA256 tokens. No JWT library is used at all.
Tests	370 test files · Jest 30 + Playwright + PGlite
Repository	github.com/Hemakotibonthada/WebSite.circuvent · 529 commits · 5 remotes · 10+ branches
Payments	Razorpay — and the webhook is a stub

The five decisions that define it

#	Decision	Consequence
1	Everything in one Next app	One deployment, one build. Also: a marketing page and an IoT fleet console share a bundle, a router and a security posture
2	No JWT, no cookies — HMAC tokens in headers	Simple, dependency-free, revocable by version. Also: entirely separate from every other Circuvent application
3	Schema created by the app at boot	Zero-friction new environments. Also: no history, no rollback, no review
4	A JSON-file store that degrades to memory	Nothing crashes when the disk is read-only. Also: ~27 modules silently lose all data on every cold start

#	Decision	Consequence
5	Write down the incident in the file that fixes it	The single most valuable convention in the repository. Quoted throughout these documents

2. What is actually in here

ONE NEXT.JS APPLICATION, FIVE PRODUCTS

1. MARKETING / PORTFOLIO ← the only part the README describes about · architecture · blog · careers · case-studies · docs domains · faq · open-source · privacy · projects · roadmap services · stack · team · terms · contact

2. E-COMMERCE STOREFRONT
shop · shop/[slug] · shop/c/[category] · shop/account
shop/invoice/[orderNo] · shop/devices
cart · checkout · track · warranty · returns-policy · shipping

3. ADMIN BACK OFFICE 83 of the 150 API routes
admin · admin/insights · admin/inventory
inventory · orders · CMS · marketing · pricing · tax · fraud
ICM incident management · App Insights telemetry clone

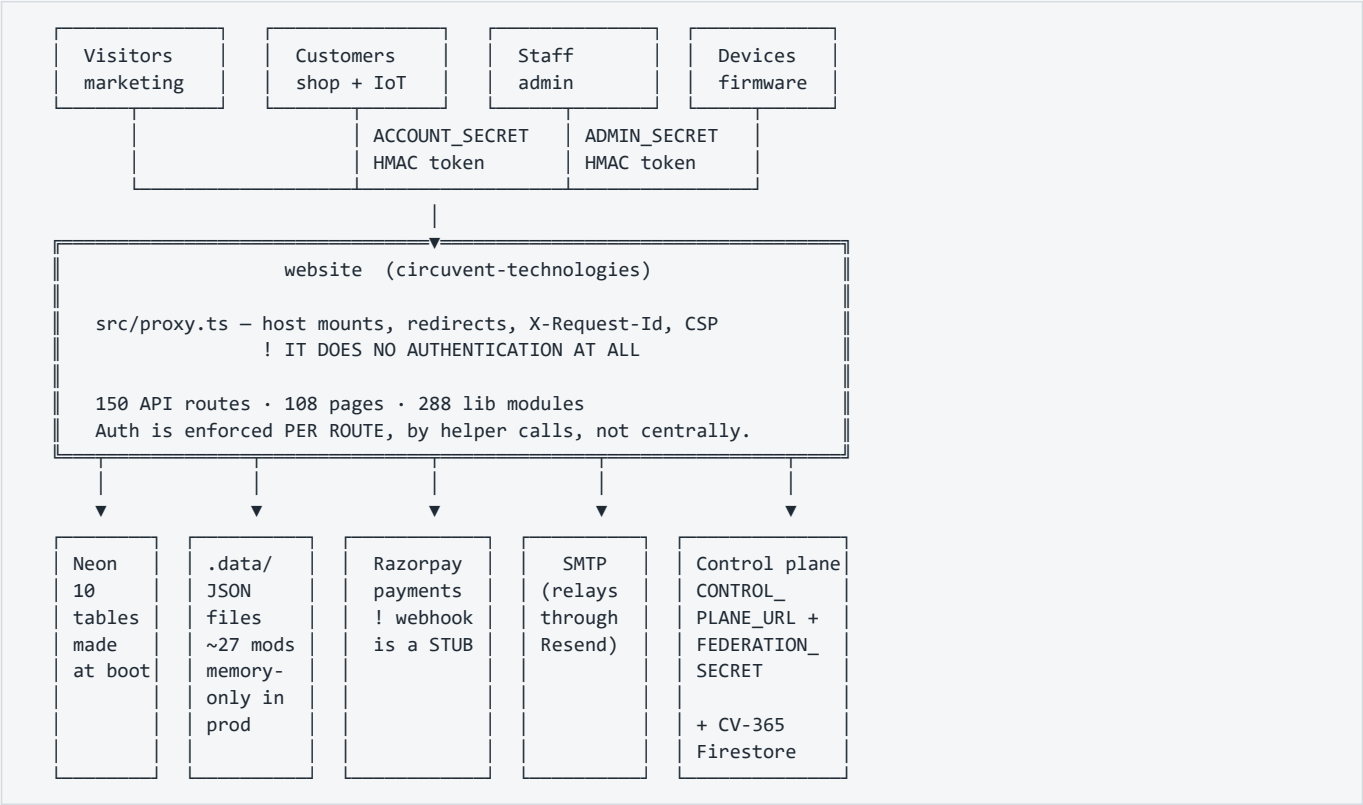
4. SMART-HOME / IoT SaaS CONSOLE the largest surface
60+ sections: admin/{access,fleet,ota,provisioning,registry,telemetry} · automation · camera · security · energy · solar
drone · anpr · attendance · kiosk · command-center · floorplan
groups · rooms · scenes · recipes · reports · widgets

5. PUBLIC DEVELOPER PORTAL
authentication · browser · commands · endpoints · errors
limits · quickstart · scopes · webhooks

Plus, outside src/: firmware/ · hardware/ · mobile/ · native/
platform/ · circuitent-platform/ – see doc 03

! THERE ARE NO NEXT.JS ROUTE GROUPS. Not one `(name)` folder.
Every top-level directory is a literal URL segment, so the five
products share one flat router with no structural boundary between
a public marketing page and a fleet-management console.

3. Topology



Mermaid diagram — renders interactively in the Markdown edition

```
C4Context
    title website - system context
    Person(vis, "Visitor", "Marketing and portfolio")
    Person(cust, "Customer", "Shop, orders, smart-home console")
    Person(staff, "Staff", "83-route admin back office")
    Person_Ext(dev, "Integrator", "Public developer portal")

    System(web, "website", "Next.js 16 - five products in one app, 428k lines")

    System_Ext(neon, "Neon Postgres", "10 tables, created at runtime, no RLS")
    System_Ext(files, "JSON file store", "~30 modules; only 3 durable")
    System_Ext(rzp, "Razorpay", "Payments - webhook is a stub")
    System_Ext(smtp, "SMTP relay", "Relays through Resend")
    System_Ext(cp, "Control plane", "Federated SSO, HMAC handoff")
    System_Ext(fb, "CV-365 Firestore", "Contact form duplicate write")
    System_Ext(iot, "IoT devices", "firmware/ - see doc 03")

    Rel(vis, web, "Browse")
    Rel(cust, web, "Shop + device control")
    Rel(staff, web, "Administer")
    Rel(dev, web, "Read API docs")
    Rel(web, neon, "neon() HTTP driver - no transactions")
    Rel(web, files, "createFileStore")
    Rel(web, rzp, "Create order, verify capture")
    Rel(web, smtp, "nodemailer")
    Rel(web, cp, "FEDERATION_SECRET")
    Rel(web, fb, "Contact form, lazily loaded")
    Rel(iot, web, "Telemetry and commands")
```

4. Module map

Figure — module and layer map

THE GATE

src/proxy.ts - host mounts, redirects, CSP - AND NO AUTHENTICATION

ROUTES

150 handlers - each enforces its own auth, five schemes coexist

AUTHORIZATION

AdminRole x AdminArea capability gate - 5 roles, ~20 areas

DOMAIN MODULES

288 lib files - icm, commerce, smarthome, telemetry, reports

STORAGE CHOICE

db.ts (10 tables) OR data-file.ts (JSON) - only 3 of ~30 durable

POSTGRES

Neon HTTP driver - no transactions, no RLS, no foreign keys

Fourteen CI steps would check this stack. Twenty-seven runs have failed to start.

src/lib - 288 modules

ADMIN BACK OFFICE	29 files	admin-*.ts - pricing, tax, CRM, fraud, vendors, SSO provisioning, password policy
SMART HOME	~30 files	device auth, cameras, energy budget, geofencing, recipes, scene scheduler
INCIDENT MANAGEMENT	14 files	icm*.ts - severity, TTA/TTM state machine, Teams notify, postmortems. icm.ts is PURE
APP INSIGHTS	9 files	a working clone of Azure Monitor's cost, usage, dependency and query telemetry
REPORTS	10 files	CSV / PDF / scheduled generation, charts
COMMERCE	16 files	shop-*, order-core, bundle-pricing, coupons, delivery-estimate, warranty, inventory, razorpay
AUTH & SECURITY	10 files	passkeys, passkey-ceremony, admin-auth, account, sso, totp, session-expiry, secrets
TELEMETRY	~20 files	health probes, synthetic checks, anomaly monitor, cron health, control-plane, guardian, tank, fan-speed, predictive maintenance, traffic, visitor tracking
MARKETING CONTENT	17 files	PURE constant exports
INFRASTRUCTURE	18 files	cache, config, csp, client-ip, data-file, db, logger, rate-limit, validation, fuzzy
DEVICE LOGIC	8 files	agri, avi, device-history, device-normalize, camera-audio, camera-relay, drone-mixer, switchboard
NOTIFICATIONS	5 files	alerts, email-log, web-push
SEO / BRAND	5 files	metadata, OG image generation

PURE modules (arguments in, values out, no I/O):

 icm.ts · session-expiry.ts · delivery-estimate.ts · device-normalize.ts
 validation.ts · fuzzy.ts · all *-data.ts · most seo/brand builders
The twelve largest files

#	File	Lines
1	src/app/smarthome/DeviceControls.tsx	4,870
2	src/app/admin/AppInsightsPanel.tsx	2,208
3	src/lib/control-plane.ts	2,182
4	src/lib/store.ts	2,071
5	src/app/admin/IcmPanel.tsx	1,823
6	src/app/smarthome/security/VehiclesPanel.tsx	1,539
7	src/lib/blog-data.ts	1,309
8	src/app/shop/account/page.tsx	1,284

#	File	Lines
9	<code>src/lib/smarthome-command-map.ts</code>	1,272
10	<code>src/components/DataVisualization.tsx</code>	1,261
11	<code>src/app/smarthome/_kit/primitives.tsx</code>	1,244
12	<code>src/app/admin/page.tsx</code>	1,225

A single 4,870-line React component is the largest file in the repository. Doc 05.

5. The 150 API routes

Group	#	Auth	Purpose
admin/	83	Staff HMAC token + requireArea role gate. Plus two exceptions: admin/external-stats uses a static ADMIN_API_KEY , and admin/availability/probe uses CRON_SECRET	The whole back office
account/*	14	Customer HMAC token for 9; none for the 5 pre-session routes (login, register, verify-otp, forgot/reset-password)	Identity and self-service
smarthome/*	10	! Not resolved — matched neither known session helper	Device alerts, cameras, prefs
shop/*	5	None — public catalogue	Bundles, products, questions, quote, reviews
devices/*	5	Customer token for claim/command; firmware/sync unresolved	Claim and command devices
orders/*	4	Customer token; track is public by order number	Order lifecycle
payments/*	3	create-order public (guest checkout) · webhook HMAC only · verify	Razorpay
ai/*	3	chat customer token; analyze/fleet unresolved	AI chat and fleet analysis
health · push · telemetry · visitors · wallet	2 each	Mostly public beacons; wallet and push/subscribe require a token	
13 singletons	13	blog, contact, coupons/validate, giftcards/redeem, github, loyalty, newsletter, notify-restock, projects, referral, returns, support, weather	
Total	150		reconciled exactly

! SEVEN ROUTES COULD NOT BE RESOLVED to any known authentication mechanism by a repository-wide sweep:

```

giftcards/redeem  referral      returns
ai/analyze        ai/fleet
devices/firmware  devices/sync
...plus all 10 smarthome/* routes

```

That means either an undiscovered check, or a genuine gap.
`giftcards/redeem` and `returns` are the ones to check first — both move money. Doc 05, D-06.

6. Authentication — home-grown, and entirely separate from the suite

A repository-wide search across src/ for
AUTH_JWT_SECRET · JWT_SECRET · jsonwebtoken · jose
returns ZERO MATCHES.

Every other Circuvent application shares one HS256 suite session.
This one does not participate in it at all.

```
// src/lib/account.ts
function sign(payload: string): string {
  return crypto.createHmac("sha256", secret()).update(payload).digest("hex");
}
export function signToken(email: string, version?: number): string {
  const payload = `${e}|${Date.now().toString(36)}|${ver}`;
  return Buffer.from(`${payload}:${sign(payload)}`).toString("base64");
}
```

Property	Detail
Algorithm	HMAC-SHA256 over `email\`
Customer key	ACCOUNT_SECRET
Staff key	ADMIN_SECRET — "Staff sessions get their own key so a leaked customer key cannot mint one."
Transport	Not cookies. No Set-Cookie anywhere in src/app/api. Tokens travel in Authorization: Bearer, x-account-token or x-admin-token
Storage	Client-side; admin tokens explicitly in sessionStorage
Revocation	tokenVersion — bump it and every prior token dies
Lifetime	24-hour absolute cap via session-expiry.ts, independent of renewal

Why tokenVersion exists

src/lib/admin-auth.ts — *"The old format was base64(email + ":" + HMAC(email))... It never expired and could not be revoked, so changing a password left every previously issued token fully valid, including one a departing employee had already copied."*

Five ways in

Path	Mechanism
Password	scrypt + timing-safe compare
Passkey	@simplewebauthn/server, single-use 5-minute challenges, cloned-authenticator detection via counter regression
Email OTP	6-digit code via crypto.randomInt, sign-up step two
Password reset	An OTP code, not a magic link, 15-minute TTL
SSO federation	verifyAgainstControlPlane() adopts a control-plane password when the local account has none

```
/** Which sign-in a credential belongs to. They must never be interchangeable. */
export type PasskeyScope = "admin" | "account";
```

Authorization — capability areas, not just roles

```
export type AdminRole = "superadmin" | "manager" | "inventory" | "orders" | "support";

export type AdminArea =
  | "overview" | "analytics" | "inventory" | "orders" | "returns" | "customers"
  | "coupons" | "support" | "staff" | "settings" | "cms" | "marketing"
  | ... | "icm" | "insights";

const ROLE_AREAS: Record<AdminRole, AdminArea[]> = { superadmin: [...], ... };
```

Admin 2FA is TOTP or email code, gated by a **TOTP_PENDING** sentinel proving the password stage passed first. The QR code is rendered **server-side**, deliberately, *so the secret is never sent to a third-party QR service*.

The gate does not gate

`src/proxy.ts` (Next 16's rename of `middleware.ts`) handles host mounts, a legacy `/shop?cat=X` → `/shop/c/x` 308 redirect, `X-Request-Id` propagation and the CSP header.

It performs no authentication whatsoever. Every route enforces its own. With at least five coexisting credential schemes and seven unresolved routes, that is the structural weakness of this design. Doc 05, D-05.

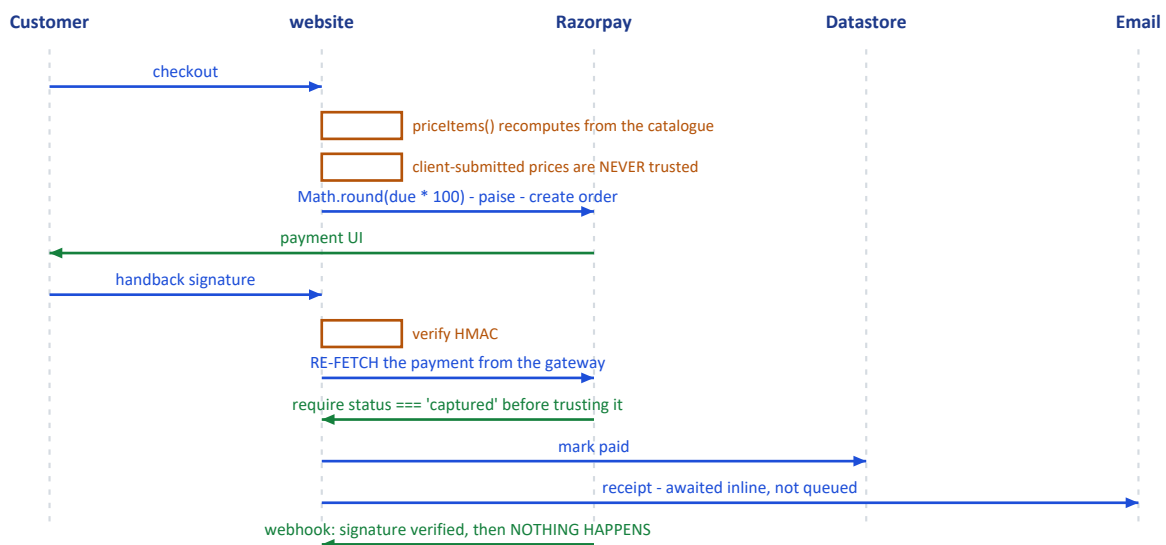
Security headers, on the other hand, are complete

```
const securityHeaders = [  
  { key: "Strict-Transport-Security", value: "max-age=63072000; includeSubDomains; preload" },  
  { key: "Content-Security-Policy", value: CSP },  
  { key: "X-Content-Type-Options", value: "nosniff" },  
  { key: "X-Frame-Options", value: "DENY" },  
  { key: "Referrer-Policy", value: "strict-origin-when-cross-origin" },  
  { key: "X-DNS-Prefetch-Control", value: "on" },  
  { key: "Permissions-Policy", value: "camera=(), microphone=(), geolocation=(self), browsing-topics=()" },  
  { key: "Cross-Origin-Opener-Policy", value: "same-origin" },  
];
```

Applied both at the edge in `proxy.ts` and globally via `headers()`, so routes the proxy matcher skips are still covered. Non-production deployments additionally get `X-Robots-Tag: noindex, nofollow`.

7. Commerce — and the webhook that does nothing

Figure — checkout and payment, end to end



Mermaid diagram — renders interactively in the Markdown edition

```
sequenceDiagram
    autonumber
    participant C as Customer
    participant W as website
    participant R as Razorpay
    participant D as Datastore

    C->>W: checkout
    W->>W: priceItems() – recomputes totals from the catalogue
    Note over W: Client-submitted prices are NEVER trusted
    W->>W: Math.round(due * 100) → paise
    W->>R: create order
    R-->>C: payment UI
    C->>W: handback signature
    W->>W: verify HMAC
    W->>R: re-fetch the payment
    Note over W,R: requires status === "captured" before trusting the amount
    W->>D: mark paid, email receipt

    R->>W: POST /api/payments/webhook
    W->>W: verify HMAC – correctly, with timingSafeEqual
    W-->>D: NOTHING. It only console.log()s.
```

THE WEBHOOK IS A STUB.

```
const expected = crypto.createHmac("sha256", secret).update(raw).digest("hex");
const valid = sigBuf.length === expBuf.length && crypto.timingSafeEqual(sigBuf, expBuf);
```

The signature check is correct and constant-time. Then the handler logs the event and returns. Its own comment admits it:

```
// Reconciliation hook: when a persistent order store exists,
// mark the order paid/failed here...
```

So any payment that completes at the gateway but whose browser never returns – a closed tab, a dropped connection, a failed redirect – is captured by Razorpay and never reconciled here. Doc 05, D-04.

Money is a hybrid. The catalogue uses `price: number; // whole INR` — a float in major units. The gateway boundary uses `amountPaise: number` — an integer in minor units, converted once with `Math.round(due * 100)`. No fractional rupee exists in the sample data, so no wrong number is demonstrable today. But nothing in the type system prevents one.

delivery-estimate.ts is a good module. Pure, no courier API, buckets Indian PIN-code prefixes into metro / remote / no-COD sets, and returns a **range** with explicit "estimate" framing rather than a firm promise.

8. Notifications and documents

Dependency	Reality
<code>nodemailer</code>	YES Real. SMTP transport for orders, OTPs, password resets, contact. Awaited inline, not queued — failures are logged and the request proceeds
<code>resend</code>	! Declared but its SDK is never imported. Mail goes out over SMTP, which <i>relays through</i> Resend. A direct Resend-API fallback was deliberately removed
<code>web-push</code>	YES Real browser push
<code>pdf-lib</code>	YES Real, server-side admin report generation
<code>qrcode</code>	YES One import site — TOTP enrolment, rendered server-side on purpose

`src/app/api/contact/route.ts` documents why the Resend path went away: "This route used to call the Resend API directly, from `onboarding@resend.dev` to a hardcoded gmail address... these sends consumed the Resend free allowance without ever appearing in the outbound counts."

And every contact submission is written twice. `ContactForm.tsx` calls `saveContactMessage()` (Firestore, in a separate CV-365 project) and `fetch("/api/contact")` (local store + SMTP) on the same submit. Two independent persistence paths for one form. Doc 05, D-12.

9. The comment convention — the best thing in this repository

Module headers name the exact incident that motivated the module. These are worth reading in full.

`src/lib/passkeys.ts` — *"The update branch used to store whatever case the caller passed, so re-registering an existing credential wrote back `Mixed@Example.COM` and every later lookup — which normalises — stopped finding it. **The passkey still existed, still verified, and belonged to nobody.** It only appeared on the second save of the same credential, which is why it survived the first run of its own test."*

`src/lib/sso.ts` — *"So on dev, a sign-in that missed locally fell through to `api.circuvent.com/auth/login`, the live fleet vouched for a real customer, and `/api/account/login` then created that customer in the dev database with a hash of their real password. **Production users could sign in to dev, and dev quietly accumulated live credentials while doing it.** The isolation guard was pointed at the wrong door."*

`next.config.ts` — *"A dev server and a production build share `.next`, and the dev server keeps rewriting it — which silently deleted `BUILD_ID` twice while auditing, so `next start` served nothing and **the audit reported a clean sweep of an empty site.**"*

`scripts/verify-icm-durability.ts` — *"Incidents were written to a JSON file that the serverless host cannot write... The next request — a cold start, or simply one routed elsewhere — began from an empty seed and rendered an empty queue. **Incidents filed weeks ago were not hidden; they were gone.**"*

`src/lib/db.ts` — *the environment guard exists because "dev.circuvent.com came to serve real customer accounts, orders and wallet balances."*

10. Design patterns actually in use

Pattern	Where
Pure core / impure shell	<code>icm.ts</code> , <code>session-expiry.ts</code> , <code>delivery-estimate.ts</code> , <code>device-normalize.ts</code> — each with an impure store or transport sibling
Stateless tokens with a version counter	Revocation without a session table
Capability-gated RBAC	<code>AdminArea</code> + <code>ROLE_AREAS</code> , not bare role checks
Fail closed in production, warn in development	<code>secrets.ts</code> — "A hardcoded fallback therefore fails open"
Signature, then re-verify	Payment capture is re-fetched from Razorpay; a forged client signature cannot credit an order
Host allow-lists, not block-lists	<code>PROD_DATA_HOSTS</code> , <code>PROD_IDENTITY_HOSTS</code>
Estimate, not promise	<code>delivery-estimate.ts</code> returns a range and says so
Degrade rather than throw	<code>createFileStore</code> — which is also the source of the largest data risk in the system

11. Health assessment

Incident documentation		9/10	best in the suite
Security headers		8/10	complete set
Passkey implementation		8/10	scoped, clone-detected
Payment capture path		7/10	re-verified server-side
Session design		6/10	versioned, 24 h cap
Auth architecture		5/10	5 schemes, no central gate
Money typing		4/10	float rupees in catalogue
Payment reconciliation		3/10	webhook is a stub
Schema management		3/10	created at boot
Documentation accuracy		2/10	README off by 10x
Tenancy / DB access ctrl		2/10	none whatsoever
Data durability		1/10	~27 modules memory-only

What is genuinely excellent: the incident-comment convention; a complete security-header set applied twice over; passkeys with scoped credentials and cloned-authenticator detection; server-side TOTP QR rendering; payment capture that re-fetches from the gateway rather than trusting a client signature; a pure, honest delivery estimator; and `verify-icm-durability.ts`, which spawns four real operating-system processes to prove a durability claim rather than mocking it.

What needs attention: roughly twenty-seven storage modules that lose all data on a serverless cold start; a payment webhook that verifies its signature and then does nothing; a README describing a tenth of the system; five coexisting credential schemes with no central gate and seven routes whose authentication could not be established; and a database with no migrations, no foreign keys and no access control of any kind.

12. Where to start reading

1. <code>src/lib/db.ts</code>	and understand <code>initDb()</code>
2. <code>src/lib/data-file.ts</code>	then count the <code>`durable: true`</code> callers
3. <code>scripts/verify-icm-durability.ts</code>	read the header
4. <code>src/lib/account.ts</code> + <code>admin-auth.ts</code>	the two session schemes
5. <code>src/lib/passkeys.ts</code>	read the header
6. <code>src/lib/sso.ts</code>	read the header
7. <code>src/proxy.ts</code>	note what it does NOT do
8. <code>src/app/api/payments/webhook</code>	note what it does NOT do
9. NOT <code>README.md</code>	it describes a different system

Next: [02_DATABASE_AND_DATA_MODELS.md](#) · [03_INTEGRATIONS_AND_ECOSYSTEM.md](#) · [04_MAINTENANCE_AND_OPERATIONS.md](#) · [05_AREAS_OF_ENHANCEMENT.md](#)

PART 2

02 · Database and Data Models

Audience: engineers and anyone responsible for not losing customer data. **Engine:** Neon Postgres via the HTTP driver · no ORM · no migrations · and a JSON-file store that most features actually use

1. The headline: schema is a side effect of the application booting

THERE ARE ZERO .sql FILES IN THIS ENTIRE REPOSITORY.

There is no ORM. No Drizzle, no Prisma, no Knex.
There is no migrations directory.
There is no schema version table.

Schema is an array of raw SQL strings in a TypeScript file, run as
CREATE TABLE IF NOT EXISTS the first time any database function is
called – which means on effectively every serverless cold start.

```
// src/lib/db.ts
const SCHEMA_STATEMENTS: string[] = [
  `CREATE TABLE IF NOT EXISTS accounts (
    email TEXT PRIMARY KEY,
    name TEXT,
    password_hash TEXT,
    ...
  ];
```

```
/** Creates the schema if it does not yet exist (idempotent, runs once). */
export function initDb(): Promise<void> {
  if (!_initPromise) return _initPromise;
  _initPromise = (async () => {
    const q = await getQuery();
    for (const stmt of SCHEMA_STATEMENTS) await q(stmt);
  })();
  return _initPromise;
}
```

Every exported **db*** function **awaits** **initDb()** before its query. **.env.example** states it plainly: *"The schema is created automatically on first run."*

WHAT THIS COSTS

- No record of when or why any column was added
- No rollback path
- No way to review a schema change in a pull request as a schema change
- The connecting role **MUST** hold DDL rights, forever – see §6
- A fresh environment gets "whatever SCHEMA_STATEMENTS says today", which is not necessarily what production actually has

WHAT IT BUYS

- A new environment needs no migration step at all. It just works.

For a site that began as a marketing page, that trade was reasonable.
For something now holding orders, wallets and passkeys, it is not.
Doc 05, D-02.

2. Ten tables

Counted by reading `SCHEMA_STATEMENTS` verbatim — `grep "CREATE TABLE"` returns 10 hits, all in `src/lib/db.ts`. This count is provably complete.

Table	Notable columns	Key	Indexes
accounts	email, name, password_hash, password_salt, phone, blocked, data JSONB	PK email	created_at
admin_users	email, name, role, active, password_hash/salt, data JSONB	PK email	role
pending_registrations	email, otp, expires, attempts, ref, data JSONB	PK email	—
store_kv	collection, key DEFAULT '_all', data JSONB, updated_at	PK (collection, key)	—
email_history	to, from_addr, subject, type, status, provider, message_id, error, body_html, meta	PK id BIGSERIAL	4: created_at DESC, type, to, status
request_metrics	endpoint, method, status, ms REAL, region	PK id BIGSERIAL	ts DESC, endpoint
camera_frames	device_id, jpeg_b64, bytes, captured_at, upload_token, token_expires	PK device_id	—
camera_audio	device_id, wav_b64, bytes, captured_at	PK id BIGSERIAL	device_id, captured_at DESC
camera_audio_session	device_id, listen_token/expires, speak_token/wav_b64/expires	PK device_id	—
page_views	path, visitor_hash (salted, daily-rotating), referrer_host, device, browser, country	PK id BIGSERIAL	ts DESC, ts+path, ts+visitor_hash

THREE CONVENTIONS WORTH STATING

1. Every entity table carries a full-fidelity `data JSONB` column, and the file header says it is "the source of truth on read". The typed columns exist only for indexing and human inspection.

2. THERE ARE NO FOREIGN KEYS. Anywhere. Not one.

3. THERE IS NO TENANT OR ORG COLUMN. Anywhere. See §6.

Y One genuinely good detail: `page_views` stores a SALTED, DAILY-ROTATING `visitor_hash` rather than an IP address or a persistent identifier. That is privacy-conscious analytics, done deliberately.

! And two that are not: `camera_frames.jpeg_b64` and `camera_audio.wav_b64` store base64 media INSIDE Postgres rows. Doc 05, D-11.

Mermaid diagram — renders interactively in the Markdown edition

```

erDiagram
    ACCOUNTS {
        text email PK
        text password_hash
        jsonb data "source of truth"
    }
    ADMIN_USERS {
        text email PK
        text role
        jsonb data
    }
    STORE_KV {
        text collection PK
        text key PK "defaults to _all"
        jsonb data "ALL rows of a collection, in ONE row"
    }
    EMAIL_HISTORY {
        bigserial id PK
        text status
    }
    PAGE_VIEWS {
        bigserial id PK
        text visitor_hash "salted, rotates daily"
    }
    CAMERA_FRAMES {
        text device_id PK
        text jpeg_b64 "base64 media in a row"
    }

    ACCOUNTS ||..|| STORE_KV : "no FK – nothing is joined"
    ADMIN_USERS ||..|| STORE_KV : "no FK"
    CAMERA_FRAMES ||..|| CAMERA_AUDIO : "no FK"

```

3. `store_kv` — 23 collections in one table, one row each

`store_kv` is a generic key-value multiplexer, reused for three unrelated purposes:

(a) THE ENTIRE SHOP – 23 collections, via `KV_COLLECTIONS`

orders	products	wallets	devices
reviews	addresses	notifyRequests	logins
coupons	tickets	returns	audit
loyalty	referrals	referralCodes	giftCards
questions	notifications	passwordResets	admin2fa
alertSettings	contactMessages	consumedPayments	

! EACH IS ****ONE ROW****, with key = `'_all'`.
 Every order in the system lives inside a single JSONB value.
 Reading one order reads all of them. Writing one order
 rewrites all of them.
 There is no index into it, and no way to page it.
 Doc 05, D-03.

(b) collection=`'user_prefs'`, key = an arbitrary user id

(c) collection=`'file_store'`, key = an arbitrary filename
 – this is where the JSON modules opt into Postgres. See §5.

Because any module can call `dbWriteFileStore("<anything>.json", ...)` with a name chosen at runtime, **the logical contents of `store_kv` cannot be enumerated statically**. The ten physical tables are knowable; what is inside them is not.

4. The client — HTTP driver, so no transactions

```
async function getQuery(): Promise<QueryFn> {
  if (_query) return _query;
  const url = process.env.DATABASE_URL;
  if (!url) throw new Error("DATABASE_URL is not set");
  assertNotProductionData(url);
  const { neon } = await import("@neondatabase/serverless");
  const client = neon(url);
  _query = (text, params = []) =>
    client.query(text, params) as Promise<Record<string, unknown>[]>;
  return _query;
}
```

Aspect	Finding
Driver	neon() HTTP only. No Pool , no Client , no neonConfig.webSocketConstructor anywhere
Pooling	None — and none is needed; the HTTP driver is stateless per call
Memoization	YES Module-scope let _query , built once per lambda instance
Test seam	YES __setQueryForTests swaps in PGlite
Transactions	None exist, and none are possible. Zero BEGIN/COMMIT/ROLLBACK in the codebase. The HTTP driver cannot hold one open

The workaround is real and reasonable — single-statement atomic SQL with a JSONB merge:

```
INSERT INTO store_kv (collection, key, data)
VALUES ($1, $2, jsonb_build_object($3::text, $4::jsonb))
ON CONFLICT (collection, key)
DO UPDATE SET data = store_kv.data || jsonb_build_object($3::text, $4::jsonb), ...
```

That is correct for the cases it covers. It is also a structural ceiling: nothing here can ever be made atomic across two tables without changing driver. Doc 05, D-10.

5. The finding that matters most: most data has no database behind it

src/lib/data-file.ts – createFileStore(), header comment:

"an in-memory working copy is the source of truth for the life of the process, and every mutation is best-effort written through to a JSON file under DATA_DIR. On read-only filesystems (serverless production without a database) writes silently stop and the module degrades to in-memory-only for that instance, instead of throwing and breaking the request."

ROUGHLY 30 MODULES ARE BUILT ON THIS.
EXACTLY THREE OPT INTO THE POSTGRES MIRROR ({ durable: true }):

icm-store.ts · admin-warranty.ts · api-failures.ts

THE OTHER ~27 ARE MEMORY-ONLY IN PRODUCTION.

On Vercel's read-only filesystem the write silently fails, the catch block swallows it, and the data lives in one lambda instance's memory until that instance recycles. Then it is gone. Not corrupted. Not stale. Gone.

What is in the non-durable ~27

admin-cms

admin-jobs

admin-seo

admin-tax

admin-telemetry

console-dev-portal

passkeys

smarthome-admin-config

smarthome-user-prefs

...and more

admin-crm

admin-macros

admin-staff-activity

admin-bulk

← 570 KB on disk

← developer portal tokens

← WebAuthn credentials

admin-currency

admin-marketing

admin-subscriptions

admin-affiliates

admin-flags

admin-report-builder

admin-surveys

admin-bundles

CMS content. CRM records. Pricing and currency. Tax configuration.
Feature flags. Marketing. Staff activity. Developer tokens. Passkeys.

None of it survives a cold start in production.

.data/ at the repository root (29 files) is simply what this same code path writes when the disk is writable — locally. It is **not tracked by git**, and is doubly ignored:

```
# local dev data fallback (email evidence log, file store)
/.data/
# Shop runtime data (orders/products/wallets)
.data/
```

Source module	Files it writes
src/lib/store.ts	shop-db.json, inventory-db.json
src/lib/data-file.ts	~20 admin-*.json, console-dev-portal.json, passkeys.json, smarthome-*.json
src/lib/email-log.ts	email-history.jsonl — the append-only fallback for email_history

6. Access control at the data layer: there is none

NO tenant or organisation column on any table.
NO per-query scoping predicate.
NO Postgres row-level security — a repository-wide search for
ROW LEVEL SECURITY · CREATE POLICY · GRANT · REVOKE
returns ZERO matches across src/ and scripts/.
NO foreign keys.

And because initDb() runs CREATE TABLE and CREATE INDEX on every cold start, the configured role MUST hold DDL rights. It cannot be a narrowly-scoped runtime role. There is no separate migration-time credential.

The only isolation that exists is a string comparison

assertNotProductionData(url) — src/lib/db.ts

Refuses to run against a host listed in PROD_DATA_HOSTS when
VERCEL_ENV is not production. A parallel PROD_IDENTITY_HOSTS guard
does the same for the federated-login control plane.

* AND THE CODE DOCUMENTS THE INCIDENT THAT CREATED IT:

"dev.circuvent.com came to serve real customer accounts, orders
and wallet balances."

BUT BOTH GUARDS ARE OPT-IN AND SHIP EMPTY in .env.example.
A new deployment that does not set them has no protection at all
against exactly the incident that motivated them. Doc 05, D-07.

It is not the suite database

Nothing in **db.ts** schema-qualifies any table — no **hrms.**, no **identity.** Every table is created unqualified in **public**. Cross-application identity is handled **out of band**, over HTTPS, via **CONTROL_PLANE_URL** + **FEDERATION_SECRET**.

Whether `DATABASE_URL` happens to point at the same Neon project as HRMS cannot be determined from committed code. But the code treats these tables as its own private, unshared, unscoped set. **From an architectural standpoint this is a separate database.**

7. Firebase — live, but narrow

firebase@12.11.0 is a real dependency, not a dead remnant.

EXACTLY ONE module imports it: `src/lib/cv365-firebase.ts`
 EXACTLY ONE caller: `src/components/ContactForm.tsx`

```
// CV-365 Firestore contact bridge – Firebase is imported lazily (dynamic
// import inside the submit path) so the heavy SDK is NOT in the initial page
// bundle; it only loads when a visitor actually submits the contact form.
const [{ initializeApp, getApps, getApp }, { getFirestore, collection, addDoc, Timestamp }] =
  await Promise.all([import("firebase/app"), import("firebase/firestore")]);
return addDoc(collection(db, "contactMessages"), { ...data, status: "new", createdAt: Timestamp.now() });
```

It writes contact-form submissions into a **separate Firebase project** so that work.circuvent.com/admin/messages can see them. It is entirely decoupled from Postgres. The 18 other files matching "firebase" are marketing pages listing a tech stack.

There is no `firestore.rules` or `database.rules.json` in this repository. So the permissiveness of that project's rules cannot be assessed here — a sibling application in this suite shipped a rules file granting root read/write to any signed-in user, and this one lives in a repository nobody in this audit can see. Doc 05, D-13.

8. PGLite — a real Postgres, in-process

`@electric-sql/pglite` is a dev dependency used in three places:

Use	What it does
<code>scripts/test-db.ts</code>	Runs the real DDL and query logic against a fresh in-memory <code>new PGLite()</code>
<code>scripts/icm-instance.ts</code>	An on-disk PGLite shared across separate OS processes — see §9
<code>src/lib/user-prefs.test.ts</code>	Unit tests for the KV merge logic

It proves `SCHEMA_STATEMENTS` is valid, executable Postgres DDL and that the query logic is correct. **It proves nothing about the live database** — not its actual state, not its permissions, not its extensions.

9. `verify-icm-durability.ts` — the best script in the repository

"ICM" is the internal incident queue.

FROM THE FILE'S OWN HEADER:

"The bug was never in the UI. Incidents were written to a JSON file that the serverless host cannot write, so ``createFileStore`` caught the failure and kept them in one lambda instance's memory. The next request — a cold start, or simply one routed elsewhere — began from an empty seed and rendered an empty queue.

Incidents filed weeks ago were not hidden; they were gone."

What makes it exceptional: it does not mock the failure. It spawns **four separate operating-system processes**, sharing no module cache and no memory, each acting as a distinct cold lambda, against one on-disk PGLite instance.

It was **executed during this audit** and passed:

```
instance A files an incident...
+ a cold instance starts with an empty queue – this was the bug
+ the store reports itself durable
+ filed as INC-0001

instance B – a different process – files another...
+ it too starts empty, sharing no memory with A
+ after loading, it sees its own and A's incident
+ A's incident survived the process that filed it
+ the id counter continued at INC-0002 rather than restarting

instance C acknowledges A's incident...
+ a later process can act on an incident it did not file

instance D reads the queue afresh...
+ both incidents are still there
+ the acknowledgement survived
+ and so did who took ownership

ALL PASSED
```

And here is the uncomfortable part. This script proves durability for one of the roughly thirty modules built on `createFileStore`. The bug it was written to catch is still live in the other twenty-seven, and nothing checks them.

10. Secrets

`src/lib/secrets.ts` handles session-signing secrets only.

```
export function lazySecret(names: string[], label: string): () => string {
  let cached: string | undefined;
  return () => (cached ??= requireSecret(names, label));
}
```

Behaviour	Detail
Minimum length	32 characters, enforced in production
Hardcoded fallback	YES None — "A hardcoded fallback therefore fails open... Production now refuses to start without a real secret"
Lazy loading	YES Deliberate, so <code>next build</code> does not fail at import time
Development	A per-process ephemeral random secret from <code>randomBytes(48)</code> , cached in memory, never persisted, warned about once
Admin bootstrap	<code>seedAdminPassword()</code> generates a random one-time password if <code>ADMIN_DEFAULT_PASSWORD</code> is unset, and prints it once
Encryption at rest	NO None — everything is read straight from <code>process.env</code>
Rotation	NO No mechanism. Rotating a secret simply invalidates every session

`scripts/secret-inventory.mjs` — a genuinely good idea

It records the path, presence, git-tracked status and SHA-256 of each credential-bearing file — for drift detection — and never the contents:

"An inventory that contains the secrets is just another copy of the secrets... which is the thing this whole exercise exists to avoid."

Environment variables the data layer needs

Variable	Purpose
<code>DATABASE_URL</code>	Neon connection. Absence silently activates the file/memory fallback
<code>PROD_DATA_HOSTS</code>	Non-production guard against a production database host
<code>PROD_IDENTITY_HOSTS</code>	The same for the control plane

Variable	Purpose
<code>CONTROL_PLANE_URL</code> , <code>FEDERATION_SECRET</code>	Federated SSO handoff
<code>DATA_DIR</code>	Overrides the <code>.data</code> location
<code>ACCOUNT_SECRET</code> , <code>ADMIN_SECRET</code>	Session-token signing
<code>ADMIN_DEFAULT_EMAIL</code> , <code>ADMIN_DEFAULT_PASSWORD</code>	Super-admin bootstrap
<code>NEXT_PUBLIC_CV365_FIREBASE_*</code> (5)	The separate Firestore project

11. Data-layer risk register

#	Finding	Sev
1	~27 of ~30 storage modules have no database backing. CMS, CRM, pricing, tax, feature flags, marketing, staff activity, 570 KB of telemetry, developer-portal tokens and passkeys are memory-only in production and vanish on every instance recycle — silently, by design of the degrade-to-memory catch block	
2	Schema is not versioned or reproducible. It exists only as an array executed at boot. No history, no rollback, no review surface	
3	Every shop collection is a single JSONB row. Reading one order reads all of them; writing one rewrites all of them. No index, no paging	
4	No RLS, no tenant column, no foreign keys, no database-level access control of any kind	
5	The two environment guards ship empty , and the code documents the incident they exist to prevent — <i>"dev.circuvent.com came to serve real customer accounts, orders and wallet balances"</i>	[HIGH]
6	The database role must hold DDL rights permanently , because <code>initDb()</code> runs <code>CREATE TABLE</code> on every cold start	[HIGH]
7	Transactions are structurally impossible with the HTTP driver	[HIGH]
8	No backup or restore story anywhere. <code>export-business-data.ts</code> exports marketing catalogue content, not customer data. Durability rests entirely on Neon's own backups, which this repository neither configures nor verifies	[HIGH]
9	**<code>.data/*.json</code> holds unencrypted PII on local disk** — customer emails, order records, tax and warranty data, staff activity	[MEDIUM]
10	Base64 media in Postgres rows — <code>camera_frames.jpeg_b64</code> and <code>camera_audio.wav_b64</code>	[MEDIUM]
11	The Firebase satellite is unauditable from here — a second datastore, with its rules in a repository this audit cannot see	[MEDIUM]
12	No key rotation for <code>ACCOUNT_SECRET</code> or <code>ADMIN_SECRET</code> ; rotating either logs everyone out	[MEDIUM]

Next: [03_INTEGRATIONS_AND_ECOSYSTEM.md](#) · Back to [01_SYSTEM_OVERVIEW.md](#)

PART 3

03 · Integrations and Ecosystem

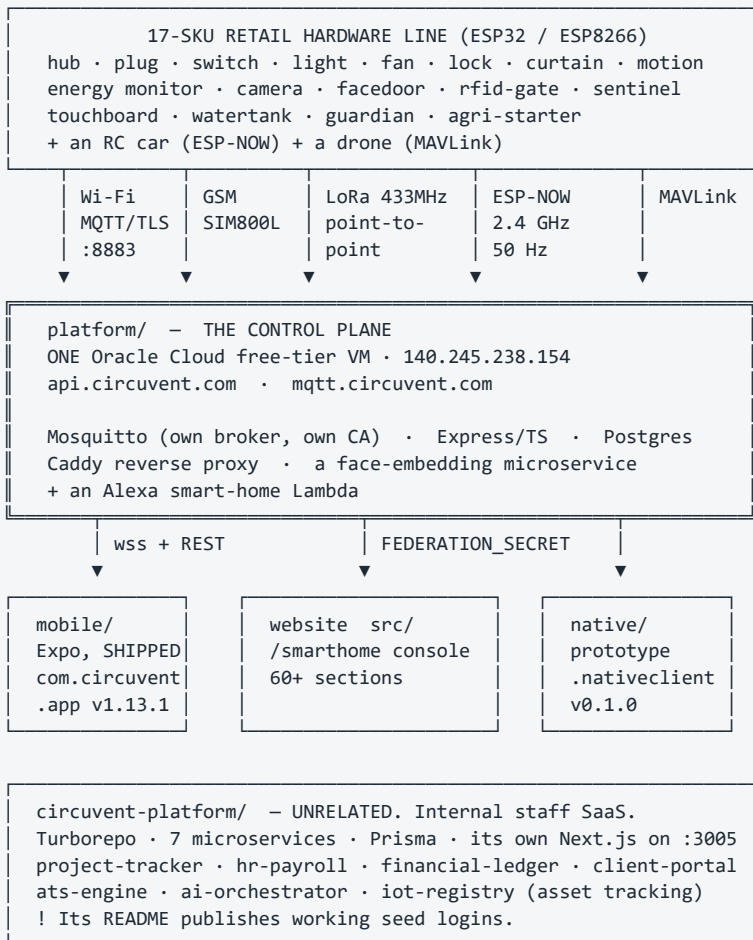
Audience: anyone who needs to know what this repository actually talks to. **The short version:** this is not a website. It is a company — a retail IoT hardware line, a self-hosted MQTT cloud on a real virtual machine, a shipped mobile app, a drone, and a separate internal HR and payroll SaaS.

1. What is actually in the repository

SIX SUB-PROJECTS SIT ALONGSIDE THE NEXT.JS APP		
Directory	Files on disk	What it actually is
firmware/	13,003	ESP32/ESP8266 firmware for a 17-product retail hardware line ! ONLY 84 FILES ARE FIRST-PARTY 99.35% is PlatformIO build cache, gitignored
mobile/	1,280	Expo React Native – SHIPPED on the Play Store, v1.13.1
circuvent-platform/	1,202	A COMPLETELY DIFFERENT PRODUCT: an internal Turborepo SaaS for project tracking, HR, payroll and a client portal
native/	999	Kotlin + Swift parity prototype. The iOS half has NEVER COMPILED.
hardware/	771	Real KiCad PCBs, Gerbers, BOMs and enclosures for 17 products
platform/	317	The self-hosted IoT control plane: Mosquitto + Node/TS + Postgres + Caddy, on ONE Oracle free-tier VM

! A file count is a lie here. `firmware/` looks like the biggest thing in the repository at 13,003 files. `git ls-files firmware` returns **84**. The rest is regenerated PlatformIO output, correctly gitignored: `"# PlatformIO build output — regenerated on build, never commit."` Anyone sizing this project by file count overstates the firmware by roughly 150x.

2. The ecosystem map



Mermaid diagram — renders interactively in the Markdown edition

```

flowchart TB
    subgraph HW["Devices - 17 SKUs"]
        WIFI["Wi-Fi devices<br/>ESP32-WROOM-32E"]
        GSM["guardian · agri-starter<br/>SIM800L cellular"]
        LORA["watertank ↔ sensor<br/>LoRa 433 MHz"]
        RC["RC car<br/>ESP-NOW, no cloud"]
        DRONE["drone-x1<br/>ESP32-S3 · MAVLink"]
    end

    CP["platform/ - control plane<br/>ONE Oracle VM · 140.245.238.154<br/>Mosquitto + Express + Postgres + Caddy"]

    subgraph CLIENTS["Clients"]
        MOB["mobile/ Expo<br/>SHIPPED v1.13.1"]
        WEBC["website /smarthome<br/>60+ sections"]
        NAT["native/ Kotlin+Swift<br/>prototype, iOS never built"]
    end

    ALEXA["Alexa smart-home<br/>Lambda"]
    IPS["circuvent-platform/<br/>internal staff SaaS<br/>UNRELATED"]

    WIFI -->|"MQTT/TLS :8883"| CP
    GSM -->|"MQTT over cellular"| CP
    LORA -->|"via watertank hub"| CP
    DRONE -->|"MAVLink → MQTT"| CP
    RC -->|"no cloud at all"| RC

    CP -->|"wss + REST"| MOB
    CP -->|"FEDERATION_SECRET"| WEBC
    CP --> NAT
    ALEXA --> CP

    style IPS fill:#3d3d3d,color:#fff
    style RC fill:#5a5a5a,color:#fff
  
```

3. platform/ VS circuvent-platform/ — a naming trap

These are two entirely different products. Neither supersedes the other. Someone skimming directory names would get this badly wrong.

	platform/	circuvent-platform/
Is	The device-facing IoT control plane	An internal staff SaaS
README title	"Circuvent Control Plane — self-hosted IoT platform"	"Circuvent Technologies — Internal Management Platform"
Users	ESP32 devices, the mobile app, the website's /smarthome pages	Circuvent employees
Structure	One Node/TS Express service + Mosquitto + Caddy + a face service, plain Docker Compose	Turborepo + pnpm, 7 microservices + a gateway + its own Next.js 14 dashboard
Database	Postgres, hand-rolled pool.query	Postgres via Prisma, shared packages/database
Ports	80/443 (Caddy), 8883 (MQTT/TLS)	3000 gateway, 3001–3008 services, 3005 web
Secrets in .env.example	Real annotated device/broker secrets, with incident post-mortems in the comments	Literal placeholders — JWT_SECRET=your-super-secret-jwt-key-change-in-production
Seed credentials	None	Published in the README: admin@circuvent.com / admin@123, engineer@123, client@123 — for an application that handles HR, payroll and a financial ledger

The only thing they share is the word "IoT": circuvent-platform/apps/services/iot-registry is an internal asset and firmware-version tracker for engineering and R&D bookkeeping, not a broker. Doc 05, D-08.

4. The device protocol

```
TRANSPORT  MQTT over TLS, port 8883, self-hosted Mosquitto
            mqtt://mqtt.circuvent.com:8883

"Everything is JSON over MQTT on our own broker."
"no third-party IoT cloud."

ENCRYPTION  Y TLS with Circuvent's OWN EMBEDDED CA
            (CIRCUVENT_DEFAULT_CA in CircuventDevice.h)
            The README notes what was removed to get here:
            "setInsecure() no longer used for MQTT"

TOPICS      cv/<id>/cmd          commands in
            cv/<id>/state       current state out
            cv/<id>/telemetry    measurements
            cv/<id>/frame        camera stills
            cv/<id>/anpr         plate reads
            cv/<id>/track        drone position (16-byte header
                                + 40-byte fixed records)
            cv/<id>/status       online/offline

LIVE APP    wss://api.circuvent.com/ws?token=<JWT>  — a separate hop

// state — AquaGuard tank controller
{ "level": 72, "pump": true, "mode": "auto",
  "startPct": 30, "stopPct": 90, "dryRun": false }

// cmd
{ "action": "set", "ch": 2, "on": true }
```

Device authentication

Y WHAT IT IS

A per-device secret. username = deviceId, password = key.
 Minted by POST /devices/provision.
 Stored BCRYPT-HASHED. The registry doc is emphatic:

"A device key cannot be looked up. Not by support, not by engineering, not by an administrator. devices.key_hash is bcrypt. There is no plaintext anywhere in the system."

Broker ACLs restrict each device to its own namespace: cv/<deviceId>/#

! WHAT IT IS NOT

Not mutual TLS. Not an X.509 client certificate. There is no hardware-backed secret storage on an ESP32. A leaked device key is enough to impersonate that device – though only within its own topic namespace, which meaningfully limits blast radius.

! AND PROVISIONING IS STILL SEMI-MANUAL

The API mints the credential; granting broker access is a separate operator step (scripts/add-device.sh, mosquito_passwd). The firmware README flags it:
 "Future: delegate broker auth to Postgres (mosquitto-go-auth) so provisioning alone grants broker access without add-device.sh."

Wi-Fi handoff during setup is separately encrypted with NaCl sealed boxes (`crypto_box_keypair` / `crypto_box_open`, via the vendored `tweetnacl.c`) – so the household Wi-Fi password never crosses the captive portal in the clear. That is a genuinely good detail.

5. Over-the-air updates — the most serious finding in this document

THERE IS NO FIRMWARE IMAGE SIGNATURE VERIFICATION.

Both OTA paths call `httpUpdate.update(otaClient, binUrl)` over a certificate-PINNED TLS connection. That authenticates the SERVER. It does not authenticate the FIRMWARE.

Searched for and NOT found anywhere in `CircuventDevice.h`:

- Ed25519 or RSA signature verification
- a hash manifest checked against an on-device public key
- ESP32 Secure Boot or signed-partition configuration in any of the 29 `platformio.ini` files

"Signed OTA" in the documentation means "downloaded over a pinned HTTPS connection." That is a materially weaker guarantee.

And the code itself already articulates the stakes, in the comment explaining why `setInsecure()` was removed:

"It used to use `setInsecure()`, which disables certificate validation entirely – anyone able to intercept that connection ... could serve arbitrary firmware and take permanent control of a board that switches MAINS RELAYS AND DOOR LOCKS."

The transport hole was closed. The integrity hole was not.

And the checklist is honest about it. `hardware/CHECKLIST.md` marks both of these unchecked:

- [] "OTA manifest endpoint (`/api/devices/firmware`) serving signed builds"
- [] "Field OTA rollout + rollback plan; key rotation policy"

Confirmed server-side: `platform/api/src/routes/` contains no `firmware.ts` or `ota.ts`. So the device's periodic pull-OTA check polls an endpoint that does not exist. Pull-OTA is dead code today. Push-OTA works, and trusts whatever URL an administrator supplies.

A shared-library fix worth noting: OTA is handled generically in `CircuventDevice.h` after a bug in which all twenty product sketches individually forgot to implement it. That is the right correction — move it to the one place, not twenty.

6. Mobile — and a collision deliberately avoided

	mobile/	native/android	native/ios
Stack	Expo ~53 · RN 0.79.6 · React 19	Kotlin + Jetpack Compose	Swift + SwiftUI via XcodeGen
App id	com.circuvent.app	com.circuvent.app.nativ eclient	com.circuvent.app.nativecli ent
Version	1.13.1, build 24	0.1.0, versionCode 1	0.1.0
Shipped ?	YES Yes — on the Play Store	NO Debug side-load only	NO Never compiled, once
Evidence	mobile/dist/ holds real signed artifacts spanning 1.1.0 → 1.12.0+, play-upload-key.json, PLAYSTORE.md	Gradle assembleDebug works	"There is no Mac in this pipeline."

* THE GOOD DECISION

A sibling repository in this suite shipped TWO mobile implementations under ONE app id, which is a real hazard. This repository used a DIFFERENT app id on purpose, and said why:

"That is deliberate twice over: the first thing anybody needs while replacing an app is to run both on one phone and compare them, and it removes any chance of a debug build replacing somebody's provisioned installation."

And native/README.md is candid rather than aspirational:

"mobile/ is what is on the Play Store and it has not been modified."
"The Swift is not compiled by anything."

7. The same rule, written four times

THE RULE: a per-device "capability table" — which fields a device type reports, which command keys control it, and critically which device types must expose NO toggle at all (a camera's only switch is `streaming`; a drone's is flight permission).

IMPLEMENTED INDEPENDENTLY IN FOUR PLACES:

TypeScript (Expo, shipping)	mobile/src/store.tsx	capabilitiesFor()
Kotlin	native/android/.../core/Capabilities.kt	
Swift	native/ios/Circuvent/Core/Capabilities.swift	
TypeScript (web console)	src/	— referenced by name in native/README

AND native/README.md NAMES THE FAILURE, TWICE OVER:

"A Home Hub reports `power2` and is commanded with {ch: 1, on: true} ... Send the state key to a device that wanted the command key and nothing errors...
THAT BUG HAS ALREADY SHIPPED TWICE — once on the web and once in the Expo app."

The only guard is tests/native-client-parity.test.ts, which diffs the Kotlin, Swift and Expo tables at test time.

Drift is CAUGHT BY AN ASSERTION, not PREVENTED BY A SHARED SCHEMA.
Doc 05, D-07.

A second, lower-severity instance: the device serial-number and QR format is defined server-side in `platform/api/src/serial.ts` and re-implemented for parsing in `mobile/src/qr.ts`.

8. The hardware

17 SKUs, sold on circuvent.com, Amazon.in and Flipkart

Home Automation Hub · AquaGuard tank controller · Smart Plug · Smart Switch · Smart Light · Smart Fan · Smart Lock · Curtain controller · Motion Sensor · Energy Monitor · Guardian (wearable SOS) · Agri Pump Starter · Touch Switchboard · FaceDoor · RFID Gate · Sentinel (gas/climate) · Load Controller

MCU: ESP32-WROOM-32E for most; ESP32-S3-WROOM-1 for the drone

hardware/ IS REAL FIRST-PARTY DESIGN SOURCE – not datasheets.

Every product except the drone has:

pcb/ .kicad_pcb · .kicad_pro · .kicad_dru · BOM.csv · SCHEMATIC.md · gerbers/ enclosure/ · listings/ · DATASHEET.md · MANUAL.md

Generated, not hand-drawn: gen-hardware.js (Node) + gen-pcb.py (Python) build real boards and Gerbers for all 17 devices from SCHEMATIC.md + BOM.csv. CHECKLIST.md flags this itself:

"the netlist is derived by the generator... not captured from a drawn schematic"

! NOT MANUFACTURED. CHECKLIST.md's own legend marks BIS/WPC/CE certification, tooling, photography and seller accounts as "requires an external vendor, lab, physical process, or account."

! 97 unconnected nets remain across the 17 boards (47 blocked on mains-isolation clearance), and the ESP32 antenna keepout was reduced from Espressif's recommended 48×21 mm to 7 mm. DRC-clean is not the same as safe to fabricate for mains products.

The drone is the least mature thing here

feat/drone-x1 · feature/drone-link · fix/drone-stale-control-plane – three long-lived branches for one physical product.

TWO COMPETING FIRMWARE ARCHITECTURES CO-EXIST:

firmware/drone-fc an IN-HOUSE flight-controller stack – motor mixer, arm/disarm, IMU loop. The datasheet warns: "This airframe carries a new flight stack."

firmware/drone-link a COMPANION COMPUTER that explicitly refuses to be a flight controller: "It sits on the airframe next to a real flight controller – ArduPilot or PX4... WHY THE CLOUD IS NEVER IN THE CONTROL LOOP... There is deliberately no 'nudge forward while I hold this button'." Bridges MAVLink → MQTT with whole-intent commands only: takeoff, goto, RTL, land.

The second design is excellent and its reasoning is right. The first contradicts it.

! AND drone-x1 IS THE ONE PRODUCT WITH NO pcb/ DIRECTORY. A DATASHEET only. Spinning propellers, no certification, no PCB source, and a documented warning not to fit props before reading the commissioning ladder. Doc 05, D-09.

9. Build and release reality

Sub-project	Build	Automated?	Evidence it has run
firmware/	PlatformIO per device	NO CHECKLIST.md : [] <i>"Build in CI (arduino-cli) + signed release binaries hosted for OTA"</i>	YES Every device has a populated .pio/build with compiled .elf/.bin — built locally at least once
platform/	tsc → dist/ , Docker Compose	NO none found	YES dist/ has 129 compiled files; deployed to a real VM with an SSH runbook
circuvent-platform/	Turborepo, pnpm, Prisma	NO none found	[MEDIUM] A real 196 KB pnpm-lock.yaml and installed modules — run locally, no deployment evidence
mobile/	EAS Build — production profile, App Bundle, autoIncrement: true	[MEDIUM] Managed cloud CI, invoked manually	YES The strongest release evidence in the repository — signed artifacts across many versions
native/android	Gradle	NO	[MEDIUM] Debug only, no release signing
native/ios	XcodeGen	NO	NO Never built
hardware/	Generators, not a software build	NO	YES Generated Gerbers and BOMs on disk; CHECKLIST.md tracks DRC pass/fail per board

10. Signing keys and a documented loss

Creds/ — at the repository root, gitignored wholesale (verified: ``git ls-files -- Creds`` returns nothing, and `.gitignore:110` carries the rule with an explanation:
 "A keystore or an SSH key committed once is in every clone and every fork forever.")

circuvent-cp.key	SSH key for the control-plane VM
circuvent-upload.jks	Play upload keystore
circuvent-upload-2026.jks	a second one
circuvent-upload.keystore	a third
upload-keystore.properties	passwords, IN PLAINTEXT
README.md	VM and key runbook

Mirrored again under mobile/credentials/, whose .gitignore says:
 "# Signing keystores — NEVER commit"

upload-keystore.properties DOCUMENTS A REAL LOSS:

"This file was overwritten on 2026-08-03 and that is how the password for circuvent-upload.jks was lost... do not delete [the .bak] until the original key is either recovered or permanently retired"

Three keystore variants plus a .bak now coexist. Losing a Play upload key is recoverable; losing an app-signing key is not.
 Doc 05, D-06.

11. Integration risk register

#	Risk	Sev
1	No firmware signature verification. OTA authenticates the transport, never the image. On devices that switch mains relays and door locks — a risk the code's own comments already articulate	
2	Pull-OTA polls an endpoint that does not exist. <code>/api/devices/firmware</code> has no route. A mechanism that appears to exist and does not is worse than none	
3	circuvent-platform/ publishes working seed logins in its README — <code>admin@123</code> and friends — for an application handling HR, payroll and a financial ledger	

#	Risk	Sev
4	Placeholder JWT secrets ship as defaults in <code>circuvent-platform/.env.example</code> , and its own README records that this exact class of misconfiguration has already happened once	[HIGH]
5	The capability table exists in four languages with no shared schema. The bug it causes has already shipped twice	[HIGH]
6	A Play upload keystore password was permanently lost, and passwords sit in plaintext beside three keystore variants	[HIGH]
7	Device auth is a shared secret, not mutual TLS, with no hardware-backed storage on the ESP32	[HIGH]
8	The whole IoT cloud is one free-tier virtual machine. No redundancy, no documented backup, no failover	[HIGH]
9	The drone has two competing firmware architectures, three branches and no PCB source — the least mature and most physically dangerous product in the line	[HIGH]
10	RF and mains-isolation compliance is unresolved — 97 unconnected nets, antenna keepout cut from 48×21 mm to 7 mm	[MEDIUM]
11	No firmware CI. Nothing builds the 29 sketches automatically; a compile break is found by a human	[MEDIUM]
12	Provisioning is semi-manual — the API mints a key, an operator must still run <code>add-device.sh</code>	[MEDIUM]

And what is genuinely well done

Its own CA and pinned TLS after removing `setInsecure()` · bcrypt-hashed device keys that nobody can look up · per-device broker ACLs · NaCl sealed boxes for the Wi-Fi handoff so a household password never crosses the captive portal in the clear · OTA fixed once in the shared library rather than twenty times in twenty sketches · a companion-computer design that refuses to put the cloud in a flight control loop · a deliberately different app id so a debug build can never replace a customer's installation · and a hardware checklist honest enough to mark its own unfinished items unchecked.

Next: [04_MAINTENANCE_AND_OPERATIONS.md](#) · Back to [02_DATABASE_AND_DATA_MODELS.md](#)

PART 4

04 · Maintenance and Operations

Audience: anyone who has to run, deploy, debug or extend this. **The finding that shapes this document:** the CI pipeline here is the most thorough in the entire Circuit suite — fourteen steps, thirteen of them hard gates, including the control plane's own test suite and a full Playwright run. **It has never executed once.**

1. The CI that has never run

```
gh run list --limit 30

27 runs on record.
27 startup_failure.
0 seconds, every one.
Spanning 2026-08-09 → 2026-08-20, across main, develop,
master and a dozen feature branches.

AND IT IS WORSE THAN IT LOOKS. Going deeper via the API:

The registered workflow "CI" (id 336804312, ci.yml, state
ACTIVE, created 2026-08-18) has ZERO RUNS EVER RECORDED
AGAINST IT.

All 27 runs are attributed instead to a synthetic, DELETED
workflow named "" / BuildFailed – GitHub's placeholder for
"could not even start". Every one returns
{"total_count": 0, "jobs": []}.

This pipeline has never executed one line of its own YAML,
even after ci.yml became syntactically valid and active.

No YAML defect was found. The billing API returned 403 on the
audit token, so the precise trigger is unconfirmed – but this is
the same signature the ATS repository shows.
```

What it would do

`.github/workflows/ci.yml` — the only workflow. Push to `main/master/feature/shopping`, PR to `main/master`. Ubuntu, Node 20, 20-minute timeout, `cancel-in-progress`. *"Deterministic, secret-free build. Real values are configured in Vercel."*

Mermaid diagram — renders interactively in the Markdown edition

```
flowchart TD
    T["push / pull_request"] --> A["1-3 checkout · node 20 · npm ci"]
    A --> B["4 npx tsc --noEmit"]
    B --> C["5 npm run lint"]
    C --> D["6 npm test --ci --runInBand"]
    D --> E["7-9 platform/api:<br/>npx ci · tsc · npm test"]
    E --> F["10 npm run test:db (PGlite)"]
    F --> G["11 npm run build"]
    G --> H["12 playwright install chromium"]
    H --> I["13 npm run test:e2e"]
    I --> J["14 upload report – if: always()"]

    style C fill:#5c4a1a,color:#fff
    style E fill:#1a3d5c,color:#fff
    style I fill:#1a3d5c,color:#fff
```

Only step 5 (`lint`) is `continue-on-error`. Everything else is a hard gate. That is stricter than most of the suite.

And the workflow narrates its own history in comments:

“The control plane is a separate package with its own dependencies and its own runner, so `npm test` at the root never touched it. That left ~290 tests — the MQTT bridge, session revocation, the device registry, the whole ANPR recognition and visit-pairing pipeline — unable to block a deploy.”

“E2E was never run in CI, which is how the sitemap assertion below stayed broken (it expected the wrong domain)... Must match the browsers `playwright.config.ts` runs. These disagreed: only chromium was installed while the config declared three projects, so every run failed on 'Executable doesn't exist' for firefox and webkit.”

Both are real fixes to real gaps. Neither has ever run.

Thirteen scripts CI never touches

Script	Status
<code>verify:secrets</code>	Not a CI gate. Only a bypassable local hook — see §2
<code>test:coverage</code>	NO not in CI
<code>verify:icm</code>	NO not in CI — the durability proof from doc 02 §9
<code>audit:contrast</code> , <code>audit:admin-theme</code> , <code>audit:perf</code> , <code>audit:live</code> , <code>audit:images</code>	NO none of the five
<code>docs:business</code> , <code>docs:business:verify</code> , <code>docs:kt</code> , <code>docs:kt:verify</code> , <code>docs:business:data</code>	NO the entire documentation pipeline

No real database in CI. `test:db` runs PGlite in-process. Production uses Neon. Neon is never touched by CI.

2. The git hook

```
package.json: "prepare": "node scripts/install-hooks.mjs"
Runs on every npm install / npm ci.

INSTALLS EXACTLY ONE HOOK:

#!/bin/sh
# Installed by scripts/install-hooks.mjs – edit that, not this.
node "$(git rev-parse --show-toplevel)/scripts/check-no-secrets.js" --staged || exit 1

Y VERIFIED PRESENT: .git/hooks/pre-commit exists, 163 bytes,
byte-identical to the template.
```

Its header explains both why hooks are installed rather than committed, and why it is a Node script:

“Hooks live in `.git/hooks`, which is not tracked, so a hook that exists on one machine does not exist on the next clone... The hook is deliberately a single node call. An earlier guard in this repo was written in bash and never ran once, because bash on the Windows box that does the builds is WSL with no distribution installed — it failed silently and the build carried on with the wrong signing key.”

```
BUT IT IS TRIVIAALLY BYPASSABLE, THREE WAYS:
git commit --no-verify
skip npm install (the hook is then never installed)
delete .git/hooks/pre-commit

And because verify:secrets is not in CI, a developer who bypasses the
hook has NOTHING stopping a secret at push time. Doc 05, D-03.
```

3. The audit scripts — and what happened when they were run

Script	What it checks	Run result
<code>check-no-secrets.js</code>	Forbidden filenames and content patterns, staged or whole-tree	<code>exit 0</code> — yes no-secrets – checked the tracked tree

Script	What it checks	Run result
<code>audit-images.mjs</code>	Every asset in public/ against a per-type size budget	exit 1 — 3 over budget: <code>logo.png</code> and <code>logo-mark.png</code> at 368 kB against an 80 kB budget, <code>icon-512.png</code> at 172 kB
<code>verify_business_docs.py</code>	The generated business documents actually contain what they claim	exit 0 — All 43 checks passed across 7 artefacts
<code>verify_kt_docs.py</code>	The same for the knowledge-transfer pack	exit 1 — 61/73 checks passed — see §5
<code>audit-code-contrast.mjs</code>	Computed CSS contrast of code surfaces, in a real browser	exit 2 — needs <code>next start -p 3199</code> first. It correctly refused to claim success: FAILED: no code/pre elements were examined — the audit proved nothing.
<code>audit-admin-theme.js</code>	Screenshots the console under every theme, no login required	not run — needs a server
<code>perf-probe.mjs</code>	Real latency, transferred bytes from the Resource Timing API, directly observed CLS	not run — needs a server
<code>audit-live.mjs</code>	Every read-only production surface; <code>--control</code> round-trips one live device command	not run — needs production

Three quotes worth keeping

`scripts/check-no-secrets.js` — *“`.gitignore` stops the files it knows about. It does not stop `git add -f`, a secret pasted into a source file, a new `.env` under a name nobody thought of, or an archive of the whole credentials directory... a secret pushed once is in every clone and every fork, and deleting the file in a later commit does not remove it from history.”*

`scripts/audit-code-contrast.mjs` — *“Written because the bug it checks for was invisible to every existing test: the stylesheet compiled, the page rendered, and the only symptom was that a human could not read it. Asserting the computed colours in a real browser is the only thing that would have caught it.”*

`scripts/verify_kt_docs.py` — *“`npm run docs:kt` printing 'ok' only proves three files were written. It does not prove the deck has slides, that the device list reached the page, or that the traps table survived being parsed out of a markdown file — and a build script in this repository has previously reported success while publishing the previous run's artifact, so 'the command succeeded' is not evidence.”*

That last one is the philosophy of this whole repository in one paragraph.

4. The documentation pipeline

FOUR PYTHON SCRIPTS. TWO BUILD, TWO VERIFY.

```

build_business_docs.py    investor deck · sales deck · company profile ·
                        business plan · new-joiner handbook ·
                        product catalogue · price list
                        → PPTX, DOCX, PDF into Docs/business/
                        SOURCE: business-data.json, itself exported
                        from the LIVE SHOP CATALOGUE by
                        scripts/export-business-data.ts

build_kt_docs.py          onboarding deck · handbook · quick reference
                        → Docs/kt/
                        SOURCE: the repository itself — device list
                        from the firmware tree, doc index from Docs/,
                        traps table parsed out of 00-start-here.md

```

Why generate rather than write:

“business documents quote prices, and prices move. A deck, a catalogue and a price list that each carry their own typed copy will disagree with the shop within a quarter... Refuses to run if the export is missing or stale rather than quietly producing documents from yesterday's prices.”

And the verifiers check content, not existence:

Pack	Asserts
Business	The company name is in every document · every priced document contains a real catalogue price · no raw unformatted price slipped past the formatter · exact slide counts · PDFs have extractable text · placeholders appear only where intended
KT	Every document names the company and the commit it was generated from · the deck has slides, speaker notes and the parity rule · the handbook lists every file in Docs/ · every device in the firmware tree appears somewhere · nothing in the KT pack carries the business pack's "live product catalogue" stamp — a claim it has no right to make

Neither is run by CI or by a hook. And the difference shows: the business pack is fresh at 43/43; the KT pack is stale at 61/73.

5. Test suite — 4,328 tests, one failing

```
npm test -- --ci --runInBand          EXIT CODE 1

Test Suites:  1 failed, 235 passed, 236 total
Tests:       1 failed, 4,327 passed, 4,328 total
Time:        49.5 s

THE FAILURE:
  src/lib/report-logo.test.ts
    > "the embedded mark matches the artwork"
      > "re-derives byte for byte from public/logo-mark-160.png"

The embedded base64 logo bytes have drifted from the PNG on disk.
A small failure — and a test that is doing exactly its job.

FILE COUNT RECONCILES EXACTLY:
tests/**                120
co-located src/**/*.*test.*  116
-----
236 files = 236 Jest suites
```

Aspect	Finding
Coverage threshold	jest.config.js has no coverageThreshold key at all. <code>test:coverage</code> exists; nothing fails on low coverage; and it is not in CI anyway
Playwright	8 specs in <code>e2e/.test-results/.last-run.json</code> says "passed" — but per the workflow's own comment <code>e2e "was never run in CI"</code> , so this is an undated local cache and should not be trusted
Untested	src/lib/db.ts — the entire database access layer. src/lib/csp.ts — which generates the Content-Security-Policy string that next.config.ts ships

Largest test files

Lines	File	Covers
443	<code>src/app/api/admin/icm/route.test.ts</code>	Incident management API
424	<code>tests/firmware-avi.test.ts</code>	Firmware AVI parsing
382	<code>src/app/api/admin/availability/probe/route.test.ts</code>	Availability probe
362	<code>src/lib/app-insights-usage.test.ts</code>	Telemetry usage reporting
355	<code>tests/camera-fps-parity.test.ts</code>	Camera FPS parity
342	<code>tests/icm.test.ts</code>	Incident state machine
336	<code>tests/drone-flight-safety.test.ts</code>	Drone flight-safety limits
324	<code>src/lib/app-insights-query.test.ts</code>	Telemetry queries
322	<code>src/app/admin/insights-charts.test.tsx</code>	Admin charts
316	<code>tests/lib/extended-utils.test.ts</code>	Utilities

6. Code quality, measured

Metric in <code>src/</code>	Count
<code>TODO / FIXME</code>	1
<code>@ts-ignore / @ts-expect-error</code>	0
<code>eslint-disable</code>	47 across 38 files
literal <code>console.log(</code>	7 across 4 files — including one in <code>payments/webhook/route.ts</code> , which is the stub
Files using <code>any</code>	9 files, 46 occurrences — <code>AnalyticsPanel.tsx</code> alone accounts for 24
<code>.ts/.tsx</code> files	971

For a 971-file tree that is a genuinely clean signal.

```
npx tsc --noEmit          EXIT 0. Clean. tsconfig strict: true.

! BUT tsconfig `exclude` covers scripts, e2e, jest.setup.tsx and
  circuvent-platform — so the scripts and e2e specs are NOT typechecked.

npm run lint              EXIT 1 — 29,687 problems
                          (5,726 errors, 23,961 warnings)

AND ALMOST NONE OF IT IS REAL:
.next-audit/              1,050 files   23,271 problems
circuvent-platform/      543 files   4,627
mobile/                  247 files   378
platform/api/dist        117 files   706 ← COMPILED OUTPUT

src/ + tests/ + scripts/ IN SCOPE      ≈705 (2.4%)

CAUSE: eslint.config.mjs's globalIgnores lists only SHALLOW patterns
— .next/**, out/**, build/**, next-env.d.ts — which do not match
nested paths in a monorepo. So ESLint is linting build output and two
unrelated sub-projects.

Fixing the ignore patterns turns a 29,687-problem wall of noise into
roughly 705 actionable items. Doc 05, D-05.
```

7. Deployment

```
// vercel.json — this is the ENTIRE file
{
  "$schema": "https://openapi.vercel.sh/vercel.json",
  "crons": [
    { "path": "/api/admin/alerts/run",          "schedule": "0 8 * * *" },
    { "path": "/api/admin/reports/send",        "schedule": "0 4 * * *" },
    { "path": "/api/smarthome/alerts/cron",      "schedule": "0 6 * * *" },
    { "path": "/api/admin/availability/probe",   "schedule": "0 5 * * *" }
  ]
}
```

Four crons. No regions, no function config, no rewrites, no headers — headers live in `next.config.ts`.

Security headers — complete, and applied twice

```
const securityHeaders = [
  { key: "Strict-Transport-Security", value: "max-age=63072000; includeSubDomains; preload" },
  { key: "Content-Security-Policy", value: CSP },
  { key: "X-Content-Type-Options", value: "nosniff" },
  { key: "X-Frame-Options", value: "DENY" },
  { key: "Referrer-Policy", value: "strict-origin-when-cross-origin" },
  { key: "X-DNS-Prefetch-Control", value: "on" },
  { key: "Permissions-Policy", value: "camera=(), microphone=(), geolocation=(self), browsing-topics=()" },
  { key: "Cross-Origin-Opener-Policy", value: "same-origin" },
];
```

Applied to `/:path*`, with `/_next/static/:path*` getting `max-age=31536000`, `immutable` and `/api/:path*` getting `no-store`, `must-revalidate`. Non-production deployments add `X-Robots-Tag: noindex, nofollow`.

! The CSP string comes from `src/lib/csp.ts` — which has no test anywhere.

Image remote patterns are scoped to exactly two hosts, with a comment that shows why: `"/** would turn the Next image optimizer into an open proxy for every tenant on res.cloudinary.com."`

Two redirects, both with a reason:

- `/fw/:file` → an R2 bucket, because *"Eighteen images, about twenty megabytes, were committed under `public/fw/` and shipped with every deployment."*
- `/developers` → `/developer`, avoiding a stale static-prerender 200 followed by a JavaScript redirect.

`distDir` is conditional on an env var, because the dev server previously deleted `BUILD_ID` from a shared `.next` and an audit silently ran against an empty site.

8. Secrets — history is clean

```
git log --all --diff-filter=A --name-only --pretty=format:
| Sort-Object -Unique
| Select-String 'creds|.env|.jks|.keystore|.key$|.pem$|secret'

MATCHES — all benign:
circuvent-platform/.env.example      platform/.env.example
Docs/11-secrets.md                  scripts/check-no-secrets.js
scripts/secret-inventory.mjs        src/lib/secrets.ts
tests/check-no-secrets.test.ts      tests/unit/secrets-lazy.test.ts

Y NO REAL .env, .jks, .keystore, .pem OR .key HAS EVER BEEN ADDED
IN THIS REPOSITORY'S HISTORY, ON ANY BRANCH.

`.gitignore` carries redundant, overlapping coverage: *.jks,
*.keystore, *.p12, *.pfx, *.pem, *.key, id_rsa, id_ed25519,
credentials/, *.vault, circuvent-vault/, Creds/ and .env*.

Docs/11-secrets.md is a full inventory and rotation document with no
values in it.
```

The gap is procedural, not historical: the scanner runs only in a bypassable local hook. Getting it into CI is a two-line change.

But the history is not clean of bloat

```
The 15 largest blobs in `git rev-list --objects --all` are ALL
firmware images — public/fw/camera-*.bin and sentinel-cam-*.bin —
at 1.12 to 1.25 MB EACH, across dozens of versions.

They are GONE from HEAD and from the index. They are PERMANENT in
history. Every clone still pays for them.

.git/ on disk is ~191 MB. Doc 05, D-10.
```

9. Documentation

`Docs/` holds 61 files — 39 numbered documents from `00-start-here.md` to `38-rc-platform.md`, plus generated `business/` and `kt/` packs.

Claim	Reality	Verdict
<code>README.md</code> : "18 routes", "50+ React components"	151 route.ts, 108 page.tsx	FALSE. The README describes an early landing-page stage and was never updated as the monorepo grew

Claim	Reality	Verdict
The KT pack is current	<code>verify_kt_docs.py</code> exit 1 — missing 5 devices (<code>rc-link</code> , <code>rc-remote</code> , <code>rccar</code> , <code>rfid-attend</code> , <code>switchboard</code>) and 10 newer documents; no commit stamp at all	STALE
<code>test-results/.last-run.json</code> says "passed"	An undated local cache; e2e "was never run in CI"	[HIGH] not evidence

And several documents check out as accurate:

Document	Status
<code>Docs/25-git-and-releases.md</code>	YES Branch table matches the real <code>ci.yml</code> triggers and branch list exactly
<code>Docs/05-databases.md</code>	YES Correctly names Neon
<code>Docs/13-maintenance.md</code>	YES "There is no monitoring stack." — an honest absence claim, still true

Notably, the sibling-repository failure mode is inverted here. Other Circuvent repositories had documents describing systems that no longer existed. This one has documents describing a fraction of a system that grew past them.

`.agents/`, `skills-lock.json` and `Prompt.txt`

Artefact	What it is
<code>.agents/skills/workflow-init/SKILL.md</code>	YES A genuine AI-coding-agent "skill" teaching how to configure the Vercel Workflow SDK — which matches the <code>workflow</code> dependency and the <code>withWorkflow()</code> wrapper actually used in <code>next.config.ts</code>
<code>skills-lock.json</code>	YES Pins that skill by <code>computedHash</code> for reproducibility
<code>Prompt.txt</code>	A verbatim AI prompt for building an entirely unrelated internal management platform. No connection to this Next.js site. An accidentally-committed scratch file

The first two are a real convention a future contributor must respect. The third is hygiene debt.

10. Repository state

```
533 commits · 2026-03-08 → 2026-08-20 · currently on `develop`

FIVE REMOTES — and only one is the working repository:
origin      Hemakotibonthada/WebSite.circuvent.git ← THE REAL ONE
hema        Hemakotibonthada/circuvent-technologies.git
vercel      Hemakotibonthada/circuvent-technologies.git (same URL)
circuvent   Circuvent-Technologies/circuvent.git
company-portal Circuvent-Technologies/Company-Portal.git (a DIFFERENT repo)

git status: Architecture_Docs/ untracked (this audit), plus four
deleted-but-unstaged screenshot scripts.

Y test-results/, node_modules/, .next/ and *.log are all UNTRACKED.
```

11. Observability

WHAT EXISTS	WHAT DOES NOT
Y /api/health	NO error tracking. No Sentry,
Y /api/health/db	Datadog, New Relic, Bugsnag or
Y /api/admin/cron-health	LogRocket in package.json.
Y api.circuvent.com/health	Only @vercel/analytics and
→ { ok, db }	speed-insights — usage and
Y /admin/health (admin token)	performance, not errors.
→ MQTT, DB, uptime	NO alerting of any kind.
Y src/lib/logger.ts —	NO log shipper or sink.
structured logging	The logger itself has NO TEST.
Y Neon point-in-time recovery	No custom backup tooling.
Docs/13-maintenance.md says it plainly, and is still correct:	
"There is no monitoring stack. The endpoints exist...	
Point any uptime service at /health."	
Nothing is pointed at /health.	

What you could not diagnose today

1. Any production error.

There is no aggregation and no alerting — only a /health endpoint nobody is polling.
2. Whether a bad deploy correlates with a broken build

— there is no CI signal at all to correlate against.
3. A database or CSP regression

— db.ts and csp.ts are the two modules with no test anywhere.
4. Whether the ~27 non-durable storage modules have lost data

— nothing reports it, by construction (doc 02 §5).
5. Whether a device is running the firmware you think it is

— there is no OTA manifest endpoint (doc 03 §5).

12. Routine maintenance

Cadence	Task
Immediately	Fix whatever prevents GitHub Actions from starting. Fourteen well-designed gates are worth nothing until then
Immediately	Add verify:secrets to CI. It is currently guarded only by a hook that --no-verify defeats
Immediately	Fix the report-logo test — regenerate the embedded bytes from public/logo-mark-160.png
This week	Fix eslint.config.mjs 's ignore patterns so npm run lint reports ~705 real problems rather than 29,687
Every deploy	Run locally what CI cannot: tsc --noEmit , npm test , npm run test:db , npm run build , npm run test:e2e
Every deploy	Also run verify:icm , verify:secrets and audit:images — none run anywhere else
After any device or Docs change	Re-run npm run docs:kt && npm run docs:kt:verify . It is currently failing at 61/73
After any price change	Re-run docs:business:data then docs:business — the documents are generated from the live catalogue precisely so they cannot drift
Monthly	Point an uptime service at /api/health and api.circuvent.com/health . Docs/13-maintenance.md has been asking for this
Before scaling	Nothing in .data/ -backed modules survives a second instance. See doc 02 §5 first

Next: [05_AREAS_OF_ENHANCEMENT.md](#) · Back to [03_INTEGRATIONS_AND_ECOSYSTEM.md](#)

PART 5

05 · Areas of Enhancement

Audience: engineering leadership. **Method:** every item is traceable to a file, a command executed during this audit, or a comment in the codebase itself.

1. Gap analysis

Dimension	Now	Target	Gap
Incident documentation			best in the suite
Security headers			complete, applied 2x
Secret history hygiene			provably clean
Verification thinking			scripts prove, not assert
Test volume			4,328 tests, 1 red
Hardware engineering			17 real board designs
Code hygiene			lint config broken
Documentation accuracy			README off by 8x
Auth architecture			5 schemes, no gate
Payment reconciliation			webhook is a stub
Schema management			created at boot
Firmware supply chain			no image signing
Data durability			~27 modules memory
Observability			nothing at all
CI actually running			0 of 27 runs

2. The four things that would keep me awake

2.1 Roughly twenty-seven storage modules lose all data on a cold start

```
src/lib/data-file.ts - createFileStore()
```

"On read-only filesystems (serverless production without a database) writes silently stop and the module degrades to in-memory-only for that instance, instead of throwing and breaking the request."

~30 modules use this. THREE pass { durable: true }:
icm-store.ts · admin-warranty.ts · api-failures.ts

THE OTHER ~27 INCLUDE:

CMS content · CRM records · pricing · currency · tax configuration
feature flags · marketing · staff activity · 570 KB of telemetry
developer-portal tokens · AND PASSKEYS

In production these live in one lambda instance's memory. When that instance recycles, the data is gone. Not stale. Not corrupted. Gone.

AND THE REPOSITORY ALREADY KNOWS. From verify-icm-durability.ts:

"Incidents were written to a JSON file that the serverless host cannot write... The next request – a cold start, or simply one routed elsewhere – began from an empty seed and rendered an empty queue. Incidents filed weeks ago were not hidden; THEY WERE GONE."

That bug was found, understood, written down, and fixed for ONE module. The excellent four-process durability test proves ICM works. Nothing checks the other twenty-seven.

2.2 Firmware has no signature verification

Devices that switch MAINS RELAYS and DOOR LOCKS accept over-the-air firmware whose only integrity guarantee is that it arrived over a certificate-pinned TLS connection.

No Ed25519. No RSA. No hash manifest against an on-device key.
No ESP32 Secure Boot in any of the 29 platformio.ini files.

The code articulates the stakes itself, explaining why `setInsecure()` was removed:

```
"anyone able to intercept that connection ... could serve arbitrary
firmware and take permanent control of a board that switches mains
relays and door locks."
```

The transport hole was closed. The integrity hole was not.

And hardware/CHECKLIST.md is honest about it – both unchecked:

```
[ ] "OTA manifest endpoint (/api/devices/firmware) serving signed builds"
[ ] "Field OTA rollout + rollback plan; key rotation policy"
```

Meanwhile the device polls that manifest endpoint on a timer.
It does not exist. `platform/api/src/routes/` has no firmware route.

2.3 The payment webhook verifies its signature, then does nothing

```
const expected = crypto.createHmac("sha256", secret).update(raw).digest("hex");
const valid = sigBuf.length === expBuf.length && crypto.timingSafeEqual(sigBuf, expBuf);
```

Correct. Constant-time. Then it `console.log()`s and returns.

```
// Reconciliation hook: when a persistent order store exists,
// mark the order paid/failed here...
```

Any payment captured at Razorpay whose browser never returns – a closed tab, a dead connection, a failed redirect – is money taken and an order never marked paid.

The checkout-side path is genuinely good: `verifyCapturedPayment()` re-fetches from Razorpay and requires `status === "captured"` before trusting the amount. A forged client signature cannot credit an order. The webhook was meant to be the safety net under that, and it is a stub.

2.4 CI has never run

27 runs. 27 `startup_failure`. 0 seconds each.

The registered "CI" workflow has ZERO runs ever attributed to it; all 27 belong to a synthetic deleted workflow. No YAML defect exists.

What is not running: a typecheck, 4,328 root tests, the control plane's own ~290 tests, a PGLite database test, a production build, and a full Playwright suite – thirteen hard gates.

The workflow's own comments describe fixing exactly the gaps that made those tests unable to block a deploy. Those fixes have never executed either.

This is the second Circuvent repository with this precise signature.

3. Technical debt log

Severity: critical · [HIGH] high · [MEDIUM] medium · low

ID	Finding	Sev	Effort
D-01	~27 storage modules are memory-only in production — CMS, CRM, pricing, tax, flags, telemetry, developer tokens and passkeys vanish on every instance recycle		L
D-02	No firmware image signature verification on devices controlling mains relays and door locks; the pull-OTA manifest endpoint does not exist at all		L
D-03	CI has never executed — 27 of 27 <code>startup_failure</code> — and <code>verify:secrets</code> is guarded only by a <code>--no-verify</code> -able local hook		S
D-04	The payment webhook is a stub. Signature verified correctly, then nothing. Payments captured without a browser return are never reconciled		M
D-05	Schema is created at runtime by <code>initDb()</code> on every cold start. No migrations, no version table, no rollback, no review surface		M
D-06	No RLS, no tenant column, no foreign keys, no database access control of any kind		L
D-07	<code>circuvent-platform/</code> publishes working seed logins (<code>admin@123</code>) in its README — for an application handling HR, payroll and a financial ledger		S
D-08	README describes a fraction of the system — "18 routes, 50+ components" against 151 routes and 108 pages, plus firmware, hardware and two native apps		S
D-09	Every shop collection is one JSONB row. Reading one order reads all of them; writing one rewrites all of them	[HIGH]	M
D-10	The capability table exists in four languages with no shared schema; the bug it causes "has already shipped twice"	[HIGH]	M
D-11	<code>npm run lint</code> reports 29,687 problems, of which ~705 are in scope. <code>globalIgnores</code> uses shallow patterns that miss nested monorepo paths	[HIGH]	S
D-12	Five coexisting credential schemes and no central gate. <code>proxy.ts</code> performs no authentication; seven routes could not be matched to any known mechanism	[HIGH]	M
D-13	No observability at all — no error tracking, no alerting, no log sink. <code>Docs/13-maintenance.md</code> says so plainly and has been ignored	[HIGH]	M
D-14	A failing test on main — <code>report-logo.test.ts</code> ; the embedded logo bytes have drifted from the PNG on disk	[HIGH]	S
D-15	The KT documentation pack is stale — <code>verify_kt_docs.py</code> exits 1 at 61/73, missing 5 devices and 10 documents, with no commit stamp	[HIGH]	S
D-16	A Play upload keystore password was permanently lost on 2026-08-03, and passwords sit in plaintext beside three keystore variants	[HIGH]	M
D-17	<code>db.ts</code> and <code>csp.ts</code> have no test anywhere — the database layer and the module that generates the shipped Content-Security-Policy	[HIGH]	M
D-18	The environment guards ship empty. <code>PROD_DATA_HOSTS</code> and <code>PROD_IDENTITY_HOSTS</code> are opt-in, and the code records the incident they exist to prevent	[HIGH]	S
D-19	The whole IoT cloud is one free-tier virtual machine — no redundancy, no failover, no documented backup	[HIGH]	L
D-20	Transactions are structurally impossible with the Neon HTTP driver	[MEDIUM]	M
D-21	Git history carries ~20 MB of firmware binaries permanently. <code>.git</code> is ~191 MB	[MEDIUM]	M
D-22	No coverage threshold in <code>jest.config.js</code> , and <code>test:coverage</code> runs nowhere	[MEDIUM]	S
D-23	Money is a float in the catalogue — <code>price: number; // whole INR</code> — with integer paise only at the gateway boundary	[MEDIUM]	M
D-24	Every contact submission is written twice, to Firestore and to the local store, from the same handler	[MEDIUM]	S
D-25	The drone has two competing firmware architectures, three branches and no PCB source	[MEDIUM]	L
D-26	<code>resend</code> is declared but its SDK is never imported	[MEDIUM]	S
D-27	13 npm scripts run nowhere automated — including all five audits and the entire documentation pipeline	[MEDIUM]	S

ID	Finding	Sev	Effort
D-28	<code>DeviceControls.tsx</code> is 4,870 lines in a single React component	[MEDIUM]	L
D-29	Three public assets exceed their size budget — <code>logo.png</code> and <code>logo-mark.png</code> at 368 kB against 80 kB		S
D-30	<code>Prompt.txt</code> is an accidentally-committed AI prompt for an unrelated project		S
D-31	Device auth is a shared secret, not mutual TLS, with no hardware-backed storage on the ESP32	[HIGH]	L
D-32	97 unconnected nets across the 17 boards, and the ESP32 antenna keepout was cut from 48×21 mm to 7 mm	[MEDIUM]	L
D-33	<code>tsconfig</code> excludes <code>scripts</code> and <code>e2e</code> — a clean <code>tsc --noEmit</code> does not cover them	[MEDIUM]	S
D-34	Firmware provisioning is semi-manual — the API mints a key, an operator still runs <code>add-device.sh</code>	[MEDIUM]	M

4. The pattern worth naming

This repository does something almost nothing else does: when it finds a bug, it writes the bug down in the file that fixes it.

```

passkeys.ts      "The passkey still existed, still verified, and
                  belonged to nobody."
sso.ts           "Production users could sign in to dev, and dev
                  quietly accumulated live credentials while doing
                  it. The isolation guard was pointed at the wrong
                  door."
next.config.ts   "silently deleted BUILD_ID twice while auditing,
                  so `next start` served nothing and the audit
                  reported a clean sweep of an empty site."
verify-icm-      "Incidents filed weeks ago were not hidden;
durability.ts    they were gone."
install-hooks.mjs "bash on the Windows box that does the builds is
                  WSL with no distribution installed – it failed
                  silently and the build carried on with the wrong
                  signing key."
db.ts            "dev.circuvent.com came to serve real customer
                  accounts, orders and wallet balances."
check-no-secrets "a secret pushed once is in every clone and every
                  fork, and deleting the file in a later commit
                  does not remove it from history."
verify_kt_docs.py "a build script in this repository has previously
                  reported success while publishing the previous
                  run's artifact, so 'the command succeeded' is
                  not evidence."
native/README    "That bug has already shipped twice – once on the
                  web and once in the Expo app."
CircuventDevice.h "could serve arbitrary firmware and take permanent
                  control of a board that switches mains relays and
                  door locks."

```

AND THE VERIFICATION SCRIPTS FOLLOW THE SAME PHILOSOPHY:
they PROVE rather than assert.

```

verify-icm-durability.ts  spawns FOUR REAL OS PROCESSES rather
                           than mocking a cold start
audit-code-contrast.mjs  reads COMPUTED CSS in a real browser,
                           because "the only symptom was that a
                           human could not read it"
perf-probe.mjs           uses the Resource Timing API for real
                           transferred bytes, and DIRECTLY OBSERVES
                           CLS rather than inferring it
verify_business_docs.py   opens the generated PPTX and asserts a
                           REAL CATALOGUE PRICE is inside it

```

The engineering instinct here is excellent.
Almost none of it is automated.

5. Phased roadmap

Mermaid diagram — renders interactively in the Markdown edition

```

gantt
  title website — improvement roadmap
  dateFormat YYYY-MM-DD
  axisFormat %b

  section Phase 1 — Turn the lights on
  Fix GitHub Actions so CI can start :p1a, 2026-01-01, 2d
  Add verify:secrets to CI :p1b, after p1a, 1d
  Fix the failing report-logo test :p1c, 2026-01-01, 1d
  Fix eslint ignore patterns :p1d, after p1c, 2d
  Rewrite README to match reality :p1e, after p1d, 3d

  section Phase 2 — Stop losing data
  Audit all 30 createFileStore users :p2a, after p1e, 5d
  Make every one durable: true :p2b, after p2a, 14d
  Extend the 4-process proof to all :p2c, after p2b, 7d

  section Phase 3 — Money
  Wire the payment webhook :p3a, after p2c, 10d
  Reconciliation job for orphan pays :p3b, after p3a, 7d
  Brand the money type :p3c, after p3b, 7d

  section Phase 4 — Firmware supply chain
  Ed25519 signing key + on-device pub :p4a, after p3c, 14d
  Verify signature before flashing :p4b, after p4a, 14d
  Build the OTA manifest endpoint :p4c, after p4b, 10d
  Rollback plan + key rotation policy :p4d, after p4c, 7d

  section Phase 5 — Foundations
  Real migrations :p5a, after p4d, 14d
  Split store_kv into real tables :p5b, after p5a, 21d
  Error tracking and alerting :p5c, after p5a, 10d
  One shared capability schema :p5d, after p5c, 14d

```

Phase 1 — Turn the lights on (about a week)

Task	Debt	Why
Fix whatever prevents Actions from starting	D-03	Thirteen hard gates are worth nothing until then
Add <code>verify:secrets</code> to CI	D-03	It is currently defeated by <code>git commit --no-verify</code>
Fix <code>report-logo.test.ts</code>	D-14	Regenerate the embedded bytes from the PNG
Fix <code>eslint.config.mjs</code> 's ignore patterns	D-11	Turns 29,687 problems into ~705 real ones
Rewrite the README	D-08	It currently describes about one-eighth of the system

Phase 2 — Stop losing data (about a month)

Enumerate every `createFileStore` caller. Decide, per module, whether its data matters. For everything that does, pass `durable: true` — the Postgres mirror already exists and is already proven to work. Then generalise `verify-icm-durability.ts` from one module to all of them; the four-process harness is already written.

This is the highest-value work in the document, because the failure is silent, and because the fix is largely a flag on a function that already supports it.

Phase 3 — Money (about a month)

Wire the webhook to actually mark orders paid or failed. Add a reconciliation job that reconciles Razorpay's captures against local orders and reports the difference. Then give money a branded integer type so a fractional rupee can never reach `Math.round(due * 100)`.

Phase 4 — Firmware supply chain (about two months)

Generate an Ed25519 signing key. Embed the public key in `CircuventDevice.h`. Verify the signature over the downloaded image before `httpUpdate.update()` commits it. Build the `/api/devices/firmware` manifest endpoint the devices are already polling. Then write the rollback plan and key-rotation policy the checklist already has unchecked boxes for.

This should arguably be Phase 2. It is ranked lower only because the data loss in Phase 2 is happening now, whereas this is a serious latent risk requiring an attacker.

Phase 5 — Foundations (a quarter)

Real migrations with a version table. Split `store_kv`'s 23 single-row collections into actual tables with actual indexes. Install error tracking and point an uptime service at the health endpoints that already exist. Replace the four hand-written capability tables with one schema and generated bindings.

6. What must not change

Y THE INCIDENT-COMMENT CONVENTION

Eleven examples are quoted in §4. It made this audit possible and it is worth more than any documentation folder.

Y VERIFICATION THAT PROVES RATHER THAN ASSERTS

Four real OS processes instead of a mocked cold start. Computed CSS in a real browser. Directly observed CLS. A generated deck opened and checked for a real catalogue price.
"the command succeeded" is not evidence."

Y THE COMPLETE SECURITY-HEADER SET, applied both at the edge and globally so routes the proxy skips are still covered.

Y IMAGE remotePatterns SCOPED TO TWO HOSTS

"/*" would turn the Next image optimizer into an open proxy for every tenant on res.cloudinary.com."

Y SEPARATE SECRETS FOR STAFF AND CUSTOMER SESSIONS

"Staff sessions get their own key so a leaked customer key cannot mint one." Plus tokenVersion, added after a departing employee's copied token stayed valid forever.

Y PASSKEY SCOPES

"Which sign-in a credential belongs to. They must never be interchangeable." Plus cloned-authenticator detection.

Y SERVER-SIDE TOTP QR RENDERING

So the secret never reaches a third-party QR service.

Y PAYMENT CAPTURE RE-FETCHED FROM THE GATEWAY

A forged client signature cannot credit an order.

Y DOCUMENTS GENERATED FROM THE LIVE CATALOGUE

"prices move... Refuses to run if the export is missing or stale."

Y A DIFFERENT APP ID FOR THE NATIVE PROTOTYPE

Deliberately avoiding the exact collision a sibling repository hit.

Y THE DRONE COMPANION-COMPUTER DESIGN

"WHY THE CLOUD IS NEVER IN THE CONTROL LOOP... There is deliberately no 'nudge forward while I hold this button'."

Y NaCl SEALED BOXES FOR THE WI-FI HANDOFF

A household password never crosses the captive portal in the clear.

Y A HARDWARE CHECKLIST HONEST ENOUGH TO LEAVE ITS OWN BOXES UNCHECKED.

7. If you only do five things

1	Fix GitHub Actions, and add verify:secrets to CI. Thirteen hard gates and 4,328 tests are currently decorative.	1 day
2	Set durable: true on every createFileStore module whose data matters. Passkeys and developer tokens are among the ~27 that vanish on every cold start.	3 weeks
3	Wire the payment webhook. It verifies its signature correctly and then does nothing.	2 weeks
4	Sign firmware images and verify the signature on-device before flashing – and build the manifest endpoint devices already poll.	6 weeks
5	Rewrite the README. It describes an eighth of this system, and it is the first thing anyone reads.	1 day

#1 and #5 together take two days and change what every future contributor sees and what every future commit is checked against.

#2 is the only item on this list where data is being lost right now.

[Back to 01_SYSTEM_OVERVIEW.md](#) · [02_DATABASE_AND_DATA_MODELS.md](#) · [03_INTEGRATIONS_AND_ECOSYSTEM.md](#) · [04_MAINTENANCE_AND_OPERATIONS.md](#)

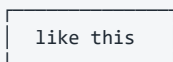
PART 6

06 · Architecture Diagram Atlas

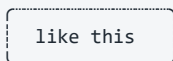
What this is. Every other document in this set explains. This one shows. It is a complete visual inventory of the repository the rest of [Architecture_Docs/](#) calls "website" — 150 API routes, 108 pages, 10 database tables, 35 file-backed stores (only 3 of them durable), five coexisting authentication schemes, and the six other things that live in this same working tree but are not part of the Next.js application at all. Nothing here is summarised away: where a list is long, the list is printed in full.

How to read it. Every numbered section gives the same picture twice: an ASCII/Unicode block that renders in a terminal, a diff or a printed page, and a Mermaid block that renders as an interactive graphic on GitHub, in VS Code and in most wikis. Read whichever one displays for you; they are redundant by design, not complementary. Sections 3 and 5 are the two that must be complete rather than representative — §3 maps every top-level thing in this working tree and §5 names all 150 routes and all 108 pages by path. Every fact below was read from the source in this repository. Where it differs from [01_SYSTEM_OVERVIEW.md](#) through [05_AREAS_OF_ENHANCEMENT.md](#), this document follows the source and says so inline — it corrects one such case in §8.3.

Legend



a component whose code lives in THIS repository (website)



a component that lives somewhere else — another repository in this same working tree, a third-party service, or a physical device



synchronous call, made and awaited



asynchronous, fire-and-forget, or "best effort"



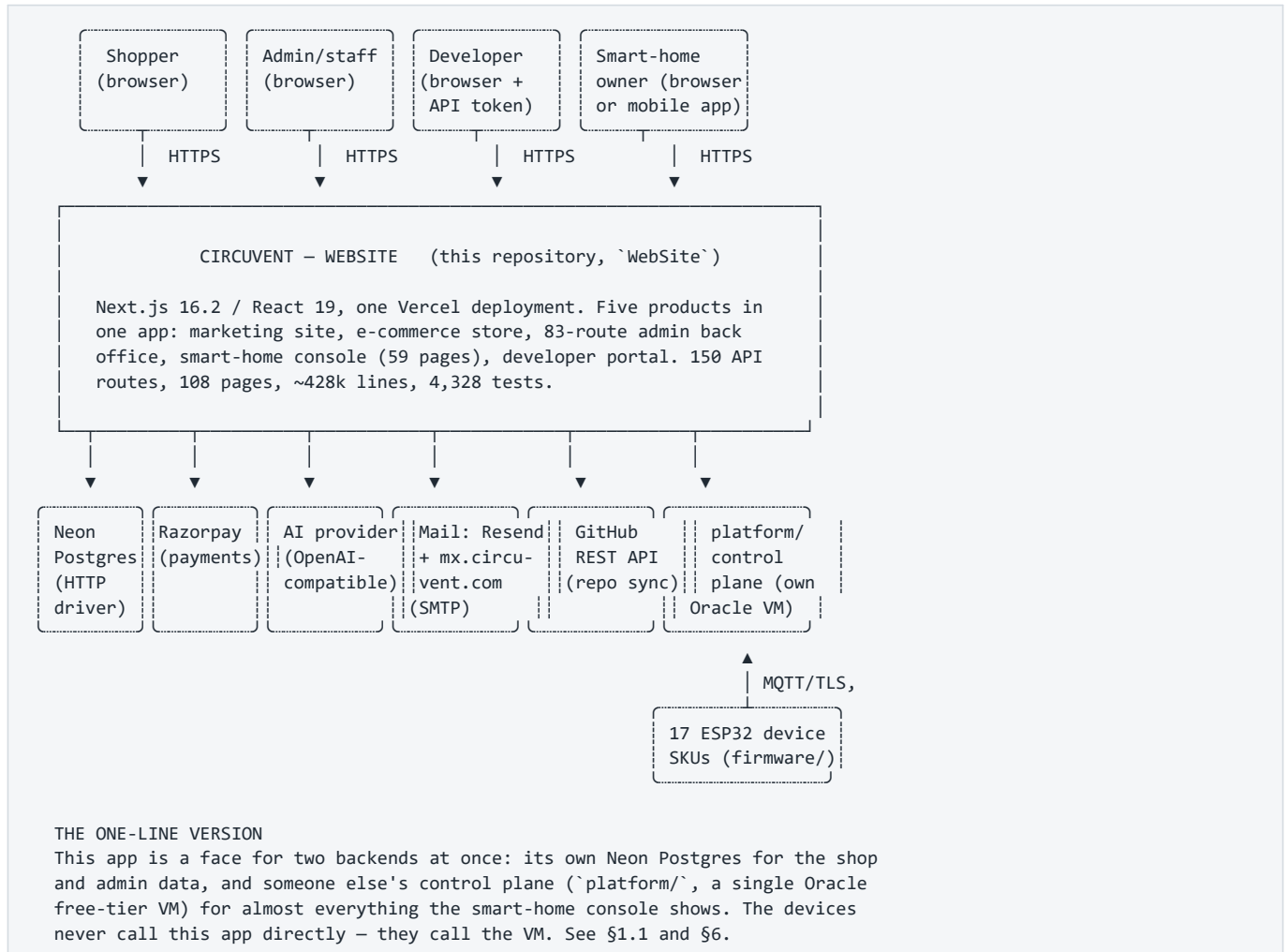
redirect, navigation, or a physical-world action (a relay switching, a lock turning)

[S] static rendered once at build time [R] route handler
[D] dynamic rendered per request, "no-store" [C] client component
[M] memory-only store — gone on cold start / redeploy
[P] Postgres-durable store — the 3 exceptions to [M], see §7.4

No [A] tag appears anywhere in this document: `grep`ing the whole of `src/` for `"use server"` returns nothing. Every mutation in this application is a fetch to a route handler, never a Next.js Server Action.

1. C4 Level 1 — System context

Six things share this working tree (full detail in §3). This section's system boundary is only the first of them: the Next.js application under [src/](#), deployed as one Vercel project. The other five — [firmware/](#), [platform/](#), [hardware/](#), [mobile/](#), [native/](#) — are separate deployables that happen to sit beside it in the same repository, and [circuvent-platform/](#) is not part of Circuvent at all (§3.1).



1.1 — what does NOT appear above, on purpose. `circuvent-platform/` (an unrelated internal HR/payroll SaaS, §3.1) has no network path to or from this system: it is not omitted for brevity, it genuinely does not connect. The mobile app (`mobile/`, shipped Expo build) also does not call this website's API — its `API_BASE` constant points at <https://api.circuvent.com>, the same `platform/` control plane the smart-home console uses (verified in `mobile/src/config.ts`). This website's only two roles in the smart-home story are: rendering the console's page shells and relaying a handful of server-side calls (AI analysis, the alerts cron, one admin probe), and brokering a one-time identity exchange so a shop customer's session can be traded for a console session (`/api/account/sso/console`, detailed in §8.4).

Mermaid diagram — renders interactively in the Markdown edition

C4Context

title Circuvent Website – system context

```
Person(shopper, "Shopper", "Browses the store, buys, tracks orders, manages a wallet")
Person(staff, "Admin / staff", "Runs the back office across 83 admin routes")
Person(developer, "Developer", "Reads docs, requests an API token, calls the public API")
Person(owner, "Smart-home owner", "Uses the console in a browser or the shipped mobile app")

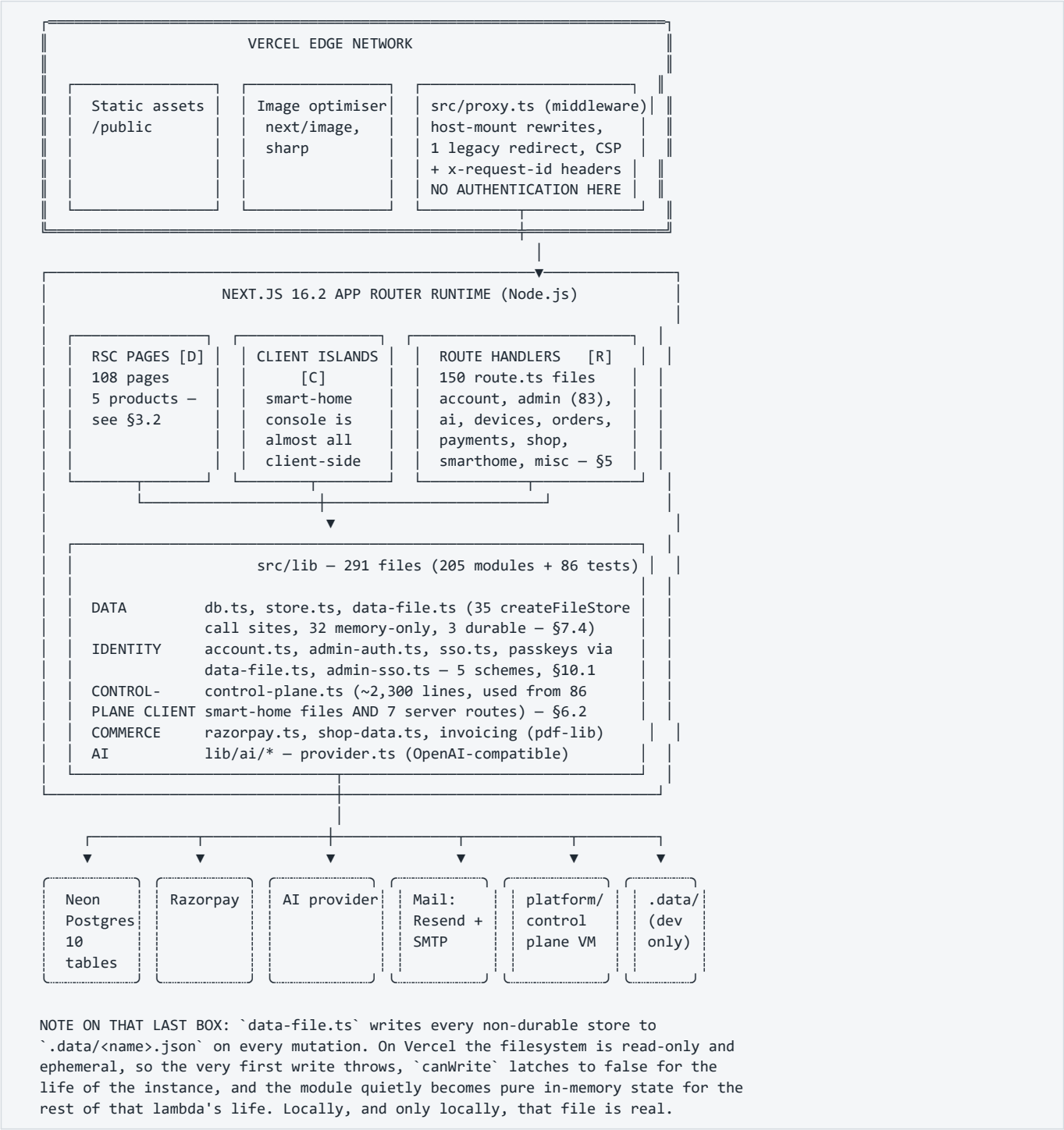
System(website, "Circuvent Website", "Next.js 16 app. 150 API routes, 108 pages. Five products in one deployment.")

SystemDb_Ext(neon, "Neon Postgres", "10 tables, HTTP driver, no transactions, created at runtime by initDb()")
System_Ext(razorpay, "Razorpay", "Payment gateway, called by raw fetch, no SDK dependency")
System_Ext(ai, "AI provider", "OpenAI-compatible chat completions endpoint")
System_Ext(mail, "Mail", "Resend API and self-hosted SMTP at mx.circuvent.com")
System_Ext(github, "GitHub API", "Live repository stats for the developer portal and site pages")
System_Ext(controlplane, "platform/ control plane", "Separate repo. One Oracle VM. Owns almost all smart-home data.")
System_Ext(devices, "ESP32 devices", "17 retail SKUs. Talk to the control plane over MQTT/TLS, never to this app.")
System_Ext(mobileapp, "Mobile app", "Shipped Expo build v1.13.1. Talks to the control plane directly, not this app.")
System_Ext(unrelated, "circuvent-platform/", "Unrelated HR/payroll SaaS in the same working tree. No connection.")

Rel(shopper, website, "Uses", "HTTPS")
Rel(staff, website, "Uses", "HTTPS")
Rel(developer, website, "Uses", "HTTPS + API token")
Rel(owner, website, "Uses for page shells and AI", "HTTPS")
Rel(owner, controlplane, "Uses directly for console data", "HTTPS + Bearer")
Rel(website, neon, "Reads and writes", "SQL over HTTP")
Rel(website, razorpay, "Creates and verifies orders", "HTTPS + webhook")
Rel(website, ai, "Sends prompts", "HTTPS")
Rel(website, mail, "Sends transactional email", "HTTPS / SMTP")
Rel(website, github, "Reads repo metadata", "HTTPS + PAT")
Rel(website, controlplane, "Relays a few server-side calls", "HTTPS + HMAC")
Rel(mobileapp, controlplane, "Uses directly", "HTTPS + WSS")
Rel(devices, controlplane, "Publishes/subscribes", "MQTT/TLS")

UpdateLayoutConfig($c4ShapeInRow="3", $c4BoundaryInRow="1")
```

2. C4 Level 2 — Containers



Mermaid diagram — renders interactively in the Markdown edition

```

flowchart TB
    subgraph edge["Vercel edge network"]
        static["Static assets<br>/public"]
        imgopt["Image optimiser<br>next/image + sharp"]
        proxy["src/proxy.ts middleware<br>host-mounts, 1 redirect, CSP,<br>x-request-id – NO AUTH"]
    end

    subgraph runtime["Next.js 16.2 App Router runtime (Node.js)"]
        pages["RSC pages · 108<br>marketing, shop, admin,<br>smart-home, developer"]
        islands["Client islands<br>smart-home console is<br>almost entirely client-side"]
        handlers["Route handlers · 150<br>account, admin x83, ai,<br>devices, orders, payments,<br>shop, smarthome, misc"]

        subgraph lib["src/lib – 291 files"]
            data["Data<br>db.ts, store.ts, data-file.ts<br>>35 createFileStore sites"]
            identity["Identity<br>account.ts, admin-auth.ts,<br>sso.ts, admin-sso.ts"]
            cp["control-plane.ts<br>>~2,300 lines<br>used client AND server side"]
            commerce["razorpay.ts, shop-data.ts,<br>pdf-lib invoicing"]
            ai["lib/ai/*<br>provider.ts"]
        end
    end

    neon[("Neon Postgres<br>10 tables, HTTP driver")]
    razorpay(["Razorpay"])
    aiprvider(["AI provider<br>OpenAI-compatible"])
    mail(["Mail<br>Resend + SMTP"])
    controlplane(["platform/ control plane<br>one Oracle VM"])
    diskfile[("`.data/*.json`<br>dev only")]

    static --> pages
    imgopt --> pages
    proxy --> handlers
    pages --> lib
    islands --> lib
    handlers --> lib
    data --> neon
    data -->|"dev fallback only"| diskfile
    commerce --> razorpay
    ai --> aiprvider
    identity --> mail
    cp <-->|"HTTPS + Bearer"| controlplane

    style proxy fill:#FEE2E2,stroke:#B91C1C
    style neon fill:#DCFCE7,stroke:#15803D
    style diskfile fill:#FEF3C7,stroke:#B45309
    style controlplane fill:#E0E7FF,stroke:#4338CA

```

3. Container map, level 3 — the complete repository

This is one of the two sections this atlas promises to make *complete*, not representative (§5 is the other). Every top-level entry below was counted with `git ls-files`, run fresh while writing this document — not copied from an older audit.

3.1 — what is actually in this repository.

```

REPOSITORY ROOT  C:\...\WebSite                                3,101 git-tracked files total
|
|— src/                                978 files  THE WEBSITE — this whole atlas is it
|   Next.js 16 App Router, one Vercel deployment, five web products bundled as one
|   app: marketing, e-commerce shop, admin back office, smart-home console,
|   developer portal. Full internal tree in §3.3.
|
|— firmware/                          89 files  30 ESP32 SKETCHES for 17 RETAIL SKUs
|   13,003 files on disk; 12,914 of those are PlatformIO's .pio/ build cache, all
|   gitignored. Only source, headers and platformio.ini are tracked. See §3.2/§11.1.
|
|— platform/                          188 files  THE IoT CONTROL PLANE — ONE ORACLE VM
|   Mosquitto (MQTT+TLS, own CA) + Express API + Postgres + Caddy, plus an Alexa
|   Lambda and a face-recognition service. Single point of failure. See §3.2/§11.2.
|
|— hardware/                          638 files  REAL KiCad PCBs — 25 product folders
|   Schematics, board layouts, Gerbers, enclosures, marketplace listings, product
|   photography. This is not a mockup: these boards are manufacturable. See §3.2.
|
|— mobile/                            279 files  SHIPPED Expo app — Play Store v1.13.1
|   Talks DIRECTLY to platform/ at api.circuitent.com — bypasses this website's own
|   API entirely (verified in mobile/src/config.ts). Includes a native Siri /
|   Shortcuts bridge module (mobile/modules/circuitent-siri/). See §3.2.
|
|— native/                            39 files  Kotlin/Swift PROTOTYPE — iOS unbuilt
|   A from-scratch rewrite attempt with deliberately different application ids
|   (...nativeclient) so it can never be confused with the shipped app. The ios/
|   half has never successfully compiled. Not a release candidate.
|
|— circuitent-platform/              585 files  UNRELATED HR / PAYROLL SaaS
|   Its own Turborepo — apps/{api-gateway,services,web}, packages/{auth,database,
|   event-bus,...}. Shares a brand prefix with everything above and NOTHING else.
|   THE NAME IS A TRAP. See 3.1.1 below.
|
|— Docs/                             55 files  internal 39-chapter knowledge base
|   00-start-here.md ... 39-witness.md — web app, control plane, MQTT protocol,
|   every device, mobile, Play Store, Siri/HomeKit, drone, ANPR, and more. Titles
|   read and listed in §3.2; contents not re-verified for this atlas.
|
|— Architecture_Docs/                11 files  audits 01-05 plus this atlas (06)
|— tests/ + e2e/                    129 files  4,328 automated tests — see §12
|— public/ + scripts/                90 files  static assets, build/ops scripts
|
|— Creds/ + mobile/credentials/      0 tracked  Android signing keystores and one
|   control-plane key sit on local disk but are gitignored (`Creds/`, `credentials/`,
|   `*.jks`, `*.keystore`, `*.key`) and confirmed absent from `git ls-files`. This is
|   correct secrets hygiene, called out here as one of this repo's few unambiguous
|   positives.

```

3.1.1 — the trap, explicitly. `circuitent-platform/` shares three characters of naming with `platform/` (this app's real IoT control plane) and a "Circuitent" brand prefix with everything else in this working tree — and nothing more. It is a completely separate Turborepo: its own `package.json`, its own `.turbo/` cache, its own `.vercel/` project link, its own `apps/{api-gateway,services,web}` and its own `packages/{audit,auth,database,event-bus,pdf-engine,shared,testing,websocket}` — a full independent `auth` package and a full independent `database` package, sharing zero source files, zero database connections and zero deployment target with the website this atlas describes. Its own `README.md` publishes seed login credentials for its own demo environment; that is a defect of **that unrelated project**, not of this one, and is out of scope beyond this warning. If a search of this working tree for "circuitent" ever lands you in `circuitent-platform/`, this paragraph is why — turn back.

[Mermaid diagram — renders interactively in the Markdown edition](#)

```

flowchart TB
    root["Working tree<br/>C:\...\WebSite<br/>3,101 git-tracked files"]

    subgraph website["src/ -- THE WEBSITE -- this atlas"]
        web["978 files<br/>5 web products, 1 Next.js app"]
    end

    subgraph fw["firmware/"]
        firm["89 tracked, approx 13,003 on disk<br/>30 ESP32 sketches, 17 retail SKUs"]
    end

    subgraph plat["platform/"]
        cp["188 files<br/>IoT control plane, 1 Oracle VM"]
    end

    subgraph hw["hardware/"]
        hard["638 files<br/>KiCad PCBs, 25 products"]
    end

    subgraph mob["mobile/"]
        mobapp["279 files<br/>shipped Expo app v1.13.1"]
    end

    subgraph nat["native/"]
        natapp["39 files<br/>Kotlin/Swift, iOS never built"]
    end

    subgraph unrelated["circuvent-platform/"]
        hr["585 files<br/>UNRELATED HR/payroll SaaS"]
    end

    root --> website
    root --> fw
    root --> plat
    root --> hw
    root --> mob
    root --> nat
    root -.->|"no code relationship"| unrelated

    mobapp -->|"talks directly to"| cp
    firm -->|"MQTT over TLS"| cp
    web -->|"HMAC federation<br/>+ a few relayed calls"| cp

    style unrelated fill:#FEE2E2,stroke:#B91C1C,stroke-dasharray: 5 5
    style website fill:#DCFCE7,stroke:#15803D

```

3.2 — inside the six non-website projects.

```

firmware/                                30 sketches (one .ino each) + 2 shared libraries
├─ CircuventDevice/                      shared base library -- Wi-Fi/MQTT/OTA client, no .ino
├─ CircuventRC/                          shared RC-vehicle library -- ESP-NOW/MAVLink, no .ino
├─ agri-starter/ anpr-cam/ aquaguard/ camera/ curtain/ drone-fc/ drone-link/
├─ energy-monitor/ facedoor/ guardian/ home-hub/ meter/ motion-sensor/
├─ rc-link/ rc-remote/ rccar/ rfid-attend/ rfid-gate/ sentinel/ smart-fan/
├─ smart-light/ smart-lock/ smart-plug/ smart-switch/ switchboard/
├─ touchboard/ touchboard-8/ watertank/ watertank-sensor/ witness/
└─ each sketch dir: <name>.ino + platformio.ini + .pio/ (cache, gitignored)
    30 sketches total; 17 of them ship as retail SKUs -- device table in §11.1.

platform/                                the IoT control plane -- one Oracle free-tier VM
├─ mosquitto/                            MQTT broker config + its own certificate authority
├─ api/src/                              Express/Node API -- devices, auth, MQTT bridge, ANPR
├─ api/dist/, node_modules/, scripts/    build output, dependencies, ops scripts
├─ caddy/                                reverse proxy / TLS termination in front of api/
├─ alexa-lambda/                         Alexa Smart Home skill Lambda -- voice control bridge
└─ face/                                face-recognition service, backs the facedoor/ SKU

hardware/                                25 product directories + one shared KiCad library
├─ lib/Circuvent.pretty/                 shared KiCad footprint library ("pretty" is KiCad's
├─                                         own naming convention for a footprint-library folder)
├─ agri-starter/ anpr-camera/ curtain/ drone-x1/ energy-monitor/ facedoor/
├─ guardian/ home-automation/ load-controller/ meter-1ch/ meter-3ch/
├─ motion-sensor/ psu-5v3v3/ psu-adapter-12v/ psu-adapter-5v/ rfid-gate/
├─ sentinel/ smart-fan/ smart-light/ smart-lock/ smart-plug/ smart-switch/
├─ touchboard/ water-tank-controller/    (plus a near-empty anpr-cam/ stub)
└─ each product dir: pcb/ (schematic + layout + Gerbers), enclosure/, images/,
    listings/ (marketplace copy) -- this is a real, manufacturable hardware line.

mobile/                                  the shipped Expo / React Native app, v1.13.1
├─ src/                                  app source -- screens mirror the smart-home console
├─ modules/circuvent-siri/               native Expo module -- Siri Shortcuts + App Intents
├─ android/                              native Android project (the Play Store release build)
├─ credentials/                          signing keystores -- gitignored, not tracked (§3.1)
├─ design-system/, assets/               shared UI tokens, images, fonts
├─ store/                                Play Store listing assets -- screenshots, icons
└─ plugins/, scripts/, dist/             Expo config plugins, build/release scripts, output

native/                                  Kotlin/Swift PROTOTYPE -- a from-scratch rewrite try
├─ android/                              Kotlin client -- different application id (nativeclient)
└─ ios/                                  Swift client -- has NEVER SUCCESSFULLY COMPILED

circuvent-platform/                      UNRELATED -- see 3.1.1. Shown only for completeness.
├─ apps/api-gateway/, apps/services/, apps/web/  its own Next.js/Node apps
├─ packages/auth/, packages/database/, packages/event-bus/  its own primitives
├─ packages/audit/, packages/pdf-engine/, packages/shared/, packages/testing/
├─ packages/websocket/
└─ docker/, .turbo/                      its own containers + build cache

```

3.3 — inside `src/` — the website's own five products.

```

src/
├── app/                                108 pages + 150 route.ts + layout/error/robots/sitemap
│   Route groups by product (page counts in §5.2, API counts in §5.1):
│   marketing (25 singleton pages: /, /about, /services, /faq, /docs, /dev, ...)
│   shop/ (6 pages: /shop, /cart, /checkout, /account/*, /wallet, /orders/[id])
│   admin/ (1 page shell driving 83 API routes -- a client-rendered app-in-a-route)
│   smarthome/ (59 pages -- the largest single product, almost entirely [C])
│   developer/ (10 pages -- docs, playground, token management)
│   projects/ (2) blog/ (2) careers/ (2) domains/ (2) case-studies, roadmap, ...
│   api/ (150 route.ts files -- full tree in §5.1)
│   .well-known/ (workflow SDK + well-known probes; excluded by proxy's matcher)
├── components/                        119 files
│   ai/ (chat/assistant widgets)      shop/ (cart, product cards, checkout UI)
│   ui/ (shared design-system primitives)  weather/ (widget used on 2 pages)
├── hooks/                            15 files      shared client-side React hooks
├── lib/                              291 files      205 modules + 86 co-located *.test.ts
│   full category breakdown in §4: data, identity, control-plane client, commerce,
│   AI, notifications, PDF/report generation, validation, rate limiting, ...
├── workflows/                        2 files      icm-postmortem.ts, icm-watch.ts --
│   Vercel Workflow SDK (beta) jobs behind the durable Incident Management feature

```

Mermaid diagram — renders interactively in the Markdown edition

```

graph LR
    subgraph src ["src/"]
        appdir["app/<br/>108 pages + 150 routes"]
        comp["components/<br/>119 files"]
        hooks["hooks/<br/>15 files"]
        lib["lib/<br/>291 files"]
        wf["workflows/<br/>2 files"]
    end

    appdir --> comp
    appdir --> lib
    comp --> hooks
    lib --> wf

    subgraph products ["5 products inside app/"]
        marketing["marketing<br/>25 pages"]
        shop["shop<br/>6 pages"]
        admin["admin<br/>1 shell + 83 routes"]
        smarthome["smarthome<br/>59 pages"]
        developer["developer<br/>10 pages"]
    end

    appdir --- marketing
    appdir --- shop
    appdir --- admin
    appdir --- smarthome
    appdir --- developer

    style smarthome fill:#E0E7FF,stroke:#4338CA
    style admin fill:#FEF3C7,stroke:#B45309

```

4. Module / library layer

`src/lib` holds 205 non-test modules (plus 86 co-located `*.test.ts` files, §12). 98 of them fall into an unambiguous group by filename prefix alone (`admin-*`, `smarthome-*`, `icm*`, `report*`, `*insights*`, `shop-*`, `control-plane*`, and the nine files under `lib/ai/`). The other 107 do not share a prefix, so they are grouped here by theme; that split was verified to be an exact partition of all 107 filenames — no module counted twice, none left out. Nothing in this section is estimated.

CATEGORY	COUNT	REPRESENTATIVE MODULES
ADMIN DOMAIN	(admin-*) 29	admin-crm, admin-cms, admin-pricing, admin-tax, admin-fraud, admin-sso, admin-vendors, admin-report-builder, admin-password-policy
SMART-HOME / DEVICE DOMAIN (30 prefixed + 30 thematic)	60	smarthome-* (30 prefixed: -dashboard, -auth, -security, -firmware, -realtime, -groups...) + 30 thematic: camera-relay, camera-audio, guardian-health/-hold, tank-health/-link, switchboard, witness, energy-advisor, predictive-maintenance, device-history/-normalize, firmware-catalog.generated, agri
OBSERVABILITY & REPORTING	21	icm* (7: incident mgmt store/bridge/notify), app-insights*/insights* (8: cost, usage, anomalies, alert rules), reports*/report-logo (6: PDF generation via pdf-lib)
SHOP / COMMERCE	(shop-* + 10) 16	shop-catalog, shop-data, shop-policy + razorpay, order-core, coupons, inventory, warranty, bundle-pricing, delivery-estimate
AI	(lib/ai/*) 9	provider (OpenAI-compatible), assistant, fleet, analysis, tools, console-identity
IDENTITY & SESSIONS	11	account, sso, passkeys, passkey-ceremony, totp, session-expiry, identity-groups
CORE DATA / INFRASTRUCTURE	16	db, store, data-file, cache, rate-limit, secrets, validation, api-handler, csp, config
CONTROL-PLANE CLIENT	3	control-plane, control-plane-health, -live
CONTENT & MARKETING DATA	28	blog-data, showcase-data, stack-data, seo, brand, animations, quiz-data, case-studies-data (mostly static arrays, no I/O)
DEVELOPER PORTAL	3	developer-api.generated, developer-docs, v1-shapes
INTEGRATIONS (misc external)	3	cv365-firebase, github-sync, weather
NOTIFICATIONS	2	web-push, useWebPush
MISC	4	console-layout, deployments, documents, utils
TOTAL MODULES	205	(+ 86 co-located *.test.ts files -- see §12)

The dependency picture below is a category-level view, not a line-by-line import audit of all 205 files: it is built from the naming convention above plus the two import greps already performed for **control-plane.ts** (86 client call sites, 7 server route call sites) and **db.ts/store.ts** (imported from nearly every domain category).

[Mermaid diagram](#) — renders interactively in the Markdown edition


```

flowchart TB
    admin["ADMIN DOMAIN<br/>29 modules"]
    smarthome["SMART-HOME / DEVICE<br/>60 modules"]
    obs["OBSERVABILITY & REPORTING<br/>21 modules"]
    commerce["SHOP / COMMERCE<br/>16 modules"]
    ai["AI<br/>9 modules"]
    identity["IDENTITY & SESSIONS<br/>11 modules"]
    infra["CORE DATA / INFRA<br/>16 modules"]
    cpclient["CONTROL-PLANE CLIENT<br/>3 modules"]
    content["CONTENT & MARKETING DATA<br/>28 modules"]
    devportal["DEVELOPER PORTAL<br/>3 modules"]
    integrations["INTEGRATIONS<br/>3 modules"]
    notify["NOTIFICATIONS<br/>2 modules"]

    admin --> infra
    admin --> identity
    admin --> obs
    admin --> notify
    smarthome --> cpclient
    smarthome --> infra
    ai --> cpclient
    ai --> integrations
    commerce --> infra
    commerce --> integrations
    obs --> infra
    obs --> notify
    identity --> infra
    devportal --> infra
    devportal --> integrations
    content --> infra

    cpclient -->|"HTTPS + HMAC / Bearer"| ext["platform/ control plane<br/>(separate repo, §6.7)"]

    style smarthome fill:#E0E7FF,stroke:#4338CA
    style cpclient fill:#E0E7FF,stroke:#4338CA
    style infra fill:#DCFCE7,stroke:#15803D
    style admin fill:#FEF3C7,stroke:#B45309

```

5. The complete route map

This is the section the spec singles out for exhaustive treatment, and the numbers explain why: **150** files under `src/app/api/**/*.route.ts` and **108** files under `src/app/**/*.page.tsx` — 258 routable paths in one Next.js App Router tree, against the Career.circuitent example's few dozen. Per the spec's own rule ("if there are more than ~60 routes, give a full ASCII tree of every route path plus tables for the most important groups"), §5.1 and §5.3 below are the complete trees — every path, no sampling — and §5.2 and §5.4 add method, auth and purpose detail on top. Every one of the 150 API routes is named explicitly at least twice: once in the tree, once in a table.

5.1 API routes — the complete tree (150 routes, 26 top-level groups)

src/app/api/	150 route.ts files
	26 top-level groups
└─ account/	14 routes
└─ addresses/route.ts	
└─ avatar/route.ts	
└─ change-password/route.ts	
└─ documents/[orderNo]/route.ts	
└─ forgot-password/route.ts	
└─ login/route.ts	
└─ notifications/route.ts	
└─ orders/route.ts	
└─ passkey/route.ts	
└─ profile/route.ts	
└─ register/route.ts	
└─ reset-password/route.ts	
└─ sso/console/route.ts	
└─ verify-otp/route.ts	
└─ admin/	83 routes
└─ 2fa/route.ts	
└─ 2fa/totp/route.ts	
└─ affiliates/route.ts	
└─ alerts/route.ts	
└─ alerts/rules/route.ts	
└─ alerts/run/route.ts	
└─ analytics/route.ts	
└─ audit/route.ts	
└─ auth/route.ts	
└─ auth/sso/callback/route.ts	
└─ auth/sso/exchange/route.ts	
└─ auth/sso/start/route.ts	
└─ auth/verify-2fa/route.ts	
└─ availability/probe/route.ts	
└─ bulk/route.ts	
└─ bundles/route.ts	
└─ cms/route.ts	
└─ coupons/route.ts	
└─ crm/route.ts	
└─ cron-health/route.ts	
└─ currency/route.ts	
└─ customers/route.ts	
└─ emails/route.ts	
└─ external-stats/route.ts	
└─ flags/route.ts	
└─ forecasting/route.ts	
└─ fraud/route.ts	
└─ giftcards/route.ts	
└─ groups/route.ts	
└─ icm/route.ts	
└─ insights/route.ts	
└─ insights-failures/route.ts	
└─ insights-query/route.ts	
└─ insights-rules/route.ts	
└─ insights-telemetry/route.ts	
└─ insights-usage/route.ts	
└─ insights/export/route.ts	
└─ integrations/route.ts	
└─ inventory/batches/route.ts	
└─ inventory/counts/route.ts	
└─ inventory/export/route.ts	
└─ inventory/locations/route.ts	
└─ inventory/meta/route.ts	
└─ inventory/movements/route.ts	
└─ inventory/purchase-orders/route.ts	
└─ inventory/reports/route.ts	
└─ inventory/settings/route.ts	
└─ inventory/suppliers/route.ts	
└─ inventory/taxonomy/route.ts	

(continued — part 2 of 3)

		inventory/transfers/route.ts	
		jobs/route.ts	
		latency/route.ts	
		macros/route.ts	
		marketing/route.ts	
		messages/route.ts	
		orders/route.ts	
		passkey/route.ts	
		password/route.ts	
		pricing/route.ts	
		privacy/route.ts	
		products/route.ts	
		questions/route.ts	
		report-builder/route.ts	
		reports/catalog/route.ts	
		reports/csv/route.ts	
		reports/data/route.ts	
		reports/pdf/route.ts	
		reports/schedules/route.ts	
		reports/schedules/run/route.ts	
		reports/send/route.ts	
		returns/route.ts	
		seo/route.ts	
		shipping/route.ts	
		staff/route.ts	
		staff-activity/route.ts	
		stats/route.ts	
		subscriptions/route.ts	
		support/route.ts	
		surveys/route.ts	
		tax/route.ts	
		traffic/route.ts	
		vendor-portal/route.ts	
		warranty/route.ts	
		ai/	3 routes
		analyze/route.ts	
		chat/route.ts	
		fleet/route.ts	
		blog/route.ts	1 route
		contact/route.ts	1 route
		coupons/	1 route
		validate/route.ts	
		devices/	5 routes
		route.ts	
		claim/route.ts	
		command/route.ts	
		firmware/route.ts	
		sync/route.ts	
		giftcards/	1 route
		redeem/route.ts	
		github/route.ts	1 route
		health/	2 routes
		route.ts	
		db/route.ts	
		loyalty/route.ts	1 route
		newsletter/route.ts	1 route
		notify-restock/route.ts	1 route

(continued — part 3 of 3)

— orders/	4 routes
— route.ts	
— cancel/route.ts	
— reorder/route.ts	
— track/route.ts	
— payments/	3 routes
— create-order/route.ts	
— verify/route.ts	
— webhook/route.ts	
— projects/route.ts	1 route
— push/	2 routes
— key/route.ts	
— subscribe/route.ts	
— referral/route.ts	1 route
— returns/route.ts	1 route
— shop/	5 routes
— bundles/route.ts	
— products/route.ts	
— questions/route.ts	
— quote/route.ts	
— reviews/route.ts	
— smarthome/	10 routes
— admin/config/route.ts	
— alerts/route.ts	
— alerts/cron/route.ts	
— camera/audio/route.ts	
— camera/frame/route.ts	
— camera/listen/route.ts	
— camera/speak/route.ts	
— camera/watch/route.ts	
— dev-portal/route.ts	
— prefs/route.ts	
— support/route.ts	1 route
— telemetry/	2 routes
— route.ts	
— failure/route.ts	
— visitors/	2 routes
— route.ts	
— stream/route.ts	
— wallet/	2 routes
— route.ts	
— topup/route.ts	
— weather/route.ts	1 route

Group sizes at a glance: **admin** (83) is 55% of the whole API surface by itself — one back office out-weighs the marketing site, the store, the smart-home console, the developer portal and every utility endpoint combined. **smarthome** (10 routes) is deceptively small: most of the console's 59 pages read and write through **platform**/'s own HTTP API on the Oracle VM, not through these 10 — they are the exceptions that need a Next.js-side proxy (camera relay, dev-portal token issuance, cron-triggered alert sweep, an admin-config panel, and a preferences store). Route *count* under-represents **smarthome**'s real footprint; \$6 traces the outbound calls that make up for it.

5.2 API routes — grouped tables

Auth-column shorthand used below (all confirmed by reading the route file or its imported verifier, not inferred from the path):

- **shop token** — HMAC-signed cookie/bearer token, `verifyToken` / `tokenFromRequest` in `src/lib/account.ts` (§10.1).
- **admin session** — `guard(req, area)` in `src/lib/admin-auth.ts`: HMAC token plus a per-area role check (§10.2).

- **device header keys** — `x-device-id` + `x-device-key` request headers, checked against the device row itself; no shared secret, no JWT.
- **control-plane bearer** — the caller's bearer token is forwarded as-is to an endpoint on `platform/` (the Oracle VM) and *that* service's answer is trusted; nothing is decoded or verified locally. Four independent call sites do this against four different control-plane endpoints (§10.7).
- **cron secret** — a single static bearer value (`CRON_SECRET`) compared with `===`; shared by whatever scheduler is configured, not per-caller.
- **none** — no authentication check in the route at all; only rate limiting (if present) restrains use.

5.2.1 `account/*` — shop customer identity (14 routes)

Path	Method(s)	Auth	Purpose
<code>account/register</code>	POST	none	create a shop account, send verification OTP
<code>account/verify-otp</code>	POST	none	confirm the OTP, activate the account
<code>account/login</code>	POST	none	password login, issues the shop token
<code>account/forgot-password</code>	POST	none	request a reset link
<code>account/reset-password</code>	POST	none (reset token)	consume the reset link
<code>account/change-password</code>	POST	shop token	change password while signed in
<code>account/profile</code>	GET, PATCH	shop token	read / edit profile fields
<code>account/avatar</code>	GET, POST, DELETE	shop token	upload / remove avatar image
<code>account/addresses</code>	GET, POST, PATCH, DELETE	shop token	address book CRUD
<code>account/orders</code>	GET	shop token	this customer's order history
<code>account/documents/[orderId]</code>	GET	shop token	invoice/receipt PDF for one order
<code>account/notifications</code>	GET, PATCH, DELETE	shop token	in-app notification inbox
<code>account/passkey</code>	GET, POST, DELETE	shop token	WebAuthn passkey register/list/remove
<code>account/sso/console</code>	POST	shop token	mint a bridge token so the smart-home console can trust an already-signed-in shop customer (§8.4)

5.2.2 `devices/*` — the split that isn't obvious from the folder name (5 routes)

Three routes are called by the **signed-in shop customer** (web or mobile); two are called by **the device itself**. Nothing in the URL signals the difference — it is only visible by opening each file.

Path	Method	Auth	Purpose
<code>devices</code>	GET	shop token	list the devices this account owns
<code>devices/claim</code>	POST	shop token	bind a purchased device's serial to this account
<code>devices/command</code>	POST	shop token	queue a command (lock, relay toggle, etc.) for a device
<code>devices/sync</code>	POST	device header keys	device pushes state/telemetry up
<code>devices/firmware</code>	GET	device header keys	device polls for a pending OTA manifest (§8.2)

5.2.3 `orders/*`, `payments/*`, `coupons/*`, `giftcards/*` — checkout (4 + 3 + 1 + 1 routes)

Path	Method	Auth	Purpose
<code>orders</code>	POST	shop token	create an order from the current cart
<code>orders/cancel</code>	POST	shop token	cancel before fulfillment
<code>orders/reorder</code>	POST	shop token	duplicate a past order into a new cart

Path	Method	Auth	Purpose
orders/track	GET	none (order no. + email)	public order-status lookup
payments/create-order	POST	shop token	open a Razorpay order for the cart total
payments/verify	POST	shop token	verify the client-side payment signature on browser return
payments/webhook	POST	Razorpay HMAC signature	server-to-server payment event — verifies the signature and stops; see §8.1
coupons/validate	POST	none	check a coupon code before checkout
giftcards/redeem	POST	shop token	apply a gift-card balance to an order

5.2.4 shop/* — storefront catalogue (5 routes)

Path	Method	Auth	Purpose
shop/products	GET	none	catalogue listing/search
shop/bundles	GET	none	multi-product bundle listing
shop/quote	POST	none	freight/installation quote request
shop/questions	GET, POST, PATCH	none / shop token	product Q&A, read public, post signed in
shop/reviews	GET, POST, PATCH	none / shop token	product reviews, read public, post signed in

5.2.5 ai/* — the assistant surface (3 routes)

Path	Method	Auth	Purpose
ai/chat	POST	shop token or admin session	one endpoint, two personas — resolveConsoleIdentity/mergePersona pick shop-customer or admin voice from whichever token is present
ai/analyze	POST	(rate-limited)	ad hoc content/data analysis helper
ai/fleet	POST	control-plane bearer	forwards the caller's own console token to GET /admin/devices on platform/ ; this route deliberately makes no access decision itself (its own source comment says so) — the control plane's answer is the only gate

5.2.6 smarthome/* — console-side proxy and utility routes (10 routes)

Four different auth styles inside one 10-route folder — the console has no single "smart-home auth" of its own:

Path	Method	Auth	Purpose
smarthome/dev-portal	GET, POST, PATCH, DELETE	control-plane bearer (verifyConsolePrincipal → /devices)	issue/list/revoke developer-portal API tokens
smarthome/admin/config	GET, POST, PATCH, DELETE	control-plane bearer (verifyOperator → /admin/me)	console-wide admin configuration panel
smarthome/prefs	GET, PUT, DELETE	control-plane bearer (verifyCaller → /rooms)	per-user console preferences (layout, units, theme)
smarthome/alerts	POST	control-plane bearer (forwarded)	create/acknowledge an alert
smarthome/alerts/cron	GET, POST	cron secret (CRON_SECRET)	scheduled sweep that evaluates alert rules
smarthome/camera/frame	POST, GET	device short-lived token / control-plane bearer	camera relay: device uploads a JPEG frame (POST), owner views it (GET)
smarthome/camera/watch	POST	control-plane bearer	arm a camera session, mint the device's short-lived upload token
smarthome/camera/listen, camera/speak	POST, GET	device short-lived token / control-plane bearer	two-way audio relay, same pattern as camera/frame
smarthome/camera/audio	POST	device short-lived token	device uploads an audio chunk

Three of these routes call three *different* control-plane endpoints ([/devices](#), [/admin/me](#), [/rooms](#)) through three separately-written verifier functions ([verifyConsolePrincipal](#) in [smarthome-auth.ts](#), [verifyOperator](#) in [admin-config.ts](#), [verifyCaller](#) in

`user-prefs.ts`) — plus `ai/fleet`'s inline check against a fourth (`/admin/devices`). No shared helper decodes a session locally; every one of these is a live round trip to the Oracle VM. See §10.7.

5.2.7 `admin/*` — the 83-route back office, partitioned into seven themes

83 routes is too many for one table. They are partitioned below by what the route is *for*, verified as an exact, non-overlapping split of all 83 files ($9 + 19 + 8 + 12 + 14 + 14 + 7 = 83$). Unless noted otherwise, auth is **admin session** — `guard(request, "<area>")` — where `<area>` is one of the role-based `AdminArea` values checked by `roleCan()` in `admin-auth.ts`.

a) Identity and session establishment (9 routes) — the routes that *are* the login, so they cannot themselves require the session they create:

Path	Method	Auth	Purpose
<code>admin/auth</code>	POST, GET	none / sets session	password login; GET reads current session
<code>admin/auth/verify-2fa</code>	POST	pending-2FA state	second factor to complete login
<code>admin/auth/sso/start</code>	GET	none	begin OIDC/PKCE redirect to <code>auth.circuvent.com</code>
<code>admin/auth/sso/callback</code>	GET	OIDC state param	receive the authorization code
<code>admin/auth/sso/exchange</code>	POST	OIDC code	exchange code for tokens, issue admin session
<code>admin/2fa</code>	GET, PUT	admin session	view/enable TOTP enrollment
<code>admin/2fa/totp</code>	POST, PUT, DELETE	admin session	TOTP secret lifecycle
<code>admin/passkey</code>	GET, POST, DELETE	admin session	WebAuthn passkey register/list/remove
<code>admin/password</code>	GET, POST	admin session	change password, view policy

b) Insights, alerts and observability (19 routes) — the admin console's own telemetry and monitoring surface, distinct from `platform/'s`:

Path	Method	Purpose
<code>admin/analytics</code>	GET	site-wide traffic/conversion analytics
<code>admin/stats</code>	GET	headline dashboard counters
<code>admin/traffic</code>	GET	request-volume time series
<code>admin/latency</code>	GET	route-latency percentiles
<code>admin/audit</code>	GET	admin action audit log
<code>admin/cron-health</code>	GET	last-run status of scheduled jobs
<code>admin/external-stats</code>	GET	third-party/integration health counters
<code>admin/forecasting</code>	GET	demand/revenue forecasting
<code>admin/availability/probe</code>	GET, POST	external reachability probe of the control-plane VM, deliberately not run from the VM itself — the source comment explains that a self-hosted prober goes silent at exactly the moment it should alert
<code>admin/alerts</code>	GET	list configured alert conditions
<code>admin/alerts/rules</code>	GET, PUT	edit alert-rule thresholds
<code>admin/alerts/run</code>	GET, POST	manually trigger an alert sweep
<code>admin/insights</code>	GET	insights dashboard root
<code>admin/insights/export</code>	GET	export insights data
<code>admin/insights-query</code>	POST, GET	ad hoc insights query
<code>admin/insights-rules</code>	GET, POST, DELETE	saved insight rule CRUD
<code>admin/insights-failures</code>	GET	failed-insight-run log
<code>admin/insights-telemetry</code>	GET, DELETE	raw telemetry feeding insights
<code>admin/insights-usage</code>	GET, POST	insights feature usage metering

c) Scheduled and exportable reports (8 routes) — includes the `groups` lookup from §5.2.7(f) as its recipient list:

Path	Method	Purpose
admin/report-builder	GET, POST, DELETE	define a custom report template
admin/reports/catalog	GET	list available report types
admin/reports/data	GET	run a report, return JSON
admin/reports/csv	GET	run a report, return CSV
admin/reports/pdf	GET	run a report, return PDF
admin/reports/schedules	GET, POST, PUT, DELETE	recurring report schedule CRUD
admin/reports/schedules/run	GET, POST	force one scheduled run now
admin/reports/send	GET, POST	email a report immediately

d) Inventory and warehouse operations (12 routes) — all under `admin/inventory/*`, all admin-session gated:

Path	Method	Purpose
inventory/batches	GET, POST, DELETE	lot/batch tracking
inventory/counts	GET, POST, PATCH	cycle-count sessions
inventory/locations	GET, POST, DELETE	warehouse/bin locations
inventory/movements	GET, POST	stock movement ledger
inventory/transfers	GET, POST, PATCH	inter-location transfers
inventory/purchase-orders	GET, POST, PATCH, DELETE	supplier purchase orders
inventory/suppliers	GET, POST, DELETE	supplier directory
inventory/taxonomy	GET, POST, DELETE	category/attribute taxonomy
inventory/meta	GET, PATCH	inventory module metadata
inventory/settings	GET, PATCH	inventory module settings
inventory/export	GET	CSV/bulk export
inventory/reports	GET	inventory-specific reports

e) Commerce and catalogue configuration (14 routes):

Path	Method	Purpose
admin/products	GET, POST, PATCH, DELETE	product catalogue CRUD
admin/bundles	GET, POST, DELETE	bundle configuration
admin/pricing	GET, POST, PATCH, DELETE	price list / rule management
admin/currency	GET, POST, DELETE	supported currencies and rates
admin/tax	GET, POST, DELETE	tax rule configuration
admin/shipping	GET, POST, DELETE	shipping method/zone configuration
admin/coupons	GET, POST, PATCH, DELETE	coupon code management
admin/giftcards	GET, POST, PATCH	gift-card issuance/balance
admin/subscriptions	GET, POST, PATCH, DELETE	recurring-order plans
admin/warranty	GET, POST, PATCH	warranty term configuration
admin/returns	GET, PATCH	returns/RMA processing
admin/orders	GET, PATCH	order management
admin/questions	GET, PATCH, DELETE	moderate product Q&A
admin/fraud	GET, POST, PATCH, DELETE	fraud-rule / risk-flag management

f) CRM, marketing, support and content (14 routes):

Path	Method	Purpose
admin/customers	GET, PATCH	customer record management
admin/crm	GET, POST, DELETE	lead/contact pipeline
admin/affiliates	GET, POST, PATCH	affiliate program management
admin/marketing	GET, POST, PATCH, DELETE	campaign configuration
admin/emails	GET	transactional/marketing email log
admin/messages	GET, PATCH	internal admin messaging
admin/support	GET, PATCH	support-ticket queue
admin/macros	GET, POST, DELETE	canned support-reply macros
admin/surveys	GET, POST	customer survey management
admin/vendor-portal	GET, POST, PATCH, DELETE	vendor-facing self-service panel
admin/cms	GET, POST, PATCH, DELETE	blog/content posts, revisions, scheduling
admin/seo	GET, POST, DELETE	SEO metadata overrides and redirects
admin/privacy	GET, POST, PATCH	data-subject access/export/delete requests
admin/groups	GET	mail-enabled distribution groups fetched live from auth.circuvent.com (<code>GET {ISSUER}/api/groups</code>) for addressing scheduled reports — this is the company's own identity domain, not the unrelated circuvent-platform/ repository (§3)

g) Platform ops and staff tooling (7 routes):

Path	Method	Purpose
admin/staff	GET, POST, PATCH, DELETE	admin user/role management
admin/staff-activity	GET	staff action activity feed
admin/integrations	GET, POST, PATCH, DELETE	third-party integration credentials/config
admin/flags	GET, POST, PATCH, DELETE	feature flags and experiments
admin/jobs	GET, POST, PATCH, DELETE	background/scheduled job registry and run log
admin/bulk	GET, POST	CSV bulk import/export for products and customers
admin/icm	GET, POST, PATCH, DELETE	incident and change management: file/track incidents, on-call rotations, postmortems

5.2.8 Public and utility routes — everything not yet named (21 routes)

The remaining single-purpose routes, none of them part of a themed group above. §5.2.1–§5.2.7 between them name 129 routes (14 account + 5 devices + 9 orders/payments/coupons/giftcards + 5 shop + 3 ai + 10 smarthome + 83 admin); this table supplies the last 21, for 150 named in total.

Path	Method	Auth	Purpose
blog	GET	none	blog post listing (public marketing site)
contact	POST	none, rate-limited	contact-form submission
github	GET	none	live GitHub repo/stat card for the marketing site
health	GET	none	liveness probe
health/db	GET	none	database-reachability probe (<code>initDb()</code> round trip)
loyalty	GET, POST	shop token	loyalty-points balance and redemption
newsletter	POST	none, rate-limited	newsletter signup
notify-restock	POST	none	"notify me" back-in-stock signup
projects	GET	none	portfolio/projects listing
push/key	GET	none	public VAPID key for Web Push

Path	Method	Auth	Purpose
push/subscribe	POST, DELETE	none	register/remove a Web Push subscription
referral	GET	shop token	referral code and reward status
returns	GET, POST	shop token	self-service return request
support	GET, POST	none / shop token	support ticket create/list
telemetry	POST	none, rate-limited	client-side telemetry beacon
telemetry/failure	POST	none, rate-limited	client-side error beacon
visitors	POST, GET	none	live-visitor counter, write and read
visitors/stream	GET	none	Server-Sent-Events stream of the same counter
wallet	GET	shop token	store-credit wallet balance
wallet/topup	POST	shop token	add funds to the wallet
weather	GET	none	weather widget data (proxies a third-party API)

That table has 21 rows, one per route (**health** and **health/db** are two distinct files, likewise **push/key+push/subscribe**, **telemetry+ telemetry/failure**, **visitors+visitors/stream**, **wallet+wallet/topup**) — the last 21 of 150, so $129 + 21 = 150$ routes named in total across §5.2.1–§5.2.8, in addition to every one of them already appearing in the §5.1 tree.

5.3 Pages — the complete tree (108 pages, 32 top-level groups)

src/app/	108 page.tsx files
	32 top-level groups
├─ about/page.tsx	1 page
├─ admin/page.tsx	1 page
├─ app/page.tsx	1 page
├─ architecture/page.tsx	1 page
├─ blog/	2 pages
│ └─ page.tsx	
│ └─ [slug]/page.tsx	
├─ careers/	2 pages
│ └─ page.tsx	
│ └─ [id]/page.tsx	
├─ cart/page.tsx	1 page
├─ case-studies/page.tsx	1 page
├─ checkout/page.tsx	1 page
├─ contact/page.tsx	1 page
├─ dev/	1 page
│ └─ controls/page.tsx	
├─ developer/	10 pages
│ └─ page.tsx	
│ └─ authentication/page.tsx	
│ └─ browser/page.tsx	
│ └─ commands/page.tsx	
│ └─ endpoints/page.tsx	
│ └─ errors/page.tsx	
│ └─ limits/page.tsx	
│ └─ quickstart/page.tsx	
│ └─ scopes/page.tsx	
│ └─ webhooks/page.tsx	
├─ docs/page.tsx	1 page
├─ domains/	2 pages
│ └─ page.tsx	
│ └─ [slug]/page.tsx	
├─ faq/page.tsx	1 page
├─ open-source/page.tsx	1 page
├─ page.tsx	1 page
├─ privacy/page.tsx	1 page
├─ projects/	2 pages
│ └─ page.tsx	
│ └─ [id]/page.tsx	
├─ returns-policy/page.tsx	1 page
├─ roadmap/page.tsx	1 page
├─ services/page.tsx	1 page
├─ shipping/page.tsx	1 page
├─ shop/	6 pages

(continued — part 2 of 3)

└─ page.tsx	
└─ [slug]/page.tsx	
└─ account/page.tsx	
└─ c/[category]/page.tsx	
└─ devices/page.tsx	
└─ invoice/[orderNo]/page.tsx	
└─ smart-home/page.tsx	1 page
└─ smarthome/	59 pages
└─ page.tsx	
└─ admin/page.tsx	
└─ admin/access/page.tsx	
└─ admin/alerts/page.tsx	
└─ admin/dashboards/page.tsx	
└─ admin/fleet/page.tsx	
└─ admin/intelligence/page.tsx	
└─ admin/latency/page.tsx	
└─ admin/ota/page.tsx	
└─ admin/platform/page.tsx	
└─ admin/provisioning/page.tsx	
└─ admin/registry/page.tsx	
└─ admin/rules/page.tsx	
└─ admin/security/page.tsx	
└─ admin/telemetry/page.tsx	
└─ anpr/page.tsx	
└─ assistants/page.tsx	
└─ attendance/page.tsx	
└─ automation/page.tsx	
└─ automations/page.tsx	
└─ away-mode/page.tsx	
└─ backup/page.tsx	
└─ benchmark/page.tsx	
└─ camera/page.tsx	
└─ cameras/page.tsx	
└─ command-center/page.tsx	
└─ developer/page.tsx	
└─ device/[id]/page.tsx	
└─ devices/page.tsx	
└─ diagnostics/page.tsx	
└─ drone/page.tsx	
└─ energy/page.tsx	
└─ energy-budget/page.tsx	
└─ firmware/page.tsx	
└─ floorplan/page.tsx	
└─ groups/page.tsx	
└─ insights/page.tsx	
└─ kiosk/page.tsx	
└─ lifecycle/page.tsx	
└─ maintenance/page.tsx	
└─ notification-rules/page.tsx	
└─ notifications/page.tsx	
└─ presence/page.tsx	
└─ profile/page.tsx	
└─ properties/page.tsx	
└─ quick-actions/page.tsx	
└─ recipes/page.tsx	
└─ reports/page.tsx	
└─ rooms/page.tsx	
└─ scene-scheduler/page.tsx	
└─ scenes/page.tsx	
└─ security/page.tsx	
└─ settings/page.tsx	
└─ solar/page.tsx	
└─ spaces/page.tsx	
└─ timeline/page.tsx	
└─ traffic/page.tsx	
└─ weather/page.tsx	

(continued — part 3 of 3)

└─ widgets/page.tsx	
└─ stack/page.tsx	1 page
└─ team/page.tsx	1 page
└─ terms/page.tsx	1 page
└─ track/page.tsx	1 page
└─ warranty/page.tsx	1 page
└─ weather/page.tsx	1 page

Note the near-collision at the top level: `smart-home/page.tsx` (one page, hyphenated) is the public **marketing** page describing the smart-home product line; `smarthome/` (59 pages, no hyphen) is the **console application** itself. Same product, two different route prefixes, one character apart — worth remembering when grepping this tree.

5.4 Pages — grouped tables

5.4.1 Marketing, storefront and singleton pages (24 pages)

Path	Purpose
<code>/ (page.tsx)</code>	home page
<code>/about</code>	company/about page
<code>/services</code>	services overview
<code>/case-studies</code>	case studies listing
<code>/team</code>	team page
<code>/stack</code>	technology-stack showcase
<code>/architecture</code>	public architecture overview (marketing framing)
<code>/smart-home</code>	smart-home product marketing page (see note above)
<code>/roadmap</code>	public product roadmap
<code>/open-source</code>	open-source projects/licensing page
<code>/docs</code>	documentation landing page
<code>/faq</code>	frequently asked questions
<code>/contact</code>	contact page (renders the <code>contact</code> API's form)
<code>/app</code>	mobile-app download/marketing page
<code>/privacy, /terms</code>	legal pages
<code>/shipping, /returns-policy, /warranty</code>	policy pages
<code>/track</code>	public order tracking UI (calls <code>orders/track</code>)
<code>/weather</code>	weather widget demo page
<code>/cart, /checkout</code>	storefront cart and checkout flow
<code>/admin</code>	admin console entry point (redirects into the guarded area)
<code>/dev/controls</code>	internal developer-only control panel

5.4.2 Content collections (8 pages)

Path	Purpose
<code>/blog, /blog/[slug]</code>	blog index and article
<code>/careers, /careers/[id]</code>	careers listing and posting detail
<code>/projects, /projects/[id]</code>	portfolio listing and project detail

Path	Purpose
<code>/domains, /domains/[slug]</code>	domain-for-sale/portfolio listing and detail

5.4.3 Storefront (6 pages)

Path	Purpose
<code>/shop</code>	catalogue home
<code>/shop/[slug]</code>	product detail
<code>/shop/c/[category]</code>	category listing
<code>/shop/devices</code>	device-specific storefront section
<code>/shop/account</code>	shop customer account area
<code>/shop/invoice/[orderNo]</code>	order invoice view

5.4.4 Developer portal (10 pages)

Path	Purpose
<code>/developer</code>	portal home
<code>/developer/quickstart</code>	getting-started guide
<code>/developer/authentication</code>	auth-scheme documentation
<code>/developer/scopes</code>	API scope reference
<code>/developer/endpoints</code>	endpoint reference
<code>/developer/commands</code>	device-command reference
<code>/developer/webhooks</code>	webhook documentation
<code>/developer/errors</code>	error-code reference
<code>/developer/limits</code>	rate-limit documentation
<code>/developer/browser</code>	in-browser API explorer

5.4.5 Smart-home console — operator-facing (45 pages)

The console's largest single area, clustered by function (all under `/smarthome/*`, all client-rendered against `platform/'s` control-plane API, not against the 10 `smarthome/*` routes in §5.2.6 except where those proxy):

Cluster	Pages
Core dashboard	<code>page (home), command-center, quick-actions, widgets, timeline</code>
Devices and rooms	<code>devices, device/[id], rooms, spaces, properties, floorplan, groups</code>
Security and access	<code>security, camera, cameras, anpr, away-mode, presence</code>
Automation	<code>automation, automations, scenes, scene-scheduler, recipes, notification-rules</code>
Energy	<code>energy, energy-budget, solar</code>
Maintenance and lifecycle	<code>maintenance, lifecycle, diagnostics, backup, firmware, benchmark</code>
Notifications and profile	<code>notifications, profile, settings</code>
Extras	<code>assistants, attendance, drone, kiosk, weather, traffic, reports, insights, developer</code>

That is $5 + 7 + 6 + 6 + 3 + 6 + 3 + 9 = 45$ operator-facing pages.

5.4.6 Smart-home console — admin sub-area (14 pages)

A second tier of role-gated pages nested under `/smarthome/admin/*`, mapping loosely to the same operator concept `verifyOperator` checks server-side:

Path	Purpose
<code>admin (index)</code>	admin area home

Path	Purpose
admin/access	operator/role access management
admin/registry	device registry management
admin/provisioning	new-device provisioning workflow
admin/fleet	fleet-wide device overview (feeds ai/fleet)
admin/ota	OTA firmware release management (§8.2)
admin/security	console-wide security settings
admin/rules	automation/alert rule administration
admin/alerts	alert configuration
admin/dashboards	custom dashboard builder
admin/intelligence	AI/analytics configuration
admin/latency	latency monitoring
admin/telemetry	telemetry stream administration
admin/platform	control-plane connection/platform settings

45 + 14 = 59, matching the tree's smarthome/ count exactly.

5.5 Route map — Mermaid mirror

Mermaid diagram — renders interactively in the Markdown edition

```
flowchart TB
    subgraph API["150 API routes"]
        acc["account/* - 14"]
        adm["admin/* - 83"]
        ai_["ai/* - 3"]
        dev["devices/* - 5"]
        ord["orders + payments\n+ coupons + giftcards - 9"]
        shp["shop/* - 5"]
        smh["smarthome/* - 10"]
        pub["21 single-purpose\nutility routes"]
    end
    subgraph PAGES["108 pages"]
        mkt["marketing + singles - 24"]
        cont["content collections - 8"]
        shopP["shop/* - 6"]
        devp["developer/* - 10"]
        smhP["smarthome/* - 59\n(45 operator + 14 admin)"]
    end

    acc --> shopP
    ord --> shopP
    shp --> shopP
    dev --> smhP
    smh --> smhP
    adm -.->|"back office, no public page tree"| adm
    ai_ --> smhP
    pub --> mkt

    style adm fill:#FEF3C7,stroke:#B45309
    style smh fill:#E0E7FF,stroke:#4338CA
    style smhP fill:#E0E7FF,stroke:#4338CA
    style dev fill:#E0E7FF,stroke:#4338CA
```

6. External contracts & integrations

Career.circuvent had one upstream (its ATS). This application has none of its network surface concentrated like that: eight independent external systems, each reached by a different src/lib module, plus one internal-but-separate system (platform/) that is also called directly by a completely different repository (mobile/) without this website ever being on that path. Every entry below was confirmed either in source or in .env.example / Docs/11-secrets.md — nothing here is inferred from the variable name

alone.

PAYMENTS	
Razorpay	raw fetch, no SDK installed. Basic auth, base64 (keyId:keySecret). api.razorpay.com/v1/orders and /v1/payments/{id}. Inbound webhook verifies its HMAC signature, then does nothing else -- §8.1.
AI	
lib/ai/provider	raw fetch, OpenAI chat-completions wire format, no vendor SDK. Default https://api.openai.com/v1, model gpt-4o-mini. AI_BASE_URL/AI_API_KEY/AI_MODEL (or OPENAI * fallbacks) repoint it at Azure OpenAI, Groq, Together, or a self-hosted Ollama/vLLM with zero code changes. aiConfigured()==false -> every caller (assistant, fleet analysis) degrades to a deterministic, model-free analysis instead of an error -- resilience by design, not by accident.
EMAIL -- three independent providers, chosen per call site	
Resend	RESEND_API_KEY -- transactional mail
SMTP / nodemailer	SMTP_HOST/PORT/USER/PASS -- transactional, 2nd path
Buttondown	BUTTONDOWN_API_KEY -- /api/newsletter signups only, the ONLY caller of this provider in the codebase
CIRCUVENT'S OWN IDENTITY DOMAIN (not this repository, §3.1)	
auth.circuvent.com	OIDC/PKCE issuer for admin SSO login (§8.5) AND the live mail-group directory read by GET /api/admin/groups for report recipients
PLATFORM -- separate repo, same working tree (§11.2, C4 in §1-2)	
CONTROL_PLANE_URL	HTTPS to 140.245.238.154. Two different auth shapes depending on caller: HMAC device sync/ command from 2 of 5 device/* routes, and a control-plane-delegated bearer token re-checked at 4 independent call sites (§10.7). mobile/ calls the SAME control plane directly and does NOT go through this website at all (§1.1).
MISCELLANEOUS THIRD PARTIES	
GitHub REST API	GITHUB_TOKEN -- live commit/star stats for the public /projects page
weather provider	lib/weather.ts -- backs /api/weather; no key variable catalogued in .env.example
CV-365 Firebase	NEXT_PUBLIC_CV365_FIREBASE_* -- relays the public contact form to work.circuvent.com/admin/messages. A THIRD legitimate circuvent.com subdomain, and still nothing to do with the unrelated circuvent-platform/ repo (§3.1).
web-push (VAPID)	browser push subscriptions, delivered directly, no third-party push relay in front of it
DECLARED IN .env.example -- NOT WIRED UP ANYWHERE	
GOOGLE_CLIENT_ID / GOOGLE_CLIENT_SECRET / GOOGLE_CALLBACK_URL appear in .env.example and Docs/11-secrets.md. `grep`ing all of src/ for either name returns nothing: there is no "Sign in with Google" route, page, or library call in this codebase. Vestigial, not a 6th scheme.	

Mermaid diagram — renders interactively in the Markdown edition


```

flowchart LR
    subgraph site["website - src/lib"]
        pay["razorpay.ts"]
        ai["ai/provider.ts"]
        mail["email senders"]
        cp["control-plane.ts<br/>+ 7 server routes"]
        gh["github-sync.ts"]
        wx["weather.ts"]
        fb["cv365-firebase.ts"]
        push["web-push.ts"]
    end

    rzp["Razorpay<br/>api.razorpay.com"]
    oai["AI endpoint<br/>OpenAI-compatible<br/>(swappable)"]
    rsd["Resend"]
    smtp["SMTP relay"]
    bd["Buttondown"]
    authcv["auth.circuvent.com<br/>OIDC + groups directory"]
    ghapi["GitHub REST API"]
    wxapi["weather provider"]
    work["work.circuvent.com<br/>(CV-365 Firebase target)"]

    subgraph plat["platform/ - separate repo, §11.2"]
        vm["140.245.238.154<br/>Express control plane"]
    end
    end
    mobileApp["mobile/ - separate repo"] -.->|"direct HTTPS,<br/>bypasses website"| vm

    pay -->|"Basic auth"| rzp
    ai -->|"Bearer, swappable base URL"| oai
    mail --> rsd
    mail --> smtp
    mail --> bd
    cp -->|"HMAC + delegated bearer,<br/>4 call sites, §10.7"| vm
    gh -->|"GITHUB_TOKEN"| ghapi
    wx --> wxapi
    fb --> work
    site -.->|"OIDC/PKCE + groups"| authcv
    push -.->|"VAPID, browser-native"| push

    style site fill:#DBEAFE,stroke:#1D4ED8
    style plat fill:#E0E7FF,stroke:#4338CA
    style mobileApp fill:#E0E7FF,stroke:#4338CA
    style authcv fill:#FEF3C7,stroke:#B45309

```

Two asymmetries are worth naming explicitly. First, **platform/** is the only external system this website both calls *and* shares with a caller it has no control over (**mobile/**) — every other integration above is this website's private dependency. Second, **GOOGLE_CLIENT_ID** is the only variable in this whole inventory that is documented in two places (**.env.example**, **Docs/11-secrets.md**) and used in zero: it is carried forward here, not silently dropped, because an atlas that only reports what is wired up would miss a real piece of the deployment's configuration surface.

7. Data model

Exactly 10 tables, all created by **initDb()** (§7.2) from one TypeScript array of DDL strings in **src/lib/db.ts** — no Prisma, no Drizzle, no **.sql** migration files anywhere in the repository. **Zero FOREIGN KEY, REFERENCES, or UNIQUE** constraints appear in any of the 10 **CREATE TABLE** statements: every **PK** below is real (a real Postgres primary key), every relationship implied by an **FK**-shaped comment is enforced only in application code. Row-level security: **none**, on any table — this database has no concept of a tenant, so there is nothing for a policy to scope.

7.1 Entity-relationship diagram

IDENTITY -- 3 dedicated, typed tables		
accounts	PK email	shop customers
admin_users	PK email	staff / admin accounts + role
pending_registrations	PK email	signup OTP, in-flight only
GENERIC KV -- 1 table, 26 logical documents (§7.3)		
store_kv	PK (collection, key) data JSONB NOT NULL	23 shop collections @ key='_all' + 3 durable file-stores (§7.4)
ANALYTICS / AUDIT -- 3 tables, append-only, no updates		
email_history	BIGSERIAL id	every email ever sent (evidence log)
request_metrics	BIGSERIAL id	per-request latency/status samples
page_views	BIGSERIAL id	anonymised visits, salt rotates daily
SMART-HOME CAMERA RELAY -- 3 tables, deliberately NOT history		
camera_frames	PK device_id	newest frame only, overwritten
camera_audio	BIGSERIAL id	rolling live-listen buffer, pruned
camera_audio_session	PK device_id	one pending listen/speak token

RLS on all 10 tables above: NONE. No policy, no tenant column, no schema separation -- a single Neon connection string sees every row in the app.

Mermaid diagram — renders interactively in the Markdown edition

```
erDiagram
    accounts {
        text email PK
        text name
        text password_hash
        text password_salt
        text phone
        boolean blocked
        jsonb data "full entity, source of truth"
    }
    admin_users {
        text email PK
        text role
        boolean active
        text password_hash
        jsonb data
    }
    pending_registrations {
        text email PK
        text otp
        bigint expires
        int attempts
        text ref
    }
    store_kv {
        text collection PK "23 shop names + 'file_store'"
        text key PK "'_all', or a filename (§7.4)"
        jsonb data
        timestampz updated_at
    }
    email_history {
        bigserial id PK
        text to
        text type
        text status
        text provider
        text message_id
    }
    request_metrics {
        bigserial id PK
        text endpoint
        text method
        int status
        real ms
    }
    page_views {
        bigserial id PK
        text path
        text visitor_hash "salted, rotates daily"
        text device
        text country
    }
    camera_frames {
        text device_id PK
        text jpeg_b64
        text upload_token
        timestampz token_expires
    }
    camera_audio {
        bigserial id PK
        text device_id
        text wav_b64
        timestampz captured_at
    }
    camera_audio_session {
        text device_id PK
        text listen_token
        text speak_token
    }
```

(continued — part 2 of 2)

```

    text speak_wav_b64
  }

```

No relationship lines connect the ten boxes above because none exist in the schema — that absence is not an omission from this diagram, it is the finding. `accounts.email` and `admin_users.email` are referenced by string from application code (sessions, the shop-console SSO bridge, audit rows) but Postgres itself enforces none of it.

7.2 `initDb()` — runtime schema, not migrations

```

cold start (new lambda instance, or first import after a deploy)
|
v
store.ts evaluates -> top-level `await bootstrap()` blocks every route
handler in this module graph until it resolves (plain ESM semantics,
no framework hook involved)
|
v
dbLayer.initDb() -- memoized into a module-level `_initPromise`, so
|                every later call from THIS instance is a free,
|                already-resolved promise for the rest of its life
|
+--> first call only: for (stmt of SCHEMA_STATEMENTS) await q(stmt)
|    10x CREATE TABLE IF NOT EXISTS ...    run sequentially, in
|    12x CREATE INDEX IF NOT EXISTS ...     the array's literal order
|
v
dbLayer.dbHydrate() -- SELECT data FROM every typed table, plus every
|                  store_kv row, assembled into one `mem` object
|
v
first run only: products/coupons seed data is generated, written into
`mem`, and flushed immediately -- the database becomes source of truth
from boot one, not from the first admin edit
|
v
route handlers now read/write `mem` only. Postgres is touched again
only by scheduleFlush() (changed collections, after the response is
built) and by revalidate() (an explicit targeted re-read, called only
from login/signup/admin routes that need cross-instance consistency)

```

Mermaid diagram — renders interactively in the Markdown edition

```

sequenceDiagram
    participant L as Lambda cold start
    participant S as store.ts (bootstrap)
    participant D as db.ts (initDb / dbHydrate)
    participant PG as Neon Postgres (HTTP driver)

    L->>S: module import graph evaluates
    S->>S: top-level await bootstrap()
    S->>D: initDb()
    alt first call in this instance
        D->>PG: CREATE TABLE IF NOT EXISTS x10
        D->>PG: CREATE INDEX IF NOT EXISTS x12
    else already memoized
        Note over D: cached _initPromise, no network call
    end
    S->>D: dbHydrate()
    D->>PG: SELECT data FROM <every table / store_kv row>;
    PG-->>D: rows
    D-->>S: Partial<DB>;
    S->>S: mem = merge(emptyDB(), data); seed products/coupons if empty
    Note over S,PG: every request now reads/writes mem only;<br/>PG is touched again solely by flush() and revalidate()

```

There is no schema-version table and no rollback path: `IF NOT EXISTS` on every statement is the entire migration strategy. A column rename would need a second, additive `CREATE TABLE IF NOT EXISTS` block, hand-written, run through this same memoized function — the same shape as every table already here.

7.3 `store_kv` — one JSONB document per collection

store_kv	PRIMARY KEY (collection, key)	data JSONB NOT NULL
23 x	collection = <shop collection name>, key = '_all'	the ENTIRE collection -- an array or an email-keyed map -- serialised as ONE JSONB value. Not one row per order, per wallet, per ticket: one row per COLLECTION, full stop. e.g. collection='orders', key='_all', data={ [orderId]: ... }
3 x	collection = 'file_store', key = <filename>	one durable createFileStore module's whole document (§7.4) e.g. collection='file_store', key='icm-incidents.json'

26 logical documents, 1 physical table. Writing a single order rewrites the WHOLE `orders` blob (read-modify-write of the entire collection). Neon's HTTP driver has no transactions (§7.2), and store.ts's flush chain only serialises writes WITHIN one instance -- two lambda instances flushing `orders` at the same moment can still race.

The 23 shop collections, named exactly as declared in `db.ts`'s `KV_COLLECTIONS` array: `orders`, `products`, `wallets`, `devices`, `reviews`, `addresses`, `notifyRequests`, `logins`, `coupons`, `tickets`, `returns`, `audit`, `loyalty`, `referrals`, `referralCodes`, `giftCards`, `questions`, `notifications`, `passwordResets`, `admin2fa`, `alertSettings`, `contactMessages`, `consumedPayments`.

Mermaid diagram — renders interactively in the Markdown edition

```

flowchart LR
    kv["store_kv<br/>1 physical table"]
    shop["23 shop collections<br/>key = '_all'<br/>orders, products, wallets,<br/>devices, reviews, coupons...<br/>(full list above)"]
    fsx["3 durable file-stores<br/>collection = 'file_store'<br/>key = filename (§7.4)"]

    shop -->|"read-modify-write<br/>the WHOLE blob"| kv
    fsx -->|"dbWriteFileStore /<br/>dbReadFileStore"| kv

    style kv fill:#F5F3FF,stroke:#6D28D9
    style shop fill:#DBEAFE,stroke:#1D4ED8
    style fsx fill:#D1FAE5,stroke:#047857

```

7.4 Storage durability — `createFileStore` and the memory-only majority

`src/lib/data-file.ts`'s `createFileStore()` is a second, entirely separate persistence mechanism from §7.1–7.3 — used by standalone feature modules that never touch `store.ts`'s `DB` interface. It tries a disk write under `DATA_DIR` (`.data/` by default) first; on Vercel's read-only production filesystem that write fails exactly once, `canWrite` latches `false` for the rest of the instance's life, and a console warning is logged. Only if the caller passed `{ durable: true }` and `DATABASE_URL` is set does the module *also* mirror into `store_kv` (§7.3) — otherwise it is memory-only from that point on, for that instance.

Counted directly from source (`createFileStore` call sites, test file excluded): **37 call sites across 35 modules. Exactly 3 modules pass `durable: true` — `admin-warranty.ts`, `api-failures.ts`, and `icm-store.ts`** (which alone accounts for 2 of the 4 durable call sites). The other **32 modules are memory-only in production**, slightly more than doc 03's rough estimate of "27 of about 30" — this atlas corrects that figure to the exact one, in the same spirit as the OTA correction in §8.3.

DURABLE (3 modules -- mirrored into store_kv, survive a cold start)	
admin-warranty.ts	warranty claims and policies
api-failures.ts	recorded upstream API failures for diagnostics
icm-store.ts	incident/change records (2 durable call sites)
MEMORY-ONLY IN PRODUCTION (32 modules -- gone on cold start/redeploy)	
Commerce config	admin-pricing, admin-currency, admin-tax, admin-bundles, admin-shipping, admin-subscriptions, admin-fraud, admin-marketing, admin-vendors, admin-affiliates
Content / CRM	admin-cms (every CMS post), admin-crm (every note and tag), admin-seo-manager, admin-macros, admin-surveys
Platform ops	admin-bulk, admin-jobs, admin-flags (every feature flag and experiment), admin-integrations (API keys, webhooks, deliveries), admin-staff-activity, admin-report-builder, reports-schedule, admin-privacy, deployments
Smart-home	admin-config, alerts-store, device-history, smarthome-dev-portal (every developer-portal token), user-prefs
Security-adjacent	telemetry-store (2 call sites), web-push (every push subscription), passkeys.ts (every registered WebAuthn credential in the whole application)

The last row is the sharpest edge: a credential class (passkeys) and a delivery mechanism (push subscriptions) that most engineers would assume are durable by category are, in this deployment, exactly as ephemeral as a cached CMS draft.

[Mermaid diagram — renders interactively in the Markdown edition](#)

```

flowchart TB
    call["createFileStore(filename, seed, opts)<br/>37 call sites, 35 modules"]
    diskTry["disk write<br/>under DATA_DIR"]
    diskOK["written to .data/*.json<br/>-- true on a persistent disk,<br/>never true on Vercel prod"]
    diskFail["canWrite = false<br/>(latched for this instance)"]
    durFlag["opts.durable === true<br/>AND DATABASE_URL set?"]
    pgMirror["mirrored into store_kv<br/>collection='file_store' ($7.3)<br/>3 modules only"]
    memOnly["in-memory only<br/>gone on cold start / redeploy<br/>32 modules"]

    call --> diskTry
    diskTry -->|"writable FS"| diskOK
    diskTry -->|"read-only FS<br/>(Vercel production)"| diskFail
    diskFail --> durFlag
    durFlag -->|"yes, 3 modules"| pgMirror
    durFlag -->|"no, 32 modules"| memOnly

    style pgMirror fill:#D1FAE5,stroke:#047857
    style memOnly fill:#FEE2E2,stroke:#B91C1C
    style diskFail fill:#FEF3C7,stroke:#B45309

```

8. Workflows, end to end

Five flows, chosen because each one crosses a trust boundary this atlas has already named: shop money, device firmware, and the two SSO bridges (§6, §10). Mermaid is the primary notation here, as it is for Career.circuvent's own §8 — a sequence is a poor fit for a monospace box, and the same information is not repeated in ASCII.

8.1 Checkout and payment — Razorpay, and the reconciliation gap

[Mermaid diagram — renders interactively in the Markdown edition](#)

```
sequenceDiagram
    autonumber
    actor U as Shopper
    participant B as Browser (checkout.js)
    participant W as website /api/payments/*
    participant R as Razorpay
    participant S as store.ts (mem + Postgres)

    U->>B: place order
    B->>W: POST create-order { items, coupon, walletApply }
    W->>W: priceItems() recomputes the total server-side
    W->>R: POST /v1/orders (Basic auth keyId:keySecret)
    R-->>W: { id: orderId }
    W-->>B: { orderId, keyId, amount }
    B->>R: open Checkout.js modal
    U->>R: pays
    alt browser JS callback fires
        R-->>B: razorpay_order_id, payment_id, signature
        B->>W: POST /api/payments/verify
        W->>W: checkoutSignatureValid(): HMAC-SHA256(order_id|payment_id)
        W->>R: GET /v1/payments/{id} -- re-fetch, never trust the request body
        R-->>W: { status: "captured", amount }
        W->>W: amount == recomputed total? consumePayment() claims the id once
        W->>S: recordOrder, adjustStock, earnPoints, flushNow()
        W-->>B: 200 { order }
    else tab closes / network dies before the callback
        Note over B,R: Razorpay HAS captured the money;<br/>the browser never comes back
        R->>W: POST /api/payments/webhook (server-to-server, independent of the browser)
        W->>W: verify x-razorpay-signature, HMAC-SHA256 over the raw body
        W->>W: console.log(event) -- the reconciliation hook is a comment, not code
        W-->>R: 200 { success: true }
        Note over W,S: NO order is ever recorded for this captured payment
    end
end
```

The happy path is genuinely well defended: the handback signature only proves the browser saw a real Razorpay response, so **verify** re-fetches the payment from Razorpay's own API, requires `status === "captured"`, compares the gateway's `amount` against a total recomputed from the catalogue (never the request body), and claims the payment id exactly once through `consumePayment` so the same triple cannot be replayed into a second order. What is missing is the second channel that exists specifically to cover the first one failing: Razorpay's webhook is configured, its signature is verified correctly, and then nothing downstream of the `console.log` ever touches `orders`, `wallets` or the emails in `lib/order-core.ts`. A captured payment whose browser never returns is captured money with no order — permanently, not until the next poll.

8.2 OTA firmware update — the pull path, and the missing signature

Mermaid diagram — renders interactively in the Markdown edition

```
sequenceDiagram
    autonumber
    participant D as ESP32 device (CircuventDevice::checkOTA)
    participant W as website GET /api/devices/firmware
    participant M as fw/manifest.json (public bucket)
    participant E as process.env OTA_<TYPE>

    loop every _otaInterval (device-side timer, ~6h)
        D->>D: WiFiClientSecure + _pinRoot() -- TLS pinned, no other CA accepted
        D->>W: GET ?type&id&ver=CV_FW_VERSION<br/>headers x-device-id, x-device-key
        W->>W: require both headers present, else 401
        W->>E: read OTA_<TYPE> (operator override)
        alt env var set
            E-->>W: "version|url"
        else env not set
            W->>M: GET manifest.json (5 min in-instance cache)
            M-->>W: { version, url, sha256? }
        end
        W->>W: compare versions as OPAQUE strings, not "is newer" --<br/>this is what makes a rollback possible
        Note over W,D: sha256, even on the rare manifest entry that<br/>carries one, is never read or returned here
        W-->>D: { version, url } or { version: current, url: "" }
        alt url present and version != current
            D->>D: _applyOta(): a second WiFiClientSecure + _pinRoot()
            D->>W: GET binUrl (same pinned TLS, no further check)
            W-->>D: firmware.bin bytes
            Note over D: httpUpdate.update() flashes the image.<br/>TLS proves the HOST; nothing proves the BINARY --<br/>no hash check and no signature anywhere in this path.
            D->>D: reboot into the new image
        end
    end
end
```

TLS pinning is real and does its job: an on-path attacker cannot answer the manifest request or the binary download with a certificate this device will accept. What pinning cannot do is prove the *content* served by the legitimate, correctly authenticated host is the build the fleet owner intended to publish — a compromised publish credential, a mis-uploaded object in the bucket, or a bug in `scripts/publish-firmware.cjs` all flash straight to the chip. Several of the 17 SKUs behind this exact code path switch mains

relays or operate a door lock ([facedoor.ino](#), [switchboard.ino](#)); an unsigned OTA channel is a materially different risk on those than on a soil-moisture sensor.

8.3 Correcting the fact base — the manifest endpoint exists

CORRECTION to 03_INTEGRATIONS_AND_ECOSYSTEM.md

Doc 03 describes the OTA pull path as polling a manifest endpoint that "DOES NOT EXIST". That is no longer accurate: this repository contains a working handler at ``src/app/api/devices/firmware/route.ts`` (§5.3, §8.2 above), which every sketch's ``CircuventDevice::checkOTA()`` calls successfully.

What doc 03 identified correctly, one layer down from where it placed it, still stands: the endpoint answers with a version and a URL only, never a signature and never an enforced hash. The real defect is image authenticity, not endpoint existence -- see §8.2.

This is the one place this atlas's own reading of the source disagrees with the fact base it was handed, per the promise in the opening paragraph. Nothing else in docs 01–05 needed correcting against `src/`; this single case is called out because the difference between "no endpoint" and "an endpoint with no signature check" changes what an engineer would fix first.

8.4 Shop → console SSO bridge

Mermaid diagram — renders interactively in the Markdown edition

```
sequenceDiagram
    autonumber
    actor U as Shopper (already signed in to the shop)
    participant B as Browser
    participant W as website POST /api/account/sso/console
    participant P as platform control plane (140.245.238.154)

    U->>B: click "open smart-home console"
    B->>W: POST (shop session cookie attached)
    W->>W: verifyToken(tokenFromRequest) -- shop HMAC session, §10.1
    W->>W: getAccount(email) -- must not be blocked or deleted
    W->>W: mintConsoleSession(): HMAC-SHA256(timestamp,email, FEDERATION_SECRET)
    W->>P: POST /auth/federated<br/>x-federation-timestamp, x-federation-signature headers
    P->>P: create a console user for this email on first visit
    P-->>W: { token, user }
    W-->>B: { success, token, user }
    Note over B,P: from here the browser calls the control plane's OWN api<br/>directly with this bearer token -- the same shape mobile/<br/>uses, bypassing this website entirely
```

The storefront and the smart-home console keep separate user tables by design, so this route is a vouching step, not a login: the website proves to itself that the caller already holds a valid shop session, then asks the control plane to open or create a console account for that same address. `FEDERATION_SECRET` never leaves this process — it signs an outbound request, it is never handed to the browser — and a second guard, `federationAllowedHere()`, stops a preview deployment that happens to hold the production secret from minting live console sessions.

8.5 Admin sign-in via Circuvent SSO (OIDC + PKCE)

Mermaid diagram — renders interactively in the Markdown edition


```
sequenceDiagram
    autonumber
    actor A as Admin
    participant B as Browser
    participant S as GET .../sso/start
    participant C as GET .../sso/callback
    participant X as POST .../sso/exchange
    participant I as auth.circuvent.com (OIDC issuer)

    A->>B: "Sign in with Circuvent"
    B->>S: GET
    S->>S: beginSso(): PKCE code_challenge, random state
    S->>B: redirect to I, Set-Cookie state+verifier (sealed, path-scoped)
    B->>I: GET /authorize?...
    A->>I: authenticates at the identity service
    I->>B: redirect to C ?code&state
    B->>C: GET ?code&state (cookie attached)
    C->>C: state must equal the cookie's state, else abort
    C->>I: POST /api/oauth/token (code_verifier)
    I->>C: { access_token, id_token }
    C->>C: role claim read from id_token -- a group grant surfaces here
    C->>I: GET /api/oauth/userinfo (bearer access_token)
    I->>C: { email, email_verified, name, picture }
    C->>C: email_verified must be true, else abort
    C->>C: provision/update admin_users -- staff-list role OR an SSO-granted<br/>role, NEVER the bare default "staff" role
    C->>C: signHandoff(email, nonce) -- 90 seconds, single use
    C->>B: redirect /admin?sso=handoff, Set-Cookie nonce
    B->>X: POST { code: handoff } (nonce cookie attached)
    X->>X: verifyHandoff(code, nonce) -- re-checks staff.active again
    X->>X: signAdminToken(email) -- the SAME HMAC session password login issues, §10.2
    X->>B: { token, role, ... }
```

Three hops, not one, and each exists for a reason found in the source comments: the token never rides in a URL (so it never sits in browser history or a proxy log), the code is single-use and expires in ninety seconds, and staff status is checked twice — once at the callback and again at the exchange — because a role can be revoked in the gap between them. The flow's real subtlety is authorisation, not authentication: the identity service will happily sign in anyone with a Circuvent account, so `roleClaimFromIdToken` only grants console access from an explicit per-console role or group grant, never from the ordinary `staff` default every employee already has.

The common thread across 8.1, 8.4 and 8.5: each cross-boundary exchange is server-verified and single-use-claimed (`consumePayment`, the SSO nonce, the OTP- like handoff code). The one exception is the payment webhook in 8.1, which has the verification but not the claim — the only flow in this section where a second, independent confirmation channel exists on paper and is not wired to anything.

9. State machines

9.1 Order lifecycle, and the return it can spawn

Mermaid diagram — renders interactively in the Markdown edition

```
stateDiagram-v2
    [*] --> placed: POST /api/payments/verify (or COD) records the order
    placed --> shipped: admin/orders marks shipped
    shipped --> delivered: admin/orders marks delivered
    placed --> cancelled: customer or admin cancels
    shipped --> cancelled: customer or admin cancels
    cancelled --> [*]
    delivered --> requested: customer opens a return within the window
    requested --> approved: admin approves
    requested --> rejected: admin rejects
    approved --> refunded
    rejected --> [*]
    refunded --> [*]
    delivered --> [*]: no return ever opened
```

`placed`, `shipped`, `delivered` and `cancelled` are the order's own `status` field; `requested`, `approved`, `rejected` and `refunded` belong to a separate row in the `returns` collection (`store.ts:262`) that references the order rather than replacing its status. Warranty eligibility (`warranty.ts`) reads the same `delivered` transition to start its own clock.

9.2 Device online/offline

Mermaid diagram — renders interactively in the Markdown edition

```
stateDiagram-v2
    [*] --> Provisioning: no Wi-Fi credentials saved -- soft AP + portal up
    Provisioning --> Connecting: credentials received via the captive portal
    Connecting --> Reconnecting: Wi-Fi association fails
    Reconnecting --> Connecting: exponential backoff, 3s doubling to a 96s cap
    Connecting --> Online: Wi-Fi + MQTT both up --<br/>publishes retained cv/id/status {"online":true}
    Online --> Online: heartbeat interval OR an immediate publish on local change
    Online --> Offline: broker detects an unclean disconnect --<br/>LWT delivers {"online":false}
    Offline --> Connecting: device reboots or Wi-Fi returns
```

"Online" and "offline" as the console displays them are entirely MQTT's own retained-message-plus-Last-Will mechanism (`CircuventDevice.h:12`); there is no separate heartbeat table on the server side that ages a device out.

9.3 Session and token lifecycle

Mermaid diagram — renders interactively in the Markdown edition

```
stateDiagram-v2
    [*] --> Issued: sign-in succeeds -- password, one-time code, passkey, or SSO
    Issued --> Valid: HMAC verifies AND tokenVersion matches account state
    Valid --> Valid: re-checked against account state on every request
    Valid --> Revoked: password reset or an admin block bumps tokenVersion
    Valid --> Expired: sign-in age exceeds the cap for this session kind
    Revoked --> [*]
    Expired --> [*]

    note right of Valid
        Shop session: 24h flat cap (account.ts), down from 30 days
        Admin session: 12h token TTL, PLUS a 24h absolute sign-in cap
        (session-expiry.ts) measured from the original sign-in, not
        the token -- a renewal cannot extend it
        Console session (platform-owned, minted at 8.4/8.5): 30-day
        access token backed by a 60-day refresh token that renews
        itself silently on a 401 -- a materially longer, differently
        shaped lifecycle this repository does not control
    end note
```

None of the three schemes issue a classic refresh token from this codebase: a shop or admin session simply stops working at its cap and the person signs in again. Only the console session, owned by `platform/`, renews itself.

9.4 CI pipeline — the intended shape, and the observed one

Mermaid diagram — renders interactively in the Markdown edition

```
stateDiagram-v2
    [*] --> Queued: push or PR against main, master, feature/shopping
    Queued --> Running: a GitHub-hosted runner claims the job
    Running --> StartupFailure: OBSERVED 27 of 27 real runs --<br/>fails before Checkout, step 1 of 14, ever executes
    Running --> Checkout
    Checkout --> SetupAndInstall: setup-node, then npm ci
    SetupAndInstall --> TypeCheck: tsc --noEmit
    TypeCheck --> Lint: continue-on-error true -- reported, never blocks
    Lint --> UnitTests: npm test --ci --runInBand
    UnitTests --> ControlPlaneChecks: platform/api install, type-check, test -- ~290 tests
    ControlPlaneChecks --> DbAdapterTest: PGLite adapter test
    DbAdapterTest --> Build: next build
    Build --> E2e: chromium install, Playwright, then upload the report (always)
    E2e --> Passed
    Passed --> [*]
    StartupFailure --> [*]: none of the 13 hard gates is ever reached
```

Fourteen named steps; `Lint` is the only one marked `continue-on-error`, so the other thirteen are hard gates in the ordinary sense — a real failure at `Unit tests`, `Control-plane tests` or `End-to-end tests` would stop a deploy. None of that machinery has been exercised: every one of the 27 recorded runs ends in `startup_failure`, the state GitHub Actions reports when the workflow never starts executing steps at all — a runner or configuration problem upstream of the YAML shown above, per

`04_MAINTENANCE_AND_OPERATIONS.md`.

10. Cross-cutting concerns

10.1 The five authentication schemes — and the edge layer that checks none of them

#	Subject	Mechanism	Session cap
1	Shop customer	HMAC-SHA256(email, issuedAt, tokenVersion); password (scrypt) or a WebAuthn passkey	24h flat from sign-in, re-checked live every request
2	Admin staff	HMAC-SHA256(email, issuedAt, tokenVersion); password, passkey, or OIDC/PKCE (8.5)	12h token TTL PLUS a 24h absolute cap from sign-in
3	Device (ESP32)	static x-device-id + x-device-key header pair, checked against the store	none -- one long-lived shared secret printed per unit
4	Console user	Authorization: Bearer, either password direct to the control plane, or bridged from #1 (8.4)	owned by platform/: 30d access + 60d refresh, self-renews
5	Scheduler (cron)	Authorization: Bearer <CRON_SECRET>, checked inline at 5 separate route files	static; the same check hand-written five separate times

Mermaid diagram — renders interactively in the Markdown edition

```

flowchart TB
    subgraph Edge["Edge proxy -- runs on almost every matched request"]
        PX["proxy.ts<br>x-request-id + Content-Security-Policy + host-mount routing<br>NO AUTHENTICATION CHECK OF ANY KIND"]
    end

    PX --> R1["shop routes"]
    PX --> R2["admin routes"]
    PX --> R3["/api/devices/*"]
    PX --> R4["SSO / console-bridge routes"]
    PX --> R5["cron-triggered routes"]

    R1 --> S1["1. Shop HMAC session<br>account.ts"]
    R2 --> S2["2. Admin HMAC session<br>admin-auth.ts"]
    R3 --> S3["3. Device header key pair<br>store.ts deviceSync()"]
    R4 --> S4["4. Console bearer token<br>control-plane.ts / sso.ts"]
    R5 --> S5["5. Cron shared-secret bearer<br>five inline checks"]

    style PX fill:#FEF2F2,stroke:#B91C1C,stroke-width:2px

```

Every route decides its own authentication for itself; nothing upstream of the route handler enforces any of the five. `proxy.ts`'s own header comment says what it is for: "request correlation + Content-Security-Policy" — nowhere does it read a session, a device key, or a bearer token. A new route that forgets to call `verifyToken` is a public route, and the platform gives no warning that this happened. `GET /api/devices/firmware` (8.2) shows the quiet end of this: the handler checks that `x-device-id/x-device-key` are *present*, then never reads either value again — scheme 3 exists in name on that one endpoint only.

10.2 Security headers — complete, and applied twice

Header	Value
Strict-Transport-Security	max-age=63072000; includeSubDomains; preload
Content-Security-Policy	built from CSP_DIRECTIVES, lib/csp.ts
X-Content-Type-Options	nosniff
X-Frame-Options	DENY
Referrer-Policy	strict-origin-when-cross-origin
X-DNS-Prefetch-Control	on
Permissions-Policy	camera=(), microphone=(), geolocation=(self), browsing-topics=()
Cross-Origin-Opener-Policy	same-origin

All eight are declared once, globally, in `next.config.ts`'s `headers()` — every response, every route, no opt-in required. The CSP is declared a *second* time, identically, inside `proxy.ts`, precisely so a path the edge matcher excludes (`_next/static`, the Workflow SDK's own callback path) still gets the policy from `next.config.ts` alone. `connect-src` allow-lists exactly the external services 6.1

names (Razorpay, Firebase, Google Analytics, Vercel); `frame-ancestors 'none'` and `object-src 'none'` are set; nothing here is partial or deferred.

10.3 Error handling — a wrapper nobody imports

lib/api-handler.ts — withApi()
Correlation id, structured request/error logging, a consistent { success:false, error, requestId } envelope, method guarding, optional rate limiting. Fully built, fully documented.
IMPORTED BY ZERO ROUTES out of roughly 150.

Every actual route
Hand-writes its own NextResponse.json({...}, {status}) for the errors it anticipated. What it did not anticipate throws.

app/error.tsx
The one Next.js route-segment boundary in the tree. Catches a throw from any page below it; shows a retry button.

app/global-error.tsx — DOES NOT EXIST
If the root layout itself throws, Next's own unstyled fallback is what a visitor sees. Nothing in this repository catches it.

A uniform, well-designed API wrapper was built and then never adopted: every one of the ~150 route handlers rolls its own response shape and its own decision about whether an unexpected throw is caught at all. Some do; a route with no top-level `try/catch` and no `withApi` simply 500s with Next's default body, carrying no `requestId` for support to search logs by.

10.4 No tenancy, and the guard against crossing it by accident

There is no tenant model anywhere in this application: one deployment, one catalogue, one `admin_users` table, one Postgres database. "Tenancy" here means only environment isolation — production data must never be reachable from a preview deployment — and that boundary has already been crossed twice, by accident, which is why two purpose-built guards now exist:

Guard	Incident it followed	Failure mode
<code>assertNotProductionData()</code> <code>+ PROD_DATA_HOSTS</code> (<code>db.ts</code>)	A preview deployment's <code>DATABASE_URL</code> pointed at prod; dev.circuvent.com served real orders and wallet balances	THROWS -- refuses to boot rather than silently serving the wrong database
<code>federationAllowedHere()</code> <code>+ PROD_IDENTITY_HOSTS</code> (<code>sso.ts</code>)	A preview build's SSO config vouched for real customers against the live fleet; dev created accounts from real password hashes	RETURNS FALSE -- looks exactly like a bad password, so a refusal never confirms an address exists

Both guards share the same shape: a comma-separated host list, checked only on *non*-production deployments, safe to leave unset, and safe to set everywhere because a hostname is not a credential. They differ in what a match does — `db.ts` throws (an empty shop is an obvious incident); `sso.ts` returns `false` (a loud federation error would leak which emails are real customers). Neither guard is a tenancy system; both are the scar tissue from not having one.

11. Deployment topology

11.1 Device and connectivity topology — 17 SKUs, five radios, one VM

Every one of the 30 firmware sketches counted in §3.2 ends up on the same box: the Oracle VM at `140.245.238.154`. Nothing in `firmware/` talks to Vercel, to Neon, or to this website at all — the split this atlas keeps repeating (website owns commerce and content; `platform/` owns every device) is physical, not just architectural. This table is the 17 retail SKUs; the paragraph below it is the other 13 tracked sketches, which are not sold.

Retail SKU (as sold)	Firmware directory(ies)	Code	Radio to the VM
Home Automation Hub	home-hub/	HUB	Wi-Fi -> MQTTS
AquaGuard tank controller	aquaguard/	AQG	Wi-Fi -> MQTTS
-- split variant --	watertank/ + watertank-sensor/	TNK	LoRa, then Wi-Fi
Smart Plug	smart-plug/	PLG	Wi-Fi -> MQTTS
Smart Switch	smart-switch/	SWT	Wi-Fi -> MQTTS
Smart Light	smart-light/	LGT	Wi-Fi -> MQTTS
Smart Fan	smart-fan/	FAN	Wi-Fi -> MQTTS
Smart Lock	smart-lock/	LCK	Wi-Fi -> MQTTS
Curtain controller	curtain/	CRT	Wi-Fi -> MQTTS
Motion Sensor	motion-sensor/	MOT	Wi-Fi -> MQTTS
Energy Monitor	energy-monitor/ (older)	NRG	Wi-Fi -> MQTTS
-- replacement, see note	meter/ (cv-em1 / cv-em3)	NRG	Wi-Fi -> MQTTS
Guardian (wearable SOS)	guardian/	GRD	SIM800L cellular
Agri Pump Starter	agri-starter/	AGR	SIM800L cellular
Touch Switchboard	touchboard/ + touchboard-8/	TCH	Wi-Fi -> MQTTS
FaceDoor	facedoor/	FCD	Wi-Fi -> MQTTS
RFID Gate	rfid-gate/ + rfid-attend/	GAT	Wi-Fi -> MQTTS
Sentinel (gas/climate)	sentinel/	SNL	Wi-Fi -> MQTTS
Load Controller	switchboard/	none*	Wi-Fi -> MQTTS

* **switchboard** has no row in `platform/api/src/serial.ts`'s `PRODUCT_CODES` table. Per `DEVICE_REGISTRY.md`'s own documented rule, a serial minted from the generic fallback code can be generated but can never be resolved back to a device type by `typeFromProductCode()` — this one retail SKU is a genuine, narrow gap in a registry that is otherwise careful about exactly this. The "replacement, see note" row is explained just below the table: **meter/** is a newer metering-IC redesign of the same Energy Monitor SKU, not a duplicate.

The other 13 tracked sketches are not sold, and reach the VM differently or not at all: **camera/** (Wi-Fi, product code **CAM**, no first-party PCB in **hardware/** — an accessory module, not a named line) and **anpr-cam/** (Wi-Fi, **ANP**, a gate camera for the ANPR/visit-pairing feature in `platform/api/src/routes/anpr.ts`, sold to businesses, not retail) both reach the VM over Wi-Fi like the SKUs above. The drone (**drone-fc/**, ESP32-S3, plus **drone-link/** and **witness/**) and the RC car (**rc-car/**, **rc-link/**, **rc-remote/**) are the two items the fact base calls out as present but not sold — see the connectivity notes below.

Wi-Fi, direct (12 of 17 SKUs). `CircuventDevice.h`'s base library joins the home network, then opens `mqtt://mqtt.circuvent.com:8883`. TLS verifies the broker's certificate against Circuvent's own CA (`O=Circuvent Technologies, CN=Circuvent Device CA`, generated once by `platform/scripts/gen-certs.sh` and embedded as `ca.crt` in every sketch that links the shared library) — a public CA is never consulted, so a device will not trust anything else on the internet that merely holds a valid certificate for `mqtt.circuvent.com`.

GSM / cellular (guardian, agri-starter). Both link a SIM800L modem over **TinyGSM** instead of a Wi-Fi radio — confirmed by `grep`, no other sketch includes **TinyGSM** or **SIM800**. A wearable SOS button and a pump starter bolted to a field motor are exactly the two device classes that cannot assume a home router is nearby. MQTTS still terminates on the same broker, over carrier data instead of Wi-Fi; the broker cannot tell the two transports apart.

LoRa 433 MHz (the AquaGuard split pair, `CvTankLink.h`). **watertank-sensor** floats in the tank and has no Wi-Fi radio at all; it speaks LoRa only, to **watertank**, which is the unit within router range. Only **watertank** ever opens a socket — the sensor's readings reach the VM by relay, one more hop than every other SKU, and the sensor cannot be claimed or diagnosed except through it.

ESP-NOW (two unrelated uses of the same radio mode, never the VM). **rc-car** <-> **rc-remote** is peer-to-peer with no broker, no account and no record in this application at all — switching it off is switching off the toy, not a claim in a database (`CircuventRC/rc-protocol.h`). Separately, **touchboard-8** <-> **switchboard** uses ESP-NOW for the *panel-to-relay* hop inside one Load Controller/Touch Switchboard installation (`CvHomeLink.h`); unlike the RC car, one side of that pair still reaches the VM afterwards over Wi-Fi/MQTT. The two uses share a radio mode and nothing else.

MAVLink (drone-fc, internal only). The flight-control loop is split across the ESP32-S3's two cores on purpose: core 1 runs a 1 kHz gyro/cascade/motor loop that "never touches Wi-Fi" (the sketch's own comment), speaking MAVLink to the flight peripherals; core 0 owns Wi-Fi and MQTT and publishes batched 40-byte position records to `cv/<id>/track` (stored in `flight_track`, per `PROTOCOL.md`, never as ordinary telemetry). MAVLink is the real-time control protocol; MQTT is the reporting path — they never touch the same core.

THE VM ITSELF -- 140.245.238.154, Oracle Cloud "Always Free" tier,
VM.Standard.A1.Flex (4 OCPU / 24 GB, ARM), one docker-compose stack,
5 containers on a single private bridge network ("internal"):

```
caddy      :80 (ACME + redirect), :443 (HTTPS + WSS)  PUBLISHED
mosquitto  :8883 (device MQTT/TLS, own CA)           PUBLISHED
            :1883 (plaintext, healthcheck only)      internal
api        :8080 Express control plane               internal, behind caddy
postgres   :5432 -- platform's OWN database, NOT Neon internal only
face       :8000 OpenCV face-embedder for FaceDoor,  internal only
            384 MB hard cap so a leak cannot OOM-kill postgres instead
```

DNS: api.circuvent.com and mqtt.circuvent.com both resolve to the same
public IP. Port 22 (SSH) is the only other opening; everything else is
the OS firewall default-deny plus one matching Oracle Cloud security list.

Mermaid diagram — renders interactively in the Markdown edition

```
flowchart TB
    subgraph skus["17 retail SKUs + camera/anpr-cam"]
        wifi["12 SKUs, direct Wi-Fi<br/>hub · plug · switch · light · fan · lock<br/>curtain · motion · energy · touch · facedoor<br/>rfid-gate · sentinel · switchboard"]
        gsm["guardian · agri-starter<br/>SIM800L / TinyGSM cellular"]
        lora["watertank-sensor --LoRa 433MHz--> watertank<br/>the AquaGuard split pair"]
    end

    subgraph notsold["Not sold (fact base calls these out)"]
        rc["rcar <--ESP-NOW--> rc-remote<br/>peer-to-peer, no cloud, no account"]
        drone["drone-fc (ESP32-S3)<br/>core1: MAVLink @1kHz, never Wi-Fi<br/>core0: Wi-Fi/MQTT --> cv/id/track"]
        panel["touchboard-8 <--ESP-NOW--> switchboard<br/>panel-to-relay hop, then Wi-Fi/MQTT"]
    end

    subgraph vm["Oracle VM 140.245.238.154 (5 containers, 1 network)"]
        caddy["caddy :80/:443<br/>reverse proxy, ACME"]
        mosq["mosquitto :8883<br/>own CA, ACL cv/&id&gt;/#"]
        api["api :8080 Express<br/>internal only"]
        pg["postgres :5432<br/>internal only, own DB"]
        face["face :8000 OpenCV<br/>internal only, 384MB cap"]
    end

    wifi -->|"MQTT :8883"| mosq
    gsm -->|"MQTT over carrier data"| mosq
    lora -->|"Wi-Fi/MQTT :8883"| mosq
    drone -->|"MQTT :8883, track topic"| mosq
    panel -->|"Wi-Fi/MQTT :8883"| mosq
    rc -->|"never reaches this VM"| mosq

    mosq --> api --> pg
    facedoor2["facedoor2[\"facedoor SKU\"] -->|\"embeddings\"| face"]
    caddy --> api

    style rc fill:#F8FAFC,stroke:#94A3B8,stroke-dasharray: 5 5
    style vm fill:#FEF3C7,stroke:#B45309
```

11.2 Hosting — five clouds, one Vercel project, one VM, zero orchestration



Mermaid diagram — renders interactively in the Markdown edition


```

flowchart TB
    gh["GitHub<br/>push / PR"] -->|"webhook"| vercel

    subgraph vercel["VERCEL -- one project, five web products"]
        direction TB
        edge["Edge: proxy.ts<br/>request-id + CSP + host-mount<br/>NO auth check"]
        fn["Serverless: 150 routes, RSC<br/>ad hoc error handling"]
        cron["4 vercel.json crons<br/>CRON_SECRET each"]
    end

    gh -->|"Actions CI, 27/27 startup_failure"| ciFail(("never once green"))

    vercel -->|"DATABASE_URL, HTTP"| neon(("Neon Postgres<br/>10 tables, $7"))
    vercel -->|"CONTROL_PLANE_URL"| vm
    vercel -->|"public GET, no creds"| r2(("Cloudflare R2<br/>firmware manifest"))
    vercel -->|"OIDC/PKCE"| authcv[("auth.circuvent.com<br/>admin SSO issuer")]
    vercel -->|"orders + webhook"| razor[("Razorpay")]

    subgraph vm["platform/ -- Oracle VM 140.245.238.154"]
        vmapi["Express :8080"]
        vmpg[("Postgres :5432<br/>NOT Neon")]
        vmmq["Mosquitto :8883"]
    end

    mob["mobile/ Expo v1.13.1<br/>Play Store"] -->|"REST + wss, direct"| vmapi
    mob -->|"assistant + home analysis<br/>SITE_URL, via Vercel"| vercel
    nat["native/ prototype<br/>iOS never built"] -->|"not deployed anywhere"| vmapi

    style ciFail fill:#FEE2E2,stroke:#B91C1C
    style edge fill:#FEE2E2,stroke:#B91C1C
    style vm fill:#FEF3C7,stroke:#B45309

```

12. The test map

Mermaid diagram — renders interactively in the Markdown edition

```

flowchart LR
    subgraph colocated["CO-LOCATED -- 116 files, next to the code they cover"]
        routes["route.test.ts<br/>API handlers -- largest is<br/>admin/icm/route.test.ts (443 lines)"]
        libts["lib/*.test.ts<br/>app-insights-usage (362)<br/>app-insights-query (324)"]
        admints["admin/*.test.tsx<br/>insights-charts.test.tsx (322)"]
    end

    subgraph dedicated["tests/** -- 120 files, cross-cutting"]
        fwsim["firmware simulation<br/>drone-flight-safety (336)<br/>camera-fps-parity (355)<br/>firmware-avi (424)"]
        icm2["icm.test.ts (342)<br/>incident state machine"]
        util2["lib/extended-utils.test.ts (316)"]
        logo["report-logo.test.ts<br/>THE ONE FAILURE TODAY"]
    end

    subgraph e2edir["e2e/ -- 8 Playwright specs"]
        pw["never run in CI --<br/>the workflow's own comment<br/>says so; .last-run.json is an<br/>undated local cache, not proof"]
    end

    subgraph untested2["UNTESTED -- confirmed by grep"]
        dbts["src/lib/db.ts<br/>the entire DB access layer"]
        cspts["src/lib/csp.ts<br/>generates the CSP header"]
    end

    style logo fill:#FEE2E2,stroke:#B91C1C
    style pw fill:#FEF3C7,stroke:#B45309
    style untested2 fill:#FEE2E2,stroke:#B91C1C

```

236 Jest suite files (120 under tests/**, 116 co-located src/**/*test.* -- the file count reconciles exactly against the runner's own report) run 4,328 tests in 49.5s. One fails today: report-logo.test.ts, because the embedded base64 logo bytes drifted from the PNG on disk -- a small, honest failure, not a flaky one. jest.config.js has no coverageThreshold key, so none of this is enforced by a number -- and the 8 e2e/ Playwright specs, the only tests that exercise a real browser, have never once run in CI (\$9.4).

13. Everything, on one page

THE CIRCUENT SUITE -- EVERYTHING IN THIS WORKING TREE, ONE PAGE

```

5 web products, 1 Next.js app -> VERCEL EDGE (proxy.ts, NO auth gate)
-> 150 routes -> Neon Postgres, HTTP driver, NO transactions, 10 tables,
    schema created at runtime by initDb() on every single call (§7.2)

website/ -- this Next.js app, all of §1-10 below:
    marketing site + e-commerce/Razorpay -- admin back office, 83 routes
    -- smart-home IoT console, 60+ sections -- public developer portal
    150 API routes, 108 pages, 428,355 loc, 4,328 tests (1 failing)
    35 file stores: 32 memory-only, 3 durable -- 5 auth schemes, and
    proxy.ts, which does not gate a single one of them

everything else living in this same tree, none of it part of website/:
firmware/    30 sketches, 17 retail SKUs, ESP32, MQTTS -> one Oracle VM
platform/    that Oracle VM -- Mosquitto + Express + its OWN Postgres
              + Caddy, none of it Vercel, none of it Neon (§11)
hardware/    real KiCad PCB designs and Gerbers for the 17 SKUs
mobile/       Expo app, SHIPPED v1.13.1 to the Play Store -- talks to
              BOTH platform/ (device control) and website (catalogue)
native/       Kotlin/Swift prototype, iOS half has NEVER COMPILED
a drone and an RC car -- present on disk, never sold to anyone
circuitent-platform/ an UNRELATED HR/payroll SaaS. Its README
                  PUBLISHES SEED LOGINS. It shares a name and nothing
                  else -- see §3 for the diagram that says so loudly

```

THE BIGGEST DEFECTS, IN ONE PLACE

```

32 of 35 file stores are memory-only in production -- CMS, CRM, pricing,
currency, tax config, feature flags, telemetry, dev-portal tokens, and
PASSKEYS all vanish on the next cold start (§7.4)
Payment webhook verifies its signature, then does nothing -- a payment
with no browser return is never reconciled (§8.1)
OTA is TLS-pinned but never checks an image hash or signature; the
manifest endpoint DOES exist, correcting 03_INTEGRATIONS (§8.2, §8.3)
proxy.ts sets the request id, the CSP header and the host rewrite --
it does not gate one single route (§10.1)
CI: 14 steps, 13 hard gates, 27 of 27 observed runs = startup_failure,
never once reaching step 1 of 14 (§9.4)
Neon's HTTP driver makes transactions impossible everywhere, always
(§7.2) -- the co-located platform/ Postgres is a different database
and does not share this limitation (§11.2)

```

```

OWNS: five customer-facing web products, a home-grown HMAC session
scheme, 35 file stores (32 of them one redeploy from empty), and
all 150 routes and 108 pages catalogued in §5.
OWNS NOT: a transaction, a signed firmware image, a reconciled webhook,
a green CI run, or any relationship to the payroll app three
folders up that happens to share half its name.

```

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